

CERTIFICATION PRACTICE · OCTOBER 2026

DP-600

Fabric Analytics Engineer Associate

Ten full-length practice test sets — 450 original questions written to the July 2026 skills outline, each set weighted the way the real exam is, with an answer key and an explanation for every question.

10

test sets

45

questions per set

100

minutes per set

450

questions with answers

Prepared for Abhilash Sahani · companion to the DP-600 Study and Question Guide

How to use these sets

Each set is one sitting of the exam. Give yourself **100 minutes**, answer every question before you look at the key, and mark anything you guessed. The answer key for each set follows it immediately, with a sub-area tag on every question so you can see where your marks went, then an explanation that says why the right answer is right and why the others fail.

Sub-area of the July 2026 skills outline	Questions in every set
1 · Maintain a data analytics solution (25-30%)	12
M1 · Implement security and governance	6
M2 · Maintain the analytics development lifecycle	6
2 · Prepare data (45-50%)	21
P1 · Get data	7
P2 · Transform data	8
P3 · Query and analyze data (Visual Query Editor, SQL, KQL, DAX)	6
3 · Implement and manage semantic models (25-30%)	12
S1 · Design and build semantic models	6
S2 · Optimize enterprise-scale semantic models	6

Every set includes a **four-question case study**, counted inside the 45, because the real exam has one and it is locked once you leave it. The question formats follow the real exam: single answer, multiple response, Yes/No hot-area grids, drag-and-drop matching, build-list sequencing, and code completion in SQL, KQL, DAX and PySpark. Across the bank: Single answer: 368 · Hot area: 37 · Multiple response: 24 · Build list: 8 · Drag and drop: 13.

Scoring

The real exam is scored 1–1000 with 700 to pass, and the score is scaled, so it does not map to a fixed number of questions. These sets use a stricter rule of thumb: **34 out of 45 (75%)**. That is the level the official practice assessment advice points to. Score multi-part items fractionally. Two sets in a row at 75% or more, on sets you haven't seen, is a good sign you are ready to book.

Order of the sets

Set	Focus	Level	Case study
Set 01	Foundations — the shape of Fabric	Level 1 · Foundation	Contoso Retail
Set 02	Foundations — stores, routes and the gold layer	Level 1 · Foundation	Fabrikam Health
Set 03	Core — reading SQL, KQL, DAX and PySpark	Level 2 · Core	Tailspin Airlines
Set 04	Core — governing what you build	Level 2 · Core	Woodgrove Bank
Set 05	Exam standard — constraints decide	Level 3 · Exam standard	Litware Energy
Set 06	Exam standard — build, release, repeat	Level 3 · Exam standard	Adventure Works Cycles
Set 07	Exam standard — shaping the gold layer	Level 3 · Exam standard	Wide World Importers
Set 08	Advanced — performance and the hard edges	Level 4 · Advanced	Proseware Telecom
Set 09	Advanced — mixed scenarios	Level 4 · Advanced	Margie's Travel
Set 10	Final mock exam	Level 5 · Final mock	Coho Winery

About these questions. Every question is original, written against the published skills outline and documented product behaviour. None is a real exam item and none is copied from a brain dump: those breach the exam agreement and can cost you the certification. Fabric ships changes every month, so where a question depends on a default or a limit (V-Order defaults, Direct Lake guardrails, preview features), check the current Microsoft Learn page before exam day.

Blueprint coverage across the bank

Sub-area	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Total
M1 · Implement security and governance	6	6	6	6	6	6	6	6	6	6	60
M2 · Maintain the analytics development lifecycle	6	6	6	6	6	6	6	6	6	6	60
P1 · Get data	7	7	7	7	7	7	7	7	7	7	70
P2 · Transform data	8	8	8	8	8	8	8	8	8	8	80
P3 · Query and analyze data	6	6	6	6	6	6	6	6	6	6	60
S1 · Design and build semantic models	6	6	6	6	6	6	6	6	6	6	60
S2 · Optimize enterprise-scale semantic models	6	6	6	6	6	6	6	6	6	6	60

Exam-day reminders the sets are built to train

- **Find the constraint before reading the options.** Look for phrases like *without duplicating the data, must support T-SQL updates, security defined in SQL must be enforced, least privilege, minimise administrative effort*.
- **"Without copying"** points to a shortcut. **"A replica of an operational database, with no pipeline"** points to mirroring.
- A **Lakehouse SQL analytics endpoint is read-only**. T-SQL INSERT, UPDATE, DELETE or stored procedures that write need a **Warehouse**.
- **Direct Lake on SQL analytics endpoint** respects SQL-defined security, supports views and falls back to DirectQuery. **Direct Lake on OneLake** supports composite models and calculated objects but has no fallback.
- OneLake security roles and RLS only restrict **Viewers** and item-permission users. Admin, Member and Contributor bypass them.
- When a Direct Lake model is slow, fix the **Delta table** first: OPTIMIZE, V-Order, fewer small files.
- Git integration is **version control**. Deployment pipelines are **promotion**. Impact analysis comes **before a breaking change**.

Set 01

Foundations — the shape of Fabric

Core vocabulary across the whole outline: items, roles, routes in, the gold layer, and the three query languages.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 1 · Foundation

12

Maintain a data analytics solution
25–30%

21

Prepare data
45–50%

12

Implement and manage semantic models
25–30%

Q1 SINGLE ANSWER

A group of store managers must open reports and read data in a workspace. They must not be able to create, edit or delete any item. Which workspace role should you assign, following least privilege?

- ☐ A Member
- ☐ B Contributor
- ☐ C Viewer
- ☐ D Admin

Q2 SINGLE ANSWER

You share a lakehouse with an analyst from another team. The analyst must query its tables with T-SQL through the SQL analytics endpoint, but must not read the underlying files with Spark. Which permissions do you grant?

- ☐ A Read only
- ☐ B Write
- ☐ C Read and ReadData (Read all SQL endpoint data)
- ☐ D Read and ReadAll (Read all Apache Spark / OneLake data)

Q3 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about Microsoft Purview sensitivity labels in Fabric, select Yes if it is true.

Statement	Yes	No
1. A sensitivity label applied to a semantic model stays on data exported from it to Excel.	☐	☐
2. Applying the Highly Confidential label stops workspace Viewers from opening the item.	☐	☐
3. Labels can be applied to lakehouses and warehouses as well as to Power BI reports.	☐	☐

Q4 SINGLE ANSWER

A data engineer wants to signal that a lakehouse is ready for others to reuse. No special approval process exists. Which action is available to anyone with write permission on the item?

- ☐ A Apply a domain to the lakehouse
- ☐ B Certify the lakehouse
- ☐ C Mark the lakehouse as Master data
- ☐ D Promote the lakehouse

Q5 SINGLE ANSWER

A semantic model has a Sales table with a column Sales[RepEmail]. Each sales rep must see only their own rows when viewing the report. Which DAX filter expression do you put on the Sales table in the role?

- ☐ A Sales[RepEmail] = USERPRINCIPALNAME()
- ☐ B Sales[RepEmail] = USERNAME()
- ☐ C Sales[RepEmail] = CUSTOMDATA()
- ☐ D Sales[RepEmail] = SELECTEDVALUE(Sales[RepEmail])

Q6 MULTIPLE RESPONSE — CHOOSE TWO

Which two source-control providers can a Fabric workspace connect to through Fabric Git integration?

- ☐ A An Azure Blob Storage container
- ☐ B GitHub
- ☐ C OneDrive for Business
- ☐ D Azure DevOps (Azure Repos)
- ☐ E A SharePoint document library

Q7 SINGLE ANSWER

You create a new deployment pipeline and accept the defaults. How many stages does it have?

- ☐ A Four: Development, Test, UAT and Production
- ☐ B Three: Development, Test and Production
- ☐ C Two: Development and Production
- ☐ D Five: Development, Test, UAT, Pre-production and Production

Q8 SINGLE ANSWER

Which statement correctly describes a Power BI template file (.pbit)?

- ☐ A It contains only the report theme colours.
- ☐ B It contains only a data-source connection, so Power BI Desktop opens pre-connected.
- ☐ C It contains the report pages, the model definition and queries, but no data.
- ☐ D It contains the report, the model definition and a full copy of the data.

Q9 BUILD LIST — SEQUENCE

You need to deploy a metadata-only change to a semantic model in a Fabric workspace using Tabular Editor. Put the required actions in order.

Select the 4 actions required and arrange them in the correct order. Not every action is used.

- ☐ A Make the change and save it to the connected model
- ☐ B Republish the .pbix file from Power BI Desktop
- ☐ C Connect Tabular Editor to the workspace address and open the model
- ☐ D Create a deployment pipeline
- ☐ E Copy the workspace connection address (powerbi://api.powerbi.com/v1.0/myorg/<workspace>)
- ☐ F Make sure the capacity's XMLA endpoint setting is Read Write

Q10 SINGLE ANSWER

You plan to rename a column in a warehouse table. Before you make the change, you must find out which semantic models and reports depend on the warehouse. What should you use?

- ☐ A The deployment pipeline compare view
- ☐ B Impact analysis on the warehouse, from lineage view or the item's details
- ☐ C The Fabric Capacity Metrics app
- ☐ D The warehouse's query activity (query insights) views

Q11 SINGLE ANSWER

Fifteen report authors keep importing the same warehouse tables into their own models, and their revenue figures don't agree. What should you do?

- ☐ A Send every author a .pbids file for the warehouse, so they all connect to the same tables.
- ☐ B Publish one endorsed shared semantic model and give the authors Build permission on it.
- ☐ C Give every author the workspace Admin role, so they can review and fix each other's models.
- ☐ D Turn on Git integration for the authors' workspace, so model changes are reviewed.

Q12 SINGLE ANSWER

A data engineering team writes PySpark. They will land CSV files, nested JSON and product images before they clean anything. Which Fabric item should they use for the raw (bronze) layer?

- ☐ A Fabric SQL database
- ☐ B Lakehouse
- ☐ C Eventhouse
- ☐ D Warehouse

Q13 SINGLE ANSWER

Your organisation has five years of Parquet files in an Azure Data Lake Storage Gen2 account. Analysts must query them from a Fabric lakehouse, and the files must not be copied. What should you create?

- ☐ A A mirrored database over the storage account
- ☐ B A pipeline with a Copy activity that loads the files
- ☐ C A Dataflow Gen2 that reads the files into a table
- ☐ D An external shortcut to the ADLS Gen2 location

Q14 SINGLE ANSWER

An Azure Cosmos DB account powers an e-commerce application. Analysts need data that's at most a few minutes old in OneLake. No ETL may be built, and no analytical load may be put on the source. What should you implement?

- ☐ A Mirroring of the Cosmos DB account into Fabric
- ☐ B A Dataflow Gen2 with scheduled refresh
- ☐ C A scheduled pipeline every five minutes
- ☐ D A DirectQuery semantic model over Cosmos DB

Q15 SINGLE ANSWER

Business analysts know Power Query well. They must build and maintain a transformation themselves, and its output must be written to a lakehouse table. Which item should they use?

- ☐ A KQL queryset with an update policy
- ☐ A Spark job definition with a lakehouse reference
- ☐ A Dataflow Gen2 with a lakehouse destination
- ☐ A notebook that writes to the lakehouse with PySpark

Q16 DRAG AND DROP — MATCH

Match each discovery need to the Fabric surface that meets it.

Choices — each may be used once, more than once, or not at all

- ☐ Capacity Metrics app
- ☐ Real-Time hub
- ☐ OneLake catalog
- ☐ Lineage view

Requirement	Choice
1. Find a certified warehouse that another team owns and request access to it	
2. Browse event streams and Azure Event Hubs sources available to the organisation	
3. See which reports are built on a specific lakehouse	

Q17 SINGLE ANSWER

A pipeline must copy tables from a SQL Server instance on the company's internal network into a lakehouse. What must be in place for the Copy activity to reach the source?

- ☐ An on-premises data gateway, and a connection that uses it
- ☐ A personal-mode gateway on a developer's laptop
- ☐ A OneLake shortcut
- ☐ Nothing; Fabric reaches internal networks directly

Q18 SINGLE ANSWER

In a Fabric Warehouse, logic must insert the day's new rows into `gold.FactSales` and then update a load-audit table, as one unit a pipeline can call. Which object should you create?

- ☐ A stored procedure
- ☐ An inline table-valued function
- ☐ A view
- ☐ A SQL analytics endpoint query

Q19 SINGLE ANSWER

A silver DataFrame `df` has nulls in `Region` and `Discount`. Nulls in `Region` must become "Unknown" and nulls in `Discount` must become 0, and no rows may be lost. Which PySpark line is correct?

- ☐ `df = df.filter("Region IS NOT NULL")`
- ☐ `df = df.fillna({"Region": "Unknown", "Discount": 0})`
- ☐ `df = df.na.drop(subset=["Region", "Discount"])`
- ☐ `df = df.dropna()`

Q20 SINGLE ANSWER

A staging table stores order dates as text in the format 2026-08-14 . The gold warehouse table needs a real date column. Which expression should the load use?

- ☐ A Keep it as text and convert it in a DAX measure
- ☐ B FORMAT(OrderDateText, 'yyyy-MM-dd')
- ☐ C CAST(OrderDateText AS date)
- ☐ D CAST(OrderDateText AS varchar(10))

Q21 SINGLE ANSWER

The source has three tables: Product, Subcategory and Category, joined in a chain. You are building the gold star schema. What should you do?

- ☐ A Put the category name on the fact table.
- ☐ B Create a bridge table between Product and Category.
- ☐ C Load the three tables separately and relate them in the semantic model.
- ☐ D Denormalise the three into a single DimProduct table with Subcategory and Category columns.

Q22 SINGLE ANSWER

You need a physical summary table in a Fabric Warehouse with one row per product category per month, created and populated in one statement. Complete the statement.

```
____ gold.CategoryMonthSales
AS
SELECT p.Category, d.MonthStart, SUM(f.Amount) AS Amount
FROM   gold.FactSales f
JOIN   gold.DimProduct p ON p.ProductKey = f.ProductKey
JOIN   gold.DimDate d    ON d.DateKey    = f.DateKey
____ p.Category, d.MonthStart;
```

- ☐ A SELECT INTO with ORDER BY
- ☐ B INSERT ... VALUES
- ☐ C CREATE VIEW ... AS SELECT ... GROUP BY
- ☐ D CREATE TABLE ... AS SELECT ... GROUP BY

Q23 SINGLE ANSWER

Every product must appear in the result, including products that have never been sold, with their sales total where one exists. Which join should you use from DimProduct to FactSales?

- ☐ A INNER JOIN
- ☐ B LEFT OUTER JOIN from DimProduct to FactSales
- ☐ C CROSS JOIN
- ☐ D RIGHT OUTER JOIN from DimProduct to FactSales

Q24 SINGLE ANSWER

In a Dataflow Gen2, a customer query has repeated rows for the same CustomerID. Each CustomerID must appear once. What should you do in the Power Query editor?

- ☐ A Select all columns and choose Remove blank rows
- ☐ B Select CustomerID and choose Remove duplicates
- ☐ C Group by every column and count the rows
- ☐ D Select CustomerID and choose Keep duplicates

Q25 SINGLE ANSWER

A business analyst who doesn't write SQL must join two warehouse tables, filter them, group the result, and save it as a view. What should they use?

- ☐ A The warehouse's Visual Query Editor
 - ☐ B A KQL queryset connected to the warehouse
 - ☐ C A notebook attached to the warehouse
 - ☐ D The DAX query view in Power BI Desktop
-

Q26 SINGLE ANSWER

Which KQL query returns the number of events per hour for the last day?

- ☐ A Events | take 24 | extend Hour = hourofday(Timestamp)
 - ☐ B SELECT COUNT(*) FROM Events GROUP BY HOUR(Timestamp)
 - ☐ C Events | where Timestamp > ago(1d) | project count() by Timestamp
 - ☐ D Events | where Timestamp > ago(1d) | summarize count() by bin(Timestamp, 1h)
-

Q27 SINGLE ANSWER

Which T-SQL query returns the ten products with the highest total revenue?

- ☐ A SELECT ProductKey, SUM(Revenue) FROM gold.FactSales GROUP BY ProductKey LIMIT 10;
 - ☐ B SELECT ProductKey, MAX(Revenue) FROM gold.FactSales GROUP BY ProductKey;
 - ☐ C SELECT TOP (10) ProductKey, Revenue FROM gold.FactSales ORDER BY Revenue DESC;
 - ☐ D SELECT TOP (10) ProductKey, SUM(Revenue) AS Rev FROM gold.FactSales GROUP BY ProductKey ORDER BY Rev DESC;
-

Q28 SINGLE ANSWER

In the DAX query view, you want the five product categories with the highest [Total Sales]. Which query is valid?

- ☐ A EVALUATE TOPN(5, SUMMARIZECOLUMNS('Product'[Category], "Sales", [Total Sales]), [Sales], DESC)
 - ☐ B TOPN(5, SUMMARIZECOLUMNS('Product'[Category], "Sales", [Total Sales]), [Sales], DESC)
 - ☐ C SELECT TOP 5 Category, [Total Sales] FROM 'Product' ORDER BY [Total Sales] DESC
 - ☐ D EVALUATE CALCULATE([Total Sales], TOPN(5, VALUES('Product'[Category]), [Total Sales]))
-

Q29 SINGLE ANSWER

In KQL, you need to keep every existing column and add a new column `TempF` computed from `TempC`. Which operator should you use?

- ☐ A summarize
- ☐ B extend
- ☐ C distinct
- ☐ D project

Q30 SINGLE ANSWER

Which surface is designed for writing and saving KQL queries against an Eventhouse?

- ☐ A A Dataflow Gen2
 - ☐ B The Visual Query Editor
 - ☐ C A KQL queryset
 - ☐ D The SQL analytics endpoint of a lakehouse
-

Q31 SINGLE ANSWER

Your gold tables are Delta tables in a Fabric lakehouse. Reports need Import-class query speed, and you don't want to schedule data refreshes that copy the data. Which storage mode should you choose?

- ☐ A Import
 - ☐ B Dual
 - ☐ C DirectQuery
 - ☐ D Direct Lake
-

Q32 SINGLE ANSWER

FactSales has OrderDateKey and ShipDateKey, and both relate to DimDate. Only the OrderDate relationship is active. A measure must total sales by ship date. Which DAX function activates the ship-date relationship inside the measure?

- ☐ A RELATED
 - ☐ B TREATAS
 - ☐ C USERELATIONSHIP
 - ☐ D CROSSFILTER
-

Q33 SINGLE ANSWER

Accounts can have several customers, and customers can hold several accounts. You need to analyse balances (on the Account side) by customer attributes. What is the recommended model design?

- ☐ A Duplicate each balance row once per account holder, so every customer gets the full balance
- ☐ B A CustomerAccount bridge table, with the bridge-to-DimAccount relationship filtering both ways
- ☐ C A direct many-to-many relationship between FactBalance and DimCustomer on CustomerName
- ☐ D Merge DimCustomer and DimAccount into one table with a row per customer-account pair

Q34 DRAG AND DROP — MATCH

Match each requirement to the modelling feature that meets it.

Choices — each may be used once, more than once, or not at all

- A** What-if parameter
- B** Dynamic format string
- C** Calculation group
- D** Field parameter

Requirement	Choice
1. Thirty measures each need PY and YoY variants without writing 60 new measures	
2. Users choose, from a slicer, whether a chart shows Revenue, Margin or Units	
3. A YoY % calculation item must show as a percentage, while the others show as currency	
4. An analyst wants to test the effect of a 0-20% price increase with a slider	

Q35 SINGLE ANSWER

Sales has columns Qty and UnitPrice, and no Amount column. Which measure correctly returns total revenue?

- A** Revenue = SUM(Sales[Qty] * Sales[UnitPrice])
- B** Revenue = SUM(Sales[Qty]) * SUM(Sales[UnitPrice])
- C** Revenue = AVERAGEX(Sales, Sales[Qty] * Sales[UnitPrice])
- D** Revenue = SUMX(Sales, Sales[Qty] * Sales[UnitPrice])

Q36 SINGLE ANSWER

A visual shows Sales by City inside a Region hierarchy. A measure must return the text "City" on city rows and "Region" on region subtotal rows. Which function tells you which level is being evaluated?

- A** ISINSCOPE
- B** ISBLANK
- C** CONTAINSSTRING
- D** ISERROR

Q37 SINGLE ANSWER

A report page is slow. You need to find which visual takes the longest and copy the DAX query it sends, without leaving Power BI Desktop. What should you use?

- A** The Capacity Metrics app
- B** Impact analysis
- C** Lineage view
- D** Performance Analyzer

Q38 SINGLE ANSWER

What does a refresh of a Direct Lake semantic model do?

- A** It frames the model: it points the model at the latest Delta table versions without copying data.
- B** It copies all rows from the Delta tables into model memory, replacing the old copy.
- C** It re-runs the Power Query transformations defined for each table.
- D** It rebuilds every calculated column and table from the source data.

Q39 MULTIPLE RESPONSE — CHOOSE TWO

You configure incremental refresh on an Import fact table. Which two items are required?

- ☐ A A deployment pipeline
- ☐ B A Direct Lake storage mode on the table
- ☐ C Date/time parameters named exactly RangeStart and RangeEnd
- ☐ D A calculated column that flags the current year
- ☐ E A filter on the fact query that uses RangeStart and RangeEnd

Q40 SINGLE ANSWER

An Import model will grow to about 25 GB after refresh in the service. What must be enabled for the model to grow past the default limit?

- ☐ A Direct Lake fallback to DirectQuery
- ☐ B Large semantic model storage format
- ☐ C Query caching on the capacity
- ☐ D Automatic page refresh on the report

Q41 SINGLE ANSWER

A Spark job appends small files to a gold Delta table every 15 minutes. After a month, Direct Lake queries are slower even though the row count has barely grown. What should you do first?

- ☐ A Switch the model to Import mode
- ☐ B Add variables to every DAX measure
- ☐ C Run OPTIMIZE on the table to compact the small files
- ☐ D Run VACUUM with a retention of 0 hours

Case study · Contoso Retail

Questions 42–45

Existing environment

Contoso Retail runs 180 stores. Sales data lands every night from an on-premises SQL Server into a Fabric lakehouse named LH_Sales, built by a PySpark team. A gold semantic model in Direct Lake mode serves 300 report users. A supplier shares weekly price files in an Amazon S3 bucket.

Today, all 300 report users are workspace Contributors. The store-performance team has noticed that the nightly load sometimes inserts the same order twice.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Requirements

1. R1. Report users must not be able to change any item in the workspace.
2. R2. Supplier price files must be queryable in Fabric without being copied.
3. R3. Duplicate orders must be removed before they reach the gold table.
4. R4. Direct Lake queries must stay fast as the nightly appends build up.

Q42 SINGLE ANSWER

Which change satisfies R1 with the least administrative effort?

- ☐ A Make the users workspace Members and turn on item sharing
- ☐ B Keep the users as Contributors and apply a Highly Confidential label
- ☐ C Change the users' workspace role to Viewer, or give them access through an app
- ☐ D Create one workspace per store and make each manager Admin

Q43 SINGLE ANSWER

Which approach satisfies R2?

- ☐ A A pipeline that copies the S3 files into LH_Sales every week
 - ☐ B A Dataflow Gen2 that loads the files into a warehouse
 - ☐ C An external shortcut in LH_Sales that points to the S3 bucket
 - ☐ D Mirroring the S3 bucket
-

Q44 SINGLE ANSWER

The PySpark team must satisfy R3 while building the silver order table. Which line removes rows that repeat the same OrderID?

- ☐ A `df.dropDuplicates(["OrderID"])`
 - ☐ B `df.dropna(subset=["OrderID"])`
 - ☐ C `df.distinct().count()`
 - ☐ D `df.orderBy("OrderID")`
-

Q45 SINGLE ANSWER

Which practice best satisfies R4?

- ☐ A Turn on auto compaction, and run OPTIMIZE on tables that are already fragmented
- ☐ B Turn off V-Order on the gold tables so the nightly writes are faster
- ☐ C Schedule the semantic model to refresh every 15 minutes during the day
- ☐ D Convert the gold tables to CSV so they're simpler to scan

End of Set 01 — answers follow on the next page.

Set 01 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	C	M1		16	1→C 2→B 3→D	P1		31	D	S1	
2	C	M1		17	A	P1		32	C	S1	
3	Yes · No · Yes	M1		18	A	P2		33	B	S1	
4	D	M1		19	B	P2		34	1→C 2→D 3→B 4→A	S1	
5	A	M1		20	C	P2		35	D	S1	
6	B, D	M2		21	D	P2		36	A	S1	
7	B	M2		22	D	P2		37	D	S2	
8	C	M2		23	B	P2		38	A	S2	
9	F → E → C → A	M2		24	B	P2		39	C, E	S2	
10	B	M2		25	A	P3		40	B	S2	
11	B	M2		26	D	P3		41	C	S2	
12	B	P1		27	D	P3		42	C	M1	
13	D	P1		28	A	P3		43	C	P1	
14	A	P1		29	B	P3		44	A	P2	
15	C	P1		30	C	P3		45	A	S2	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: C M1 · Implement security and governance

Viewer is the only workspace role that can't create or change items. Contributor, Member and Admin can all create and edit content. Member and Admin can also manage access, which goes further still.

Q2 Answer: C M1 · Implement security and governance

ReadData lets a user query the item through the SQL analytics endpoint. ReadAll gives OneLake access to the files, so Spark and other engines can read them, which the stem rules out. Read alone shows the item's metadata but lets the user query nothing. Write grants everything.

Q3 Answer: Yes · No · Yes M1 · Implement security and governance

Labels classify content and protect exported files. They carry through to Excel, PowerPoint and PDF exports. They don't grant or remove access inside Fabric: that's done by workspace roles, item permissions and OneLake security. Labels apply across Fabric items, including lakehouses, warehouses, semantic models, reports and dataflows.

Q4 Answer: D M1 · Implement security and governance

Promoted is the endorsement level any user with write permission can apply. Certified and Master data are limited to users the Fabric administrator authorises. Domains are assigned to workspaces, not items, and are not an endorsement.

Q5 Answer: A M1 · Implement security and governance

USERPRINCIPALNAME() returns the signed-in user's UPN (normally their email) in the Power BI service, so the filter matches each rep's own rows. USERNAME() can return DOMAIN\user in Desktop, which is why UPN is preferred. SELECTEDVALUE reads the filter context, not the user's identity. CUSTOMDATA comes from the connection string, which is an embedding scenario.

Q6 Answer: B, D M2 · Maintain the analytics development lifecycle

Fabric Git integration connects a workspace to a branch in an Azure DevOps or GitHub repository. SharePoint, OneDrive and Blob Storage store files, but they aren't Git providers and give you no commit history or pull requests.

Q7 Answer: B M2 · Maintain the analytics development lifecycle

The default is three stages, named Development, Test and Production. You can customise a pipeline to have anywhere from 2 to 10 stages and rename them.

Q8 Answer: C M2 · Maintain the analytics development lifecycle

A .pbix holds everything except the data: report layout, model, measures and queries, plus any parameters to fill in when it's opened. A file that holds only a connection is a .pbids. A file that also holds the data is a .pbix.

Q9 Answer: F → E → C → A M2 · Maintain the analytics development lifecycle

XMLA write needs the Read Write setting on the capacity. Then you connect an external tool using the workspace connection address and save the change straight to the model. Republishing the .pbix isn't needed, and it would overwrite the reports too.

Q10 Answer: B M2 · Maintain the analytics development lifecycle

Impact analysis lists the downstream items and reports that depend on an item, and lets you notify their contacts. That's the standard answer for "before a breaking change". Capacity Metrics shows compute use. Pipeline compare shows differences between stages. Query insights shows which queries ran.

Q11 Answer: B M2 · Maintain the analytics development lifecycle

A shared semantic model gives everyone one governed set of measures and relationships. Build permission lets authors create their own reports on top of it, and endorsement helps them find it and trust it. A .pbids file only shares a connection, so each author would still build their own model.

Q12 Answer: B P1 · Get data

A lakehouse has a Files area for unstructured and semi-structured data as well as Delta tables, and Spark is its main interface. A warehouse holds structured tables only. An eventhouse is built for event and time-series data. A SQL database is an operational (OLTP) store.

Q13 Answer: D P1 · Get data

A shortcut is a pointer, so the data stays where it is and nothing is copied. ADLS Gen2 is a supported external shortcut target. A pipeline and a dataflow both copy the data. Mirroring replicates operational databases, not files in storage.

Q14 Answer: A P1 · Get data

Mirroring keeps a near-real-time replica of supported operational databases, Cosmos DB among them, in OneLake as Delta. There's no pipeline to build or maintain. A five-minute pipeline is ETL. DirectQuery puts query load straight onto the application's database.

Q15 Answer: C P1 · Get data

Dataflow Gen2 is low-code Power Query with explicit destinations, including lakehouses, warehouses and KQL databases, so analysts can own it. Notebooks and Spark job definitions need code. A KQL queryset only queries data. It doesn't load it.

Q16 Answer: 1→C 2→B 3→D P1 · Get data

The OneLake catalog is where you discover and govern stored data items across the tenant. The Real-Time hub is where you discover streaming data and events. Lineage view shows upstream and downstream dependencies. The Capacity Metrics app reports on compute, not data.

Q17 Answer: A P1 · Get data

Sources on a private network are reached through the on-premises data gateway, using a connection set up on that gateway. A personal-mode gateway supports only Power BI semantic model refresh for one user, so it isn't for production pipelines. A shortcut doesn't target a SQL Server database.

Q18 Answer: A P2 · Transform data

A stored procedure can run DML (INSERT, UPDATE, DELETE) across several tables, and a pipeline can call it with the Stored procedure activity. Views and table-valued functions return data but can't change it. A lakehouse SQL analytics endpoint is read-only.

Q19 Answer: B P2 · Transform data

fillna with a dictionary replaces nulls column by column and keeps every row. The other three options remove the rows that have nulls, which the requirement forbids.

Q20 Answer: C P2 · Transform data

Convert the type in the gold transformation so it's stored once in the correct type for every consumer. CAST to date handles ISO-format text. Casting to varchar leaves it as text. FORMAT returns text. Leaving the conversion to DAX pushes it downstream, where every query pays for it.

Q21 Answer: D P2 · Transform data

Flattening a snowflaked dimension into one table is standard star-schema practice. It removes relationship hops in the semantic model and makes the model simpler to use. Putting attributes on the fact breaks its grain. A bridge table is for many-to-many relationships, which this isn't.

Q22 Answer: D P2 · Transform data

CREATE TABLE AS SELECT (CTAS) creates and fills a table in one statement, and the GROUP BY produces one row per category per month. A view isn't a physical table. INSERT ... VALUES inserts literal rows, so it can't aggregate.

Q23 Answer: B P2 · Transform data

A LEFT OUTER JOIN keeps every row from the left table (DimProduct), with NULL sales for products that have no matching fact rows. An INNER JOIN drops the unsold products. A CROSS JOIN pairs every product with every fact row.

Q24 Answer: B P2 · Transform data

Remove duplicates on the key column keeps the first row for each CustomerID. Remove blank rows deals with empty rows, not repeated ones. Grouping by every column would only merge rows that are identical in every column.

Q25 Answer: A P3 · Query and analyze data

The Visual Query Editor is a no-code, Power Query-style canvas in the warehouse or SQL analytics endpoint. It generates SQL and can save the result as a view. Notebooks and querysets need code. The DAX query view works against semantic models, not the warehouse.

Q26 Answer: D P3 · Query and analyze data

where filters the rows, summarize ... by groups and aggregates, and bin() puts timestamps into fixed buckets. project selects columns but can't aggregate. The SELECT option is SQL, not KQL. take 24 just returns 24 arbitrary rows.

Q27 Answer: D P3 · Query and analyze data

The correct query aggregates per product, sorts by the total, and keeps ten. LIMIT isn't T-SQL. TOP (10) without GROUP BY returns the ten largest individual rows, which could include the same product twice, rather than the top products by total. MAX returns the largest single row per product, with no top-10 limit.

Q28 Answer: A P3 · Query and analyze data

A DAX query starts with EVALUATE and must return a table. Here TOPN wraps a SUMMARIZECOLUMNS grouping. The SELECT option is SQL. The TOPN without EVALUATE is an expression, not a query. EVALUATE CALCULATE(...) returns a scalar, which EVALUATE can't output.

Q29 Answer: B P3 · Query and analyze data

extend adds computed columns and keeps all the existing ones. project returns only the columns you list. summarize aggregates the rows. distinct removes duplicate rows.

Q30 Answer: C P3 · Query and analyze data

A KQL queryset is the Fabric item for writing, running and saving KQL queries against KQL databases. The Visual Query Editor and the SQL analytics endpoint generate or use T-SQL. A Dataflow Gen2 loads and transforms data.

Q31 Answer: D S1 · Design and build semantic models

Direct Lake reads Delta-Parquet column data from OneLake straight into the VertiPaq engine when a query needs it. Its refresh (framing) only updates metadata, so no data is copied. Import copies the data on each refresh. DirectQuery sends every query to the source. Dual only means something inside a composite model.

Q32 Answer: C S1 · Design and build semantic models

USERRELATIONSHIP, used as a CALCULATE modifier, turns on an inactive relationship for that calculation only. That's the standard pattern for a role-playing date dimension. CROSSFILTER changes the filter direction. TREATAS applies a filter through a virtual relationship. RELATED fetches a column from the one side of a relationship.

Q33 Answer: B S1 · Design and build semantic models

A many-to-many relationship between dimensions is modelled with a bridge (factless) table. The filter from DimCustomer has to flow through the bridge to DimAccount and on to the fact, which is why one relationship is set to filter in both directions. Duplicating fact rows inflates totals. Merging the tables destroys the grain.

Q34 Answer: 1→C 2→D 3→B 4→A S1 · Design and build semantic models

Calculation groups apply reusable logic (via `SELECTEDMEASURE`) to any measure. Field parameters let users swap which field a visual uses. Dynamic format strings change formatting based on a DAX expression. A what-if parameter creates a disconnected table of values for scenario analysis.

Q35 Answer: D S1 · Design and build semantic models

`SUMX` is an iterator: it works out `Qty × UnitPrice` for each row and then adds up the results. Multiplying two totals gives a wrong answer whenever there's more than one row. `SUM` only takes a single column reference, so `SUM(Qty * UnitPrice)` is invalid. `AVERAGEX` averages instead of summing.

Q36 Answer: A S1 · Design and build semantic models

`ISINSCOPE('Geo'[City])` is `TRUE` when the current row is grouped by City, so it tells hierarchy levels and subtotals apart. That's one of the information functions the skills outline names. The other options test values, not grouping levels.

Q37 Answer: D S2 · Optimize enterprise-scale semantic models

Performance Analyzer records each visual's duration, split into DAX query, visual display and other. Copy query then gives you the DAX to tune in the DAX query view. The other tools work at the item or capacity level, not the visual level.

Q38 Answer: A S2 · Optimize enterprise-scale semantic models

Direct Lake refresh is called framing, and it only touches metadata. Data is loaded into memory later, when queries need it, through a step called transcoding. That's why a Direct Lake refresh takes seconds and uses very little capacity.

Q39 Answer: C, E S2 · Optimize enterprise-scale semantic models

Incremental refresh needs the two reserved parameters and a filter that uses them, and the filter should fold back to the source. Direct Lake tables don't use incremental refresh. A calculated column would break folding.

Q40 Answer: B S2 · Optimize enterprise-scale semantic models

Without large semantic model storage format, a model in the service is capped at a small size limit. With it turned on, the model can grow up to the capacity's limit. The Desktop upload limit (10 GB) still applies to the .pbix you publish, and the model then grows through refresh. The other settings don't affect size.

Q41 Answer: C S2 · Optimize enterprise-scale semantic models

Lots of small files mean more column segments for Direct Lake to transcode. Compacting them with `OPTIMIZE` (then relying on auto compaction) fixes the problem at the table. `VACUUM` removes unreferenced files but doesn't improve how active files are laid out. A zero-hour retention also breaks time travel and can affect readers part-way through a query.

Q42 Answer: C M1 · Implement security and governance

Contributors can create and edit items. Viewer, or consuming content through an app, gives read-only access. A label grants and removes nothing. Member gives even more rights. One workspace per store means a lot more to administer.

Q43 Answer: C P1 · Get data

Amazon S3 is a supported external shortcut target. A shortcut gives access in place, with no copy. Copy activities and dataflows copy the data. Mirroring is for operational databases, not object storage.

Q44 Answer: A P2 · Transform data

`dropDuplicates` with a subset removes rows that repeat the key, even when other columns differ. `distinct()` only removes rows that are identical in every column, and adding `count()` returns a number instead of a DataFrame. `dropna` removes rows with nulls. `orderBy` only sorts.

Q45 Answer: A S2 · Optimize enterprise-scale semantic models

Direct Lake performance depends on how the Delta files are laid out. Auto compaction keeps file counts healthy after each write, and OPTIMIZE fixes existing fragmentation. Refreshing more often only re-frames metadata. CSV can't be read by Direct Lake. V-Order speeds up reads, so turning it off makes reports slower.

Foundations — stores, routes and the gold layer

Choosing the right item and the right ingestion route, and the T-SQL and PySpark you need to shape gold tables.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 1 · Foundation

12 Maintain a data analytics solution 25-30%	21 Prepare data 45-50%	12 Implement and manage semantic models 25-30%
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Q1 SINGLE ANSWER

A team lead must be able to add colleagues to a workspace as Contributors and share individual items. They must not be able to delete the workspace or manage other admins. Which role is the least privileged one that meets the requirement?

- ☐ A Member
- ☐ B Admin
- ☐ C Viewer
- ☐ D Contributor

Q2 SINGLE ANSWER

A lakehouse holds a Files/legal/contracts folder. Only the Legal group may read that folder, but every analyst may read all the tables. The analysts are workspace Viewers. What should you configure?

- ☐ A A row-level security role in the semantic model that filters on the folder path
- ☐ B A separate workspace for the contracts with Legal as Viewers
- ☐ C A OneLake security role on the folder for the Legal group, with the default reader role edited to exclude it
- ☐ D Object-level security in Tabular Editor on the contracts table

Q3 SINGLE ANSWER

In a Fabric Warehouse, members of the Analysts role must query `dbo.Employee` but must not see the `Salary` or `NationalID` columns. Which statement meets the requirement?

- ☐ A `GRANT SELECT ON dbo.Employee (EmployeeID, Name, Department, HireDate) TO Analysts;`
- ☐ B `GRANT SELECT ON dbo.Employee TO Analysts WITH GRANT OPTION;`
- ☐ C `DENY SELECT ON dbo.Employee TO Analysts;`
- ☐ D `ALTER TABLE dbo.Employee DROP COLUMN Salary, NationalID;`

Q4 SINGLE ANSWER

The customer lakehouse must be flagged as the single authoritative source for customer data across the organisation. Which endorsement badge is designed for this?

- ☐ A Certified
- ☐ B Master data
- ☐ C Featured
- ☐ D Promoted

Q5 HOT AREA — YES / NO FOR EACH STATEMENT

Your tenant has grown to 900 workspaces. For each statement about Fabric domains, select Yes if it is true.

Statement	Yes	No
1. A domain groups workspaces by business area, such as Finance or Supply Chain.	☐	☐
2. Domain admins can be delegated some settings without becoming Fabric administrators.	☐	☐
3. Assigning a workspace to a domain grants domain admins read access to all of its data.	☐	☐

Q6 SINGLE ANSWER

Support staff query a warehouse table that contains customer email addresses. They must see values like aXXX@XXX.com instead of the real address, while a privileged group sees the full value. What should you implement?

- ☐ A Dynamic data masking on Email with the email() function, plus UNMASK for the privileged group
- ☐ B Row-level security with a predicate function on the Email column
- ☐ C A Highly Confidential sensitivity label on the warehouse
- ☐ D Column-level security that denies SELECT on Email to support staff

Q7 HOT AREA — YES / NO FOR EACH STATEMENT

A developer saves a report as a Power BI project (.pbip). For each statement, select Yes if it is true.

Statement	Yes	No
1. The semantic model definition can be stored in TMDL, a human-readable text format.	☐	☐
2. The report definition can be stored in the PBIR format, as text files.	☐	☐
3. The project folder contains the imported data, so it can be opened offline with full data.	☐	☐

Q8 SINGLE ANSWER

In a deployment pipeline, a report named Sales exists in Development and a separately created report also named Sales exists in Test. They were never paired. What happens when you deploy from Development to Test?

- ☐ A Deployment fails with a name conflict.
- ☐ B The two reports are paired, then merged.
- ☐ C A second report named Sales is created in Test.
- ☐ D The existing Test report is overwritten.

Q9 SINGLE ANSWER

Your workspace is connected to the main branch of a GitHub repository. A developer needs an isolated workspace to work on a feature branch without affecting the shared workspace. What should they do?

- ☐ A Export every item as a .pbix and re-import it
 - ☐ B Use Branch out to another workspace from the source control pane
 - ☐ C Create a deployment pipeline with a Feature stage
 - ☐ D Turn off Git integration while they work
-

Q10 SINGLE ANSWER

Analysts often connect to the same certified warehouse and get the server address wrong. You want them to open Power BI Desktop already pointed at the correct source. What should you distribute?

- ☐ A A .pbix file
 - ☐ B A .pbix file containing data
 - ☐ C A .pbids file
 - ☐ D A deployment rule
-

Q11 SINGLE ANSWER

A semantic model in a Fabric workspace was edited through the XMLA endpoint with Tabular Editor. What is a consequence you should plan for?

- ☐ A The model can no longer be downloaded as a .pbix file.
 - ☐ B The workspace is disconnected from its Git branch.
 - ☐ C The reports connected to it must be republished.
 - ☐ D The model's storage mode switches to DirectQuery.
-

Q12 SINGLE ANSWER

The finance systems team writes T-SQL. They need stored procedures that insert, update and delete rows in gold tables, with transactions that span tables. Which item should they build the gold layer in?

- ☐ A A warehouse
 - ☐ B A lakehouse, using its SQL analytics endpoint
 - ☐ C A KQL queryset
 - ☐ D An eventhouse
-

Q13 SINGLE ANSWER

A gold table in a lakehouse in the Finance workspace must be readable from a lakehouse in the Marketing workspace. Nobody may maintain a copy, and schema changes must show up automatically. What should you create?

- ☐ A A mirrored database of the Finance lakehouse
- ☐ B A nightly Copy activity into the Marketing lakehouse
- ☐ C An internal shortcut in the Marketing lakehouse to the Finance table
- ☐ D A Dataflow Gen2 into the Marketing lakehouse

Q14 SINGLE ANSWER

A user can open Lakehouse B, which contains a shortcut to a table in Lakehouse A, but gets an access error when reading the shortcut from a notebook. What is the most likely reason?

- ☐ A Both lakehouses are assigned to the same capacity.
- ☐ B Shortcuts can only be read through the SQL endpoint.
- ☐ C The shortcut expired after 30 days without use.
- ☐ D The user has no permission on the table in Lakehouse A.

Q15 SINGLE ANSWER

Telemetry is ingested into an Eventhouse. A Direct Lake semantic model must report on the same data without a separate copy pipeline. What should you do?

- ☐ A Create a shortcut from the Eventhouse to a warehouse
- ☐ B Turn on OneLake availability for the KQL database or table
- ☐ C Build a DirectQuery semantic model on the KQL database
- ☐ D Export the KQL tables to CSV in a lakehouse every hour

Q16 SINGLE ANSWER

A pipeline must copy 50 tables from a source database. The list of tables is kept in a control table. Which pattern should you use?

- ☐ A A single Copy activity configured with 50 sources
- ☐ B Fifty pipelines, each with one Copy activity, run in parallel
- ☐ C A Lookup on the control table, then a ForEach running a parameterised Copy
- ☐ D A Dataflow Gen2 with one query and destination per table

Q17 HOT AREA — YES / NO FOR EACH STATEMENT

You are creating shortcuts in a lakehouse. For each statement, select Yes if it is true.

Statement	Yes	No
1. A shortcut in the Tables folder must be created at the top level, not inside a subfolder.	☐	☐
2. A shortcut in the Files folder can be created at any folder level and can point to any file format.	☐	☐
3. Deleting a shortcut also deletes the data at the target location.	☐	☐

Q18 SINGLE ANSWER

Complete the query so that only the most recent row per OrderID is returned.

```
WITH x AS (  
    SELECT *, ROW_NUMBER() OVER (PARTITION BY OrderID  
                                ORDER BY ModifiedAt DESC, LoadID DESC) AS rn  
    FROM stg.Orders)  
SELECT * FROM x  
____ ;
```

- ☐ A WHERE rn = 1
- ☐ B WHERE rn IS NULL
- ☐ C HAVING COUNT(*) = 1
- ☐ D WHERE rn > 1

Q19 SINGLE ANSWER

A product's colour changes from time to time. The business only ever reports on the current colour and doesn't need history. Which dimension handling is appropriate?

- ☐ A Create a new dimension table each month
- ☐ B Store the colour on the fact table
- ☐ C Slowly changing dimension Type 2: add a new row with validity dates
- ☐ D Slowly changing dimension Type 1: overwrite the attribute in place

Q20 SINGLE ANSWER

Which key design is best for joining a gold dimension to a very large fact table?

- ☐ A A string key made by joining the source system code and the business key
- ☐ B No key; join on the customer name
- ☐ C An integer surrogate key generated during the load
- ☐ D A GUID from the source system

Q21 SINGLE ANSWER

In a notebook, you need to add a column `Margin` equal to `Revenue - Cost` to a `DataFrame`. Which line is correct?

- ☐ A `df = df.drop("Revenue", "Cost")`
- ☐ B `df = df.select("Margin")`
- ☐ C `df = df.withColumn("Margin", F.col("Revenue") - F.col("Cost"))`
- ☐ D `df = df.groupBy("Revenue").agg(F.sum("Cost"))`

Q22 SINGLE ANSWER

Complete the PySpark that produces total Amount and order count per Region.

```
summary = (spark.read.table("silver_orders")
           _____ )
```

- ☐ A `.groupBy("Region").agg(F.sum("Amount").alias("Amount"), F.count("*").alias("Orders"))`
- ☐ B `.orderBy("Region").agg(F.sum("Amount"), F.count("*"))`
- ☐ C `.withColumn("Amount", F.sum("Amount")).withColumn("Orders", F.count("*"))`
- ☐ D `.select("Region", F.sum("Amount"), F.count("*"))`

Q23 SINGLE ANSWER

A Dataflow Gen2 reads a large SQL source, but the business only needs the last two years. Where should the date filter go to cut the most data movement?

- ☐ A In a DAX measure that ignores older dates
- ☐ B As a report-level filter in the semantic model
- ☐ C As the last step, after the merge and pivot
- ☐ D As an early step that folds back to the source

Q24 DRAG AND DROP — MATCH

Match each Power Query transformation to the requirement it meets in a Dataflow Gen2.

Choices — each may be used once, more than once, or not at all

- ☐ A Unpivot columns
- ☐ B Group by
- ☐ C Append queries
- ☐ D Merge queries
- ☐ E Pivot column

Requirement	Choice
1. Add the customer's segment from a lookup table to each order row	
2. Stack the January, February and March extracts, which share the same columns, into one table	
3. Produce one row per region with total sales	
4. Turn the month values in a column into separate columns	

Q25 SINGLE ANSWER

An `Amount` column was read from CSV as a string. It must become a decimal with two decimal places before it's written to the silver table. Which expression is correct?

- ☐ A `F.lit("Amount").cast("int")`
- ☐ B `F.col("Amount").substr(1, 2)`
- ☐ C `F.col("Amount").cast("decimal(18,2)")`
- ☐ D `F.col("Amount").alias("decimal")`

Q26 SINGLE ANSWER

Which T-SQL expression returns a running total of Revenue by date?

- ☐ A `LAG(Revenue) OVER (ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)`
- ☐ B `SUM(Revenue) OVER (PARTITION BY OrderDate ORDER BY OrderDate)`
- ☐ C `RANK() OVER (ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)`
- ☐ D `SUM(Revenue) OVER (ORDER BY OrderDate ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)`

Q27 SINGLE ANSWER

Which KQL query returns the five devices with the most error events?

- ☐ A `Logs | where Level == "Error" | take 5 | project DeviceId, Level`
- ☐ B `Logs | where Level == "Error" | summarize Errors = count() by DeviceId | top 5 by Errors desc`
- ☐ C `Logs | where Level == "Error" | top 5 by DeviceId desc`
- ☐ D `Logs | where Level == "Error" | project DeviceId, count() | limit 5`

Q28 SINGLE ANSWER

In the DAX query view you want to test a new measure definition against the model without saving it. Which query structure lets you do that?

- ☐ A `EVALUATE MEASURE Sales[Test] = ... SUMMARIZECOLUMNS('Date'[Year], "Test", [Test])`
- ☐ B `ALTER MODEL ADD MEASURE Sales[Test] = ... ; EVALUATE [Test]`
- ☐ C `DEFINE MEASURE Sales[Test] = ... EVALUATE SUMMARIZECOLUMNS('Date'[Year], "Test", [Test])`
- ☐ D `CREATE MEASURE Sales[Test] = ... ; SELECT [Test] FROM 'Date'`

Q29 SINGLE ANSWER

In KQL, you need to add the device's site name from the `Devices` table to each row of `Telemetry`, keeping only rows that match. Which operator should you use?

- ☐ A `extend SiteName = Devices.SiteName`
- ☐ B `join kind=inner Devices on DeviceId`
- ☐ C `mv-expand Devices`
- ☐ D `union Devices`

Q30 SINGLE ANSWER

A warehouse named `WH_Gold` and a lakehouse named `LH_Raw` are in the same workspace. From a query in `WH_Gold`, you need to join to the `orders` table in `LH_Raw` without copying it. Which reference works?

- ☐ A `external_table('orders')`
- ☐ B `OPENROWSET('LH_Raw', 'orders')`
- ☐ C `[LH_Raw]::orders`
- ☐ D `LH_Raw.dbo.orders`

Q31 SINGLE ANSWER

You mark a table as the model's date table. What must its date column contain?

- ☐ A Unique, contiguous dates covering every date in the facts
- ☐ B Unique text values in the format `yyyy-mm-dd`
- ☐ C Only the dates on which sales actually happened
- ☐ D Exactly one row per month, on the first day

Q32 SINGLE ANSWER

In a composite model, a large fact uses `DirectQuery` and an aggregation table uses `Import`. A shared dimension relates to both. Which storage mode should the dimension use?

- ☐ A Dual
- ☐ B Direct Lake
- ☐ C `DirectQuery`
- ☐ D `Import`

Q33 SINGLE ANSWER

Which measure returns each product category's share of sales across all categories, ignoring any category filter while still respecting other filters, such as year?

- ☐ A `DIVIDE([Sales], CALCULATE([Sales], ALLSELECTED('Date')))`
 - ☐ B `DIVIDE([Sales], CALCULATE([Sales], ALL(Sales)))`
 - ☐ C `DIVIDE([Sales], CALCULATE(SUM(Sales[Amount])))`
 - ☐ D `DIVIDE([Sales], CALCULATE([Sales], REMOVEFILTERS('Product'[Category])))`
-

Q34 SINGLE ANSWER

A Direct Lake model over a gold lakehouse must also include a small Import table of sales targets loaded from Excel. Which Direct Lake variant supports this?

- ☐ A Direct Lake on OneLake
 - ☐ B Direct Lake on the SQL analytics endpoint
 - ☐ C Both variants
 - ☐ D Neither variant
-

Q35 SINGLE ANSWER

A card must show the selected region's name, or "All regions" when zero or several regions are selected. Which expression is the most concise?

- ☐ A `SELECTEDVALUE('Geo'[Region], "All regions")`
 - ☐ B `VALUES('Geo'[Region])`
 - ☐ C `MAX('Geo'[Region])`
 - ☐ D `FIRSTNONBLANK('Geo'[Region], 1)`
-

Q36 SINGLE ANSWER

In a development workspace, any query that a Direct Lake on SQL model can't answer in Direct Lake mode should fail, so the problem gets noticed. Which `DirectLakeBehavior` setting should you choose?

- ☐ A Automatic
 - ☐ B `DirectLakeOnly`
 - ☐ C `DirectQueryOnly`
 - ☐ D `ImportOnly`
-

Q37 SINGLE ANSWER

A semantic model must read gold data through SQL views that join and filter several warehouse tables. Which Direct Lake variant can read the views?

- ☐ A Direct Lake on OneLake
- ☐ B Direct Lake on the SQL analytics endpoint
- ☐ C Both variants equally
- ☐ D Neither; views need Import mode

Q38 SINGLE ANSWER

Which rewrite usually improves performance most for `CALCULATE([Sales], FILTER(Sales, Sales[Channel] = "Online"))` ?

- ☐ A `CALCULATE([Sales], FILTER(ALL(Sales), Sales[Channel] = "Online"))`
- ☐ B `IFERROR([Sales], 0)`
- ☐ C `SUMX(FILTER(Sales, Sales[Channel] = "Online"), [Sales])`
- ☐ D `CALCULATE([Sales], Sales[Channel] = "Online")`

Q39 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about V-Order in Fabric, select Yes if it is true.

Statement	Yes	No
1. V-Order is a write-time optimisation of Parquet files that speeds up reads by engines such as Direct Lake.	☐	☐
2. V-Order can be turned on for a table with the table property <code>delta.parquet.vorder.enabled</code> .	☐	☐
3. V-Order makes writes faster than normal Parquet writes.	☐	☐

Q40 SINGLE ANSWER

An incremental refresh policy refreshes the last 10 days of a large fact, but most days have no changes. What reduces wasted refresh work?

- ☐ A Increase the incremental window to 60 days
- ☐ B Turn on Detect data changes using a LastModified column
- ☐ C Turn off query folding on the source query
- ☐ D Turn on automatic page refresh for the report

Q41 SINGLE ANSWER

A Direct Lake on OneLake model frames a table whose row count is over the capacity's guardrail. What happens?

- ☐ A Queries fall back to DirectQuery automatically, more slowly.
- ☐ B The model switches that table to Import mode.
- ☐ C The capacity scales up to the next SKU automatically.
- ☐ D Framing fails, and the model can't be queried until the table fits the guardrails.

Existing environment

Fabrikam Health runs 60 clinics. Appointment data lives in an Azure SQL Database that backs the booking application. The database team won't allow analytical queries against it. A Fabric warehouse, WH_Clinic, holds the gold layer and is maintained by T-SQL developers.

The team works in three workspaces (Dev, Test and Prod) connected by a deployment pipeline. Finance keeps monthly visit targets in an Excel file on SharePoint.

Requirements

1. R1. Appointment data must reach OneLake within minutes, with no ETL pipeline to build.
2. R2. Analysts must be able to list visits per clinic for the last 7 days with T-SQL.
3. R3. The semantic model must point at WH_Clinic in each stage's own workspace after deployment, without manual edits.
4. R4. The semantic model must combine the gold tables, in Direct Lake mode, with the Excel targets.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Q42 SINGLE ANSWER

Which approach satisfies R1?

- ☐ A Create a DirectQuery semantic model over the database
- ☐ B Mirror the Azure SQL Database into Fabric
- ☐ C Create an external shortcut to the Azure SQL Database
- ☐ D Build a pipeline that copies the data every five minutes

Q43 SINGLE ANSWER

Which query satisfies R2 in WH_Clinic?

- ☐ A FactVisit | where VisitDate > ago(7d) | summarize count() by ClinicID
- ☐ B SELECT ClinicID, COUNT(*) AS Visits FROM gold.FactVisit WHERE VisitDate >= DATEADD(day, -7, CAST(GETDATE() AS date)) GROUP BY ClinicID;
- ☐ C SELECT ClinicID, COUNT(*) FROM gold.FactVisit GROUP BY ClinicID HAVING VisitDate >= GETDATE() - 7;
- ☐ D SELECT ClinicID, COUNT(*) FROM gold.FactVisit WHERE VisitDate > ago(7d) GROUP BY ClinicID;

Q44 SINGLE ANSWER

Which configuration satisfies R3?

- ☐ A A .pbids file for each stage's warehouse
- ☐ B Republishing the model from Desktop to each stage's workspace
- ☐ C Three Git branches, one per stage, merged on each release
- ☐ D Deployment rules on Test and Prod that repoint the model's data source

Q45 SINGLE ANSWER

Which design satisfies R4?

- ☐ A A Direct Lake on SQL endpoint model with an Import table for the targets
- ☐ B A Direct Lake on OneLake model with an Import table for the targets
- ☐ C An Import model for every table, dropping Direct Lake
- ☐ D Two semantic models, one for gold and one for targets

End of Set 02 — answers follow on the next page.

Set 02 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	A	M1		16	C	P1		31	A	S1	
2	C	M1		17	Yes · Yes · No	P1		32	A	S1	
3	A	M1		18	A	P2		33	D	S1	
4	B	M1		19	D	P2		34	A	S1	
5	Yes · Yes · No	M1		20	C	P2		35	A	S1	
6	A	M1		21	C	P2		36	B	S2	
7	Yes · Yes · No	M2		22	A	P2		37	B	S2	
8	C	M2		23	D	P2		38	D	S2	
9	B	M2		24	1→D 2→C 3→B 4→E	P2		39	Yes · Yes · No	S2	
10	C	M2		25	C	P2		40	B	S2	
11	A	M2		26	D	P3		41	D	S2	
12	A	P1		27	B	P3		42	B	P1	
13	C	P1		28	C	P3		43	B	P3	
14	D	P1		29	B	P3		44	D	M2	
15	B	P1		30	D	P3		45	B	S1	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: A M1 · Implement security and governance

Member can add users with Member, Contributor or Viewer roles and can share items. Only Admin can delete the workspace or manage other admins. Contributors can create and edit content but can't manage access.

Q2 Answer: C M1 · Implement security and governance

OneLake security roles can be scoped to folders and tables inside an item. That's the file and folder-level access control in the skills outline. The default reader role has to stop granting the folder, or everyone keeps access. RLS and OLS work inside semantic models, not on lake folders. A separate workspace duplicates structure for no benefit.

Q3 Answer: A M1 · Implement security and governance

Column-level security in the warehouse is done with a GRANT on a list of columns. The analysts can query the granted columns, and any query that references Salary or NationalID fails. DENY on the whole table blocks everything. WITH GRANT OPTION grants every column and lets them pass the permission on. Dropping the columns destroys the data.

Q4 Answer: B M1 · Implement security and governance

Master data marks an item as the authoritative source for a core business entity, such as customers or products. Certified shows that an item meets quality standards. Promoted is a self-service signal. Like Certified, Master data can only be applied by users the administrator authorises.

Q5 Answer: Yes · Yes · No M1 · Implement security and governance

Domains are a governance and discovery layer above workspaces, with delegated administration for domain-level settings. They don't grant data access. Workspace roles, item permissions and OneLake security still decide who can see data.

Q6 Answer: A M1 · Implement security and governance

Dynamic data masking hides column values in query results without changing the stored data, and users with UNMASK see the real values. Column-level security blocks the column entirely, so support staff would get an error instead of a masked value. RLS filters rows, not values. Labels classify and protect exports.

Q7 Answer: Yes · Yes · No M2 · Maintain the analytics development lifecycle

A .pbip saves definitions as text: TMDL for the model and PBIR for the report. That's what makes diffs and pull-request review work. The data isn't part of the definition. A local cache file is excluded from source control, and data is reloaded by refresh.

Q8 Answer: C M2 · Maintain the analytics development lifecycle

Deployment pipelines overwrite only paired items. Items are paired when a workspace is assigned to a stage, or when content is first deployed. An unpaired item with the same name doesn't get overwritten, so you end up with a duplicate.

Q9 Answer: B M2 · Maintain the analytics development lifecycle

Branch out creates a new branch and connects a new (or existing) workspace to it, giving the developer an isolated copy that they later merge through a pull request. Deployment pipelines promote content between stages. They don't give you feature isolation.

Q10 Answer: C M2 · Maintain the analytics development lifecycle

A .pbids (Power BI data source) file holds only a connection, so Desktop opens with the source already chosen. A .pbir is a full template without data. A .pbix includes the data. Deployment rules apply to pipeline stages.

Q11 Answer: A M2 · Maintain the analytics development lifecycle

Once a model has been changed through the XMLA endpoint, you can't download it as a .pbix any more. From then on, you manage it as source (a .pbip or TMDL in Git, or a .bim) and deploy it with tools. The other options don't happen.

Q12 Answer: A P1 · Get data

Only the warehouse supports full T-SQL DML and DDL with multi-table transactions. The lakehouse SQL analytics endpoint is read-only for tables. Eventhouses are for event and time-series data. A queryset only runs queries.

Q13 Answer: C P1 · Get data

An internal shortcut points to data in another Fabric item, in any workspace, with no copy and no sync job, and it reflects schema changes. Copies need maintaining and go out of date. Mirroring is for external operational databases.

Q14 Answer: D P1 · Get data

Reading through an internal shortcut needs permission on both the shortcut location and the target. The user's own identity is checked against the target. Shortcuts don't expire, and Spark can read them.

Q15 Answer: B P1 · Get data

OneLake availability writes the Eventhouse data to OneLake as Delta tables, which Direct Lake can read. That's the skill named "Implement OneLake integration for Eventhouse and semantic models". Exporting to CSV duplicates the data and loses the Delta format. DirectQuery gives up Direct Lake.

Q16 Answer: C P1 · Get data

This is the standard metadata-driven pattern. Lookup returns the list, and ForEach runs a Copy activity with the source and destination parameterised from each item, so one pipeline handles any number of tables. A single Copy activity can't take 50 sources.

Q17 Answer: Yes · Yes · No P1 · Get data

Tables-folder shortcuts sit at the top level and are meant for Delta tables. Files-folder shortcuts can go at any depth and point to any format. Deleting a shortcut only removes the pointer. Deleting or moving the target is what breaks the shortcut.

Q18 Answer: A P2 · Transform data

ROW_NUMBER numbers the rows in each OrderID partition from newest to oldest, with the tie-breaker making it deterministic. Keeping $m = 1$ leaves exactly one row, the latest, per order. $m > 1$ returns the duplicates instead.

Q19 Answer: D P2 · Transform data

Type 1 overwrites the value and keeps no history, which is right when only the current value matters. Type 2 keeps history by adding versioned rows, which isn't needed here. Putting attributes on the fact breaks the star schema.

Q20 Answer: C P2 · Transform data

Integer surrogate keys are compact, compress well in both Delta and VertiPaq, and keep the gold layer independent of source keys. Composite string keys and GUIDs have high cardinality and are wide. GUID isn't a supported data type in Direct Lake either.

Q21 Answer: C P2 · Transform data

withColumn adds or replaces a column using a column expression (F is pyspark.sql.functions). select("Margin") fails because the column doesn't exist yet. groupBy aggregates. drop removes columns.

Q22 Answer: A P2 · Transform data

groupBy followed by agg is the PySpark aggregation pattern, and you can compute several aggregates in one pass. Selecting Region next to aggregates without groupBy fails. orderBy only sorts, so agg after it returns one total row with no Region. Aggregate functions can't be used in withColumn without a window.

Q23 Answer: D P2 · Transform data

Putting filters early, in steps that fold, pushes them to the source as a WHERE clause, so only the needed rows leave the database. Filtering later moves all the data first. Report filters and measures don't reduce what's loaded at all.

Q24 Answer: 1→D 2→C 3→B 4→E P2 · Transform data

Merge is a join, so it adds columns from another table. Append is a union, so it adds rows. Group by aggregates. Pivot turns values into columns, and unpivot does the opposite.

Q25 Answer: C P2 · Transform data

cast changes the column's data type, and decimal(18,2) keeps two decimal places exactly. alias only renames. lit creates a constant, here the literal text "Amount". substr extracts characters.

Q26 Answer: D P3 · Query and analyze data

A windowed SUM with a frame that starts at UNBOUNDED PRECEDING and ends at the CURRENT ROW gives a running total. Using ROWS rather than RANGE avoids surprises when dates repeat. Partitioning by OrderDate restarts the sum on every date. RANK and LAG don't accept a frame clause, and neither adds anything up.

Q27 Answer: B P3 · Query and analyze data

Filter first, then summarize the count per device, then top 5 by the count. top 5 by DeviceId sorts by the ID, not by errors. take returns arbitrary rows. project can't aggregate.

Q28 Answer: C P3 · Query and analyze data

DEFINE MEASURE declares a measure that exists only for that query, and EVALUATE returns a table that uses it. That's how you test changes in the DAX query view before updating the model. The other syntaxes don't exist in DAX queries.

Q29 Answer: B P3 · Query and analyze data

join kind=inner matches rows on the key and returns only the matches. union stacks tables. extend adds computed columns from the same row. mv-expand expands arrays into rows.

Q30 Answer: D P3 · Query and analyze data

Within a workspace, warehouses and SQL analytics endpoints support cross-database queries with three-part names (database.schema.table). external_table() is KQL. The other syntaxes are invalid here.

Q31 Answer: A S1 · Design and build semantic models

A date table needs one row per day, with no duplicates or gaps, covering every date the facts use. Time-intelligence functions depend on that. Including only sales dates leaves gaps. The column must have the Date data type.

Q32 Answer: A S1 · Design and build semantic models

Dual lets the engine use the dimension from memory when the query hits the Import aggregation, and as DirectQuery when the query goes to the detail fact. That avoids weak (limited) relationships and keeps queries that hit the aggregation in memory.

Q33 Answer: D S1 · Design and build semantic models

REMOVEFILTERS on the Category column clears only that filter, so the denominator is all categories in the current year or other context. ALL(Sales) removes every filter on the fact, including year. CALCULATE(SUM(...)) keeps the category filter, so the share is always 1. ALLSELECTED('Date') changes the date filter, not the category filter.

Q34 Answer: A S1 · Design and build semantic models

Composite models (Direct Lake tables plus Import tables) are supported with Direct Lake on OneLake, not with Direct Lake on the SQL analytics endpoint.

Q35 Answer: A S1 · Design and build semantic models

SELECTEDVALUE returns the value when exactly one is in the filter context, and otherwise returns the alternate result. VALUES returns a table and errors in a card when several regions are selected. FIRSTNONBLANK and MAX return one arbitrary value when several are selected.

Q36 Answer: B S2 · Optimize enterprise-scale semantic models

DirectLakeOnly turns off fallback to DirectQuery, so unsupported queries fail and regressions show up. Automatic (the default) quietly falls back to DirectQuery. DirectQueryOnly forces every query to use DirectQuery. ImportOnly isn't a valid value.

Q37 Answer: B S2 · Optimize enterprise-scale semantic models

Direct Lake on the SQL analytics endpoint can use SQL views, although queries against non-materialised views fall back to DirectQuery. Direct Lake on OneLake reads Delta tables directly and can't connect to views.

Q38 Answer: D S2 · Optimize enterprise-scale semantic models

A Boolean filter on a column is applied to that column only and becomes a simple predicate for the storage engine. FILTER over the whole fact table builds the full table. FILTER(ALL(Sales)) also removes existing filters, which changes the result. SUMX with a measure triggers context transition on every row.

Q39 Answer: Yes · Yes · No S2 · Optimize enterprise-scale semantic models

V-Order sorts, distributes and compresses Parquet data when it's written, so reads are faster. Writes cost more, about 15% by Microsoft's figure. You can set it per session, per table or per write. Its default state has changed over time, so set it on purpose for read-heavy tables.

Q40 Answer: B S2 · Optimize enterprise-scale semantic models

Detect data changes checks a date/time column, such as LastModified, per partition and refreshes only the partitions whose maximum value has changed. A wider window does more work. Query folding is required for incremental refresh to work efficiently, so it shouldn't be turned off.

Q41 Answer: D S2 · Optimize enterprise-scale semantic models

Direct Lake on OneLake has no DirectQuery fallback, so going over a guardrail causes a failure, not a slowdown. Direct Lake on the SQL analytics endpoint can fall back to DirectQuery if fallback is enabled. That difference is why table maintenance matters so much for the OneLake variant.

Q42 Answer: B P1 · Get data

Mirroring replicates supported operational databases into OneLake in near real time with no pipeline to build. It reads the database's change data rather than running analytical queries against it. Shortcuts point at storage, not at Azure SQL databases. A five-minute pipeline is ETL.

Q43 Answer: B P3 · Query and analyze data

DATEADD on today's date gives the cutoff, and GROUP BY counts per clinic. ago() is a KQL function. The pipe syntax is KQL. HAVING filters groups and can't reference a non-aggregated column that isn't grouped.

Q44 Answer: D M2 · Maintain the analytics development lifecycle

Deployment rules (data source or parameter rules) are set on the target stage. They change where the deployed item points, so you promote the same model each time. Git branches don't rewrite connections. Republishing manually is what R3 rules out.

Q45 Answer: B S1 · Design and build semantic models

Only Direct Lake on OneLake supports composite models that mix in Import tables. The SQL endpoint variant can't be combined with Import tables. Two models split the experience, and importing everything gives up Direct Lake.

Set 03

Core — reading SQL, KQL, DAX and PySpark

The same syllabus with more code on the page: you must read all four languages fluently, even if you only write one.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 2 · Core

12

Maintain a data analytics solution
25-30%

21

Prepare data
45-50%

12

Implement and manage semantic
models
25-30%

Q1 MULTIPLE RESPONSE — CHOOSE TWO

Which two workspace roles can manage OneLake security roles on a lakehouse?

- ☐ A Contributor
- ☐ B Member
- ☐ C Viewer
- ☐ D Admin
- ☐ E Any user with the Build permission

Q2 BUILD LIST — SEQUENCE

You must make sure each regional manager sees only their own region's rows when querying `dbo.Sales` in a Fabric Warehouse through T-SQL. Put the required actions in order.

Select the 3 actions required and arrange them in the correct order. Not every action is used.

- ☐ A Create a schema to hold the security objects
- ☐ B Create a security policy that adds the function as a FILTER PREDICATE on `dbo.Sales`, with `STATE = ON`
- ☐ C DENY SELECT on `dbo.Sales` to the managers
- ☐ D Add dynamic data masking to the Region column
- ☐ E Create an inline table-valued function that returns 1 when the row's Region matches the user's region

Q3 HOT AREA — YES / NO FOR EACH STATEMENT

A OneLake security role with a row filter is defined on a lakehouse. For each user, select Yes if the row filter restricts what they see.

Statement	Yes	No
1. A user with the workspace Viewer role who is a member of the security role	☐	☐
2. A user with the workspace Contributor role	☐	☐
3. A user with no workspace role who was given access to the lakehouse through item sharing and is a member of the security role	☐	☐

Q4 SINGLE ANSWER

The compliance team requires every new or edited Power BI and Fabric item to carry a sensitivity label before it's saved. What should you configure?

- ☐ A A Purview label policy with mandatory labelling for Fabric
- ☐ B Certification required in the tenant settings
- ☐ C A OneLake security role on every workspace
- ☐ D Workspace-level RLS with a label filter

Q5 SINGLE ANSWER

A data governance team must be the only people who can certify items. What must happen first?

- ☐ A A sensitivity label named Certified is published to the team
- ☐ B An admin enables certification in tenant settings for the team's security group
- ☐ C Each workspace admin certifies items, then the team reviews them
- ☐ D The team gets the Contributor role in every workspace

Q6 HOT AREA — YES / NO FOR EACH STATEMENT

A workspace is connected to a Git branch. For each statement about the source control pane, select Yes if it is true.

Statement	Yes	No
1. Commit pushes changes made in the workspace to the connected branch.	☐	☐
2. Update brings changes from the connected branch into the workspace.	☐	☐
3. Git integration promotes items from a Test workspace to a Production workspace with deployment rules.	☐	☐

Q7 MULTIPLE RESPONSE — CHOOSE TWO

Which two can a deployment rule change when content is deployed to a target stage?

- ☐ A The DAX of a measure
- ☐ B A semantic model's data source connection
- ☐ C A parameter value in a semantic model or dataflow
- ☐ D The visuals on a report page
- ☐ E The RLS role membership

Q8 BUILD LIST — SEQUENCE

You need to put an existing workspace under version control in Azure DevOps. Put the actions in order. Select the 4 actions required and arrange them in the correct order. Not every action is used.

- ☐ A Create a deployment pipeline and assign the workspace
- ☐ B Export each report as a .pbit file
- ☐ C Open the workspace settings and choose Git integration
- ☐ D Choose Azure DevOps and sign in
- ☐ E Connect and sync, then commit the workspace items to the branch
- ☐ F Select the organisation, project, repository, branch and folder

Q9 SINGLE ANSWER

Impact analysis on a lakehouse shows 14 downstream semantic models and reports. You want their owners to know about a schema change scheduled for next week. Which built-in option helps?

- ☐ A Notify contacts from the impact analysis pane
- ☐ B Add the owners to the workspace as Admins
- ☐ C Apply a sensitivity label
- ☐ D Create a deployment pipeline

Q10 DRAG AND DROP — MATCH

Match each reuse requirement to the asset that meets it.

Choices — each may be used once, more than once, or not at all

- ☐ A .pbix file
- ☐ B Shared semantic model
- ☐ C .pbids file
- ☐ D .pbix template
- ☐ E .pbip project

Requirement	Choice
1. Teams need a starting report with the corporate theme, measures and layout, but no data	
2. Analysts must open Desktop already pointed at the approved warehouse	
3. Forty reports must use one governed set of measures and relationships	
4. A model's definition must be diffable in pull requests	

Q11 HOT AREA — YES / NO FOR EACH STATEMENT

A capacity's XMLA endpoint is set to Read only. For each task, select Yes if it is possible.

Statement	Yes	No
1. Run DAX queries against a semantic model from DAX Studio	☐	☐
2. Deploy a changed table partition from Tabular Editor	☐	☐
3. Connect SQL Server Management Studio and browse the model's metadata	☐	☐

Q12 SINGLE ANSWER

Security engineers need to search terabytes of firewall logs with free-text filters and time-window aggregations, with results in seconds. Which store fits best?

- ☐ A Warehouse
- ☐ B Fabric SQL database
- ☐ C Eventhouse (KQL database)
- ☐ D Lakehouse Files area with CSV

Q13 SINGLE ANSWER

A new line-of-business application needs a transactional relational database inside Fabric, and its data should be available in OneLake for analytics with no extra work. Which item should you create?

- ☐ A lakehouse with its SQL analytics endpoint
 - ☐ A warehouse with stored procedures
 - ☐ A SQL database in Fabric
 - ☐ An Eventhouse with update policies
-

Q14 SINGLE ANSWER

A user tries to create an external shortcut to an ADLS Gen2 account through an existing cloud connection and is refused. Other users can create shortcuts through the same connection. What is the most likely cause?

- ☐ ADLS Gen2 isn't a supported target
 - ☐ The user doesn't have permission to use that cloud connection
 - ☐ The lakehouse already has a shortcut
 - ☐ Shortcuts can only be created by tenant administrators
-

Q15 SINGLE ANSWER

A transformation merges 2 billion clickstream rows with session data and needs custom Python logic and unit tests. Which tool should you use?

- ☐ A notebook (PySpark)
 - ☐ A Dataflow Gen2
 - ☐ A KQL queryset
 - ☐ The Visual Query Editor
-

Q16 MULTIPLE RESPONSE — CHOOSE THREE

Which three are supported targets for an external OneLake shortcut?

- ☐ Amazon S3
 - ☐ An Azure SQL Database table
 - ☐ Google Cloud Storage
 - ☐ Dataverse
 - ☐ A Salesforce object
-

Q17 SINGLE ANSWER

Point-of-sale events arrive in Azure Event Hubs. They must be routed with no code into an Eventhouse table and filtered on the way in. Which item should you use?

- ☐ A Spark job definition
- ☐ A Dataflow Gen2
- ☐ A deployment pipeline
- ☐ An Eventstream

Q18 SINGLE ANSWER

You created a cloud connection to a source database. A colleague must use it in their pipelines without ever seeing the credentials. What should you do?

- (A) Send them the credentials so they can create their own connection
- (B) Give them the Admin role on your workspace
- (C) Create a .pbids file that points at the database
- (D) Add the colleague to the connection with the User role

Q19 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about a lakehouse's SQL analytics endpoint, select Yes if it is true.

Statement	Yes	No
1. You can create views on the endpoint.	☐	☐
2. You can run INSERT and UPDATE statements against the lakehouse's Delta tables through the endpoint.	☐	☐
3. Spark notebooks or Dataflow Gen2 are the usual ways to write data to the lakehouse tables.	☐	☐

Q20 SINGLE ANSWER

In a lakehouse notebook you upsert changes into the Delta table `dim_customer`. Which clause inserts customers that don't exist in the target yet?

```
from delta.tables import DeltaTable
tgt = DeltaTable.forName(spark, "dim_customer")
(tgt.alias("t")
 .merge(updates.alias("s"), "t.CustomerID = s.CustomerID")
 .whenMatchedUpdateAll()
 _____
 .execute())
```

- (A) `.whenMatchedDelete()`
- (B) `.whenNotMatchedBySourceDelete()`
- (C) `.whenNotMatchedInsertAll()`
- (D) `.whenMatchedUpdateAll()`

Q21 SINGLE ANSWER

In PySpark, you need to keep only the latest row per OrderID based on `ModifiedAt`. Complete the expression.

```
w_rn = _____
latest = df.withColumn("rn", w_rn).filter("rn = 1").drop("rn")
```

- (A) `F.max("ModifiedAt").over(Window.partitionBy("OrderID"))`
- (B) `F.rank().over(Window.orderBy(F.col("ModifiedAt").desc()))`
- (C) `F.row_number().over(Window.partitionBy("ModifiedAt").orderBy("OrderID"))`
- (D) `F.row_number().over(Window.partitionBy("OrderID").orderBy(F.col("ModifiedAt").desc()))`

Q22 SINGLE ANSWER

A Dataflow Gen2 imports a text column holding dates in day/month/year order, such as 31/03/2026. Changing the type to Date gives errors. What should you do?

- ☐ A Leave the column as text and convert it in the semantic model
- ☐ B Split the column by the / delimiter and keep only the first part
- ☐ C Change the type to Date, then remove the rows that have errors
- ☐ D Use Change type > Using locale, with a day/month/year locale such as English (UK)

Q23 SINGLE ANSWER

You are designing a new fact table for order lines in the gold layer. What should you decide first?

- ☐ A The visuals the first report will use
- ☐ B The grain: what exactly one row represents
- ☐ C The number of partitions in the semantic model
- ☐ D The DAX measures that will reference it

Q24 SINGLE ANSWER

Some fact rows arrive with a CustomerID that doesn't exist in DimCustomer yet. Reports must not lose these sales. What is the recommended gold-layer handling?

- ☐ A Move them to a separate fact table for unmatched rows
- ☐ B Delete the fact rows until the customer arrives
- ☐ C Leave the fact's customer key NULL for now
- ☐ D Map them to an Unknown member in DimCustomer, such as key -1

Q25 SINGLE ANSWER

Which T-SQL creates a view that the semantic model can read, exposing only completed orders with a derived Year column?

- ☐ A `CREATE VIEW gold.vCompletedOrders AS SELECT o.*, YEAR(o.OrderDate) AS OrderYear FROM gold.FactOrders o WHERE o.Status = 'Completed';`
- ☐ B `CREATE PROCEDURE gold.vCompletedOrders AS SELECT * FROM gold.FactOrders WHERE Status = 'Completed';`
- ☐ C `CREATE TABLE gold.vCompletedOrders AS SELECT * FROM gold.FactOrders;`
- ☐ D `ALTER TABLE gold.FactOrders ADD OrderYear AS YEAR(OrderDate);`

Q26 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about the Visual Query Editor, select Yes if it is true.

Statement	Yes	No
1. It's available on a warehouse and on a lakehouse's SQL analytics endpoint.	☐	☐
2. It generates T-SQL that you can view.	☐	☐
3. It can only run KQL queries.	☐	☐

Q27 SINGLE ANSWER

Which KQL query shows the distinct number of users per day as a line chart?

- (A)** PageViews | summarize count() by UserId, bin(Timestamp, 1d) | render timechart
- (B)** PageViews | summarize Users = dcount(UserId) by bin(Timestamp, 1d) | render timechart
- (C)** PageViews | distinct UserId, bin(Timestamp, 1d) | render piechart
- (D)** PageViews | project Users = dcount(UserId), Timestamp | render timechart

Q28 SINGLE ANSWER

A query must return each category's revenue and its share of all revenue, in one pass. Which expression gives the share?

```
SELECT Category,  
       SUM(Revenue) AS Revenue,  
       _____ AS ShareOfTotal  
FROM gold.FactSales f JOIN gold.DimProduct p ON p.ProductKey = f.ProductKey  
GROUP BY Category;
```

- (A)** SUM(Revenue) * 1.0 / SUM(SUM(Revenue)) OVER ()
- (B)** SUM(Revenue) OVER (PARTITION BY Category)
- (C)** RANK() OVER (ORDER BY SUM(Revenue))
- (D)** SUM(Revenue) / COUNT(*)

Q29 SINGLE ANSWER

Which DAX query returns sales by month for 2026 only?

- (A)** EVALUATE SUMMARIZECOLUMNS('Date'[Month], TREATAS({2026}, 'Date'[Year]), "Sales", [Total Sales])
- (B)** SUMMARIZECOLUMNS('Date'[Month], "Sales", [Total Sales]) WHERE 'Date'[Year] = 2026
- (C)** EVALUATE FILTER([Total Sales], 'Date'[Year] = 2026)
- (D)** EVALUATE CALCULATE([Total Sales], 'Date'[Year] = 2026)

Q30 SINGLE ANSWER

A warehouse query must label each order as Small (under 100), Medium (100 to 999) or Large (1,000 or more). Which construct should you use?

- (A)** A CROSS APPLY to a table-valued function
- (B)** A recursive CTE
- (C)** A PIVOT
- (D)** A CASE expression

Q31 SINGLE ANSWER

Budgets are stored at the Category × Month grain, while the Product dimension is at product grain. You must filter budgets by Product[Category] without creating a new table. What should you do?

- (A)** Merge the Budget table into the Sales fact
- (B)** A one-to-many relationship from Budget to Product on Category
- (C)** A many-to-many relationship on Category, filtering from Product to Budget
- (D)** Duplicate each budget row for every product in the category

Q32 SINGLE ANSWER

Complete the calculation item for year-to-date so it works with any measure.

```
-- Calculation group: Time Intelligence  
-- Calculation item: YTD  
_____
```

- ☐ A TOTALYTD([Sales], 'Date'[Date])
- ☐ B CALCULATE(SELECTEDMEASURENAME(), DATESYTD('Date'[Date]))
- ☐ C CALCULATE(SELECTEDMEASURE(), ALL('Date'))
- ☐ D CALCULATE(SELECTEDMEASURE(), DATESYTD('Date'[Date]))

Q33 SINGLE ANSWER

You create a field parameter in Power BI Desktop to let users switch between three measures. What does Desktop generate?

- ☐ A A calculation group with one item per field
- ☐ B A table built with GENERATESERIES
- ☐ C A calculated table that references each field with NAMEOF()
- ☐ D Three bookmarks and a bookmark navigator

Q34 SINGLE ANSWER

Which measure gives each product a rank by revenue among the products visible in the visual, with ties sharing a rank and no gaps?

- ☐ A INDEX(1, ALLSELECTED('Product'[Name]), ORDERBY([Revenue], DESC))
- ☐ B RANK(DENSE, ALLSELECTED('Product'[Name]), ORDERBY([Revenue], DESC))
- ☐ C ROWNUMBER(ALL('Product'[Name]), ORDERBY([Revenue], DESC))
- ☐ D RANK(SKIP, ALL(Sales), ORDERBY(Sales[Qty]))

Q35 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about relationships in a semantic model, select Yes if it is true.

Statement	Yes	No
1. Setting every relationship to filter in both directions can create ambiguous filter paths and slow queries down.	☐	☐
2. A model can contain several relationships between the same two tables, but only one can be active.	☐	☐
3. Relationships in a star schema normally go from the fact table (one side) to the dimension (many side).	☐	☐

Q36 SINGLE ANSWER

Month names in a visual sort alphabetically (April, August, December, ...). What should you do?

- ☐ A Set Sort by column on MonthName to MonthNumber
- ☐ B Change MonthName to the Date data type
- ☐ C Create a calculated table of months
- ☐ D Rename the months with numeric prefixes

Q37 SINGLE ANSWER

You want to turn on query scale-out (read-only replicas) for a busy semantic model. What is a prerequisite?

- ☐ A The model must use DirectQuery mode
- ☐ B The model must use large storage format
- ☐ C The model must not have RLS roles
- ☐ D The workspace must be Git-connected

Q38 SINGLE ANSWER

A measure calculates [Sales] for the prior year three times in a long IF expression. What is the simplest way to improve both performance and readability?

- ☐ A Turn the prior-year logic into a calculated column
- ☐ B Store the prior-year value in a VAR and reuse it
- ☐ C Make the Date relationship bidirectional
- ☐ D Wrap each prior-year call in IFERROR

Q39 SINGLE ANSWER

An Import model is far larger than expected. VertiPaq Analyzer shows dozens of hidden LocalDateTable tables. What should you change?

- ☐ A Turn off Auto date/time and use one shared date table
- ☐ B Switch every date table to Dual storage mode
- ☐ C Relate each LocalDateTable to the main date table
- ☐ D Turn on large semantic model storage format

Q40 SINGLE ANSWER

An ETL pipeline writes several gold tables one after another. Reports must never show a mix of old and new tables halfway through the load. What should you configure for the Direct Lake model?

- ☐ A Set DirectLakeBehavior to DirectQueryOnly while the load runs
- ☐ B Switch the model to Import with a nightly refresh
- ☐ C Turn off automatic updates and refresh the model from the pipeline at the end
- ☐ D Turn on automatic page refresh every minute during the load

Q41 SINGLE ANSWER

For good Direct Lake performance, what row-group size does Microsoft's table-maintenance guidance recommend for gold Delta tables?

- ☐ A Over 1 billion rows per row group
- ☐ B Exactly one row group per file regardless of size
- ☐ C Fewer than 10,000 rows per row group
- ☐ D Roughly 1 to 16 million rows per row group

Existing environment

Tailspin Airlines streams aircraft status messages into an Eventhouse, KQL_Ops. Bookings land in a lakehouse, LH_Bookings, through a PySpark pipeline, and a warehouse, WH_Gold, holds the dimensional model. An Import semantic model has DAX RLS roles that filter by station (airport).

All 120 station managers have the workspace Member role so they could “see everything quickly” during the launch. The flight-status report page has 28 visuals and takes 20 seconds to load.

Requirements

1. R1. Station managers must see only their own station's data in reports.
2. R2. Missing tail numbers in WH_Gold must be shown as 'UNKNOWN'.
3. R3. Operations needs the latest status message per aircraft from KQL_Ops.
4. R4. The flight-status page must load faster without losing information.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Q42 SINGLE ANSWER

The RLS roles are set up correctly, but managers still see every station. What must you change to satisfy R1?

- ☐ A Republish the model so the roles take effect
- ☐ B Apply a Confidential sensitivity label to the model
- ☐ C Change the managers from Member to Viewer, or to app access
- ☐ D Add the managers to a second, stricter RLS role

Q43 SINGLE ANSWER

Which T-SQL expression satisfies R2 in a view over the aircraft dimension?

- ☐ A TRIM(ISNULL(TailNumber, ''))
- ☐ B COALESCE(TailNumber, 'UNKNOWN')
- ☐ C NULLIF(TailNumber, 'UNKNOWN')
- ☐ D IIF(ISNUMERIC(TailNumber) = 1, TailNumber, NULL)

Q44 SINGLE ANSWER

Which KQL query satisfies R3?

- ☐ A AircraftStatus | distinct TailNumber, Status
- ☐ B AircraftStatus | top 1 by Timestamp desc
- ☐ C AircraftStatus | summarize arg_max(Timestamp, *) by TailNumber
- ☐ D AircraftStatus | summarize max(Timestamp) by Status

Q45 MULTIPLE RESPONSE — CHOOSE TWO

Which two actions best address R4?

- ☐ A Add a slicer for every column
- ☐ B Cut the number of visuals, for example by combining cards into one multi-row card or moving detail to a drill-through page
- ☐ C Use Performance Analyzer to find the slowest visuals and tune their DAX
- ☐ D Turn off query caching
- ☐ E Make every relationship bidirectional

End of Set 03 — answers follow on the next page.

Set 03 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	B, D	M1		16	A, C, D	P1		31	C	S1	
2	A → E → B	M1		17	D	P1		32	D	S1	
3	Yes · No · Yes	M1		18	D	P1		33	C	S1	
4	A	M1		19	Yes · No · Yes	P2		34	B	S1	
5	B	M1		20	C	P2		35	Yes · Yes · No	S1	
6	Yes · Yes · No	M2		21	D	P2		36	A	S1	
7	B, C	M2		22	D	P2		37	B	S2	
8	C → D → F → E	M2		23	B	P2		38	B	S2	
9	A	M2		24	D	P2		39	A	S2	
10	1→D 2→C 3→B 4→E	M2		25	A	P2		40	C	S2	
11	Yes · No · Yes	M2		26	Yes · Yes · No	P3		41	D	S2	
12	C	P1		27	B	P3		42	C	M1	
13	C	P1		28	A	P3		43	B	P2	
14	B	P1		29	A	P3		44	C	P3	
15	A	P1		30	D	P3		45	B, C	S2	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: B, D M1 · Implement security and governance

Admin and Member can manage OneLake security. Contributor can read and write data but can't manage OneLake security. Viewers and Build-permission users are the people those roles restrict.

Q2 Answer: A → E → B M1 · Implement security and governance

Warehouse RLS is predicate-based. You create an inline table-valued function as the predicate, in a dedicated security schema (recommended practice), then bind it to the table with CREATE SECURITY POLICY ... ADD FILTER PREDICATE. Dynamic data masking and DENY on the table are different tools: one masks values and the other blocks the whole table.

Q3 Answer: Yes · No · Yes M1 · Implement security and governance

Admin, Member and Contributor already have full read and write access to the workspace's data, so OneLake security roles don't restrict them. The filter applies to Viewers and to users who get access through item permissions.

Q4 Answer: A M1 · Implement security and governance

A Purview label policy that makes labelling mandatory forces users to choose a label when they create or edit content. Certification is an endorsement setting. OneLake security and RLS control access to data, not classification.

Q5 Answer: B M1 · Implement security and governance

Certification is turned on in the tenant settings, where the administrator names who may certify. Without that, nobody can certify. Workspace roles alone don't grant the right to certify, and a label is unrelated to endorsement.

Q6 Answer: Yes · Yes · No M2 · Maintain the analytics development lifecycle

Commit sends workspace changes to the branch, and Update applies branch changes to the workspace. Promoting items between stages with deployment rules is what deployment pipelines do. Git keeps the history and handles review, and pipelines handle promotion.

Q7 Answer: B, C M2 · Maintain the analytics development lifecycle

Deployment rules are data source rules and parameter rules, plus default lakehouse rules for notebooks. They let one artefact point at each stage's own sources. They don't edit DAX, role membership or report layout.

Q8 Answer: C → D → F → E M2 · Maintain the analytics development lifecycle

The connection is made in the workspace settings: pick the provider, choose the repo, branch and folder, then connect. After the first sync you commit to put the existing items into the branch. Deployment pipelines and .pbix files have nothing to do with setting up Git.

Q9 Answer: A M2 · Maintain the analytics development lifecycle

The impact analysis pane can email the contacts of the affected items. It's built for exactly this, before a breaking change. The other options don't communicate anything.

Q10 Answer: 1→D 2→C 3→B 4→E M2 · Maintain the analytics development lifecycle

A template carries the definition without data. A .pbids carries only a connection. A shared model is a live, governed dependency. A project stores definitions as text (TMDL and PBIR) for source control. A .pbix bundles everything, data included.

Q11 Answer: Yes · No · Yes M2 · Maintain the analytics development lifecycle

Read lets client tools connect, browse metadata and run queries. Any change to the model, such as deploying, processing partitions or editing roles, needs the XMLA endpoint set to Read Write.

Q12 Answer: C P1 · Get data

Eventhouses are built for high-volume, append-only log and time-series data. Columnar indexing and KQL operators like has, contains and bin make text search and time bucketing fast. Warehouses suit structured dimensional data. SQL databases are operational OLTP stores.

Q13 Answer: C P1 · Get data

SQL database in Fabric is the operational (OLTP) database. Its data is replicated into OneLake as Delta automatically, ready for analytics. A warehouse is an analytical store, not a database for an application's day-to-day transactions.

Q14 Answer: B P1 · Get data

External shortcuts sign in through a cloud connection, and only users who have permission on that connection can bind shortcuts to it. ADLS Gen2 is a supported target. Creating shortcuts doesn't need a tenant admin.

Q15 Answer: A P1 · Get data

Very large volumes, custom code and testability all point to Spark notebooks. Dataflow Gen2 is great for low-code, analyst-owned work at moderate scale. The Visual Query Editor and querysets only query data. They don't run transformation pipelines.

Q16 Answer: A, C, D P1 · Get data

External shortcuts point at storage: ADLS Gen2, Blob Storage, Amazon S3 and S3-compatible storage, Google Cloud Storage, Dataverse, OneDrive or SharePoint, and on-premises locations through a gateway. Relational tables and SaaS objects are reached through mirroring or ingestion, not shortcuts.

Q17 Answer: D P1 · Get data

An Eventstream takes in streaming sources such as Event Hubs, applies no-code transformations and routes the events to destinations including Eventhouse and lakehouse. Dataflows are batch tools, and deployment pipelines handle ALM.

Q18 Answer: D P1 · Get data

Connections are shareable objects with roles: Owner, User, and User with resharing. A User can choose the connection in their own items without seeing the stored credentials. Workspace roles don't govern connections.

Q19 Answer: Yes · No · Yes P2 · Transform data

The SQL analytics endpoint is read-only for table data, although you can create views and set permissions there. Lakehouse tables are written with Spark, with Dataflow Gen2 or pipelines that use a lakehouse destination, or with other tools that write Delta. T-SQL writes need a warehouse.

Q20 Answer: C P2 · Transform data

In a Delta MERGE, the not-matched clause handles source rows that have no match in the target, which here means new customers to insert. whenMatchedUpdateAll updates existing customers. NotMatchedBySource handles target rows that are missing from the source.

Q21 Answer: D P2 · Transform data

row_number over a window partitioned by the key and ordered newest-first gives 1 to the latest row. Then you filter rn == 1. A window without a partition by OrderID ranks across all orders. Partitioning by ModifiedAt groups the wrong rows. max over the partition returns a timestamp, not a row number to filter on.

Q22 Answer: D P2 · Transform data

Change type using locale tells Power Query how to read the text, so 31/03/2026 is understood as 31 March. Removing error rows loses data. Leaving it as text pushes the problem downstream.

Q23 Answer: B P2 · Transform data

Declaring the grain is the first step in dimensional design. It decides which dimensions apply and which measures are additive. Everything else follows from it.

Q24 Answer: D P2 · Transform data

An Unknown (or inferred) member keeps the rows joinable, so totals stay correct and the relationship stays valid. NULL keys turn into a blank member and can break Direct Lake relationship rules. Deleting the rows understates sales.

Q25 Answer: A P2 · Transform data

A view encapsulates the filter and the derived column and is always current. A CTAS creates a copy with neither. A procedure can't be selected from like a table. A computed column changes the table itself and doesn't filter anything.

Q26 Answer: Yes · Yes · No P3 · Query and analyze data

The Visual Query Editor is a no-code, Power Query-style canvas over warehouses and SQL analytics endpoints. It generates SQL you can inspect, and you can save the result as a view. KQL is written in queriesets.

Q27 Answer: B P3 · Query and analyze data

dcount gives a distinct count, bin groups by day, and render timechart draws a line over time. count() counts page views, not distinct users. distinct returns the unique values themselves, not a count. project can't aggregate.

Q28 Answer: A P3 · Query and analyze data

Inside a GROUP BY query, SUM(SUM(Revenue)) OVER () applies a window over the grouped rows, which gives the grand total. Dividing each group's sum by it gives the share. Multiplying by 1.0 avoids integer division. The other options give an average, a rank or a per-category total.

Q29 Answer: A P3 · Query and analyze data

SUMMARIZECOLUMNS accepts filter tables as arguments, and TREATAS({2026}, 'Date'[Year]) is one. FILTER needs a table, not a measure. CALCULATE returns a scalar. DAX queries have no WHERE clause.

Q30 Answer: D P3 · Query and analyze data

CASE WHEN ... THEN ... ELSE ... END is the standard way to bucket values in a SELECT. PIVOT turns rows into columns. The other two are much more complex than this needs.

Q31 Answer: C S1 · Design and build semantic models

A many-to-many relationship (many-to-many cardinality) lets you relate facts stored at a higher grain to a dimension on a non-unique column. Filtering in a single direction keeps it predictable. Duplicating budget rows inflates totals, and merging facts with different grains breaks both.

Q32 Answer: D S1 · Design and build semantic models

Calculation items use SELECTEDMEASURE() so the same logic applies to whatever measure is in the visual, and DATESYTD gives the year-to-date dates. A hard-coded [Sales] only works for one measure. SELECTEDMEASURENAME returns text. ALL('Date') removes the date filter instead of applying YTD.

Q33 Answer: C S1 · Design and build semantic models

Field parameters are calculated tables with rows like ("Revenue", NAMEOF([Revenue]), 0). The slicer drives which field the visual shows. GENERATESERIES is what what-if parameters use.

Q34 Answer: B S1 · Design and build semantic models

RANK with DENSE gives tied values the same rank with no gaps. ALLSELECTED limits the ranking to what the user can see. ROWNUMBER never ties. INDEX returns a row, not a rank. Ranking over all Sales rows by quantity answers a different question.

Q35 Answer: Yes · Yes · No S1 · Design and build semantic models

Use bidirectional filtering sparingly, because it can cause ambiguity and expensive queries. You can have several relationships between two tables, with one active and the rest used through USERELATIONSHIP. In a star schema the dimension is the one side and the fact is the many side.

Q36 Answer: A S1 · Design and build semantic models

Sort by column orders a text column by another column, here the month number. Prefixes are a workaround that spoils the labels. MonthName isn't a date.

Q37 Answer: B S2 · Optimize enterprise-scale semantic models

Query scale-out requires large semantic model storage format on the model. RLS and Git integration are unaffected.

Q38 Answer: B S2 · Optimize enterprise-scale semantic models

A variable is evaluated once in its context and then reused, so the calculation doesn't run three times. IFERROR hides errors and blocks some optimisations. Calculated columns are stored per row and can't respond to filters the way a measure does.

Q39 Answer: A S2 · Optimize enterprise-scale semantic models

Auto date/time creates a hidden date table for every date column, which inflates the model. Turning it off and using a single marked date table removes them. Large model format raises the size limit but doesn't fix the bloat, and relating or changing the storage mode of the hidden tables doesn't remove them.

Q40 Answer: C S2 · Optimize enterprise-scale semantic models

With automatic updates on, the model can reframe while the load is still running. Turning them off and triggering a refresh (framing) at the end of the pipeline makes all the tables move to their new versions together. That keeps reports consistent without copying any data.

Q41 Answer: D S2 · Optimize enterprise-scale semantic models

Microsoft's guidance aims for row groups of about 1 to 16 million rows, with large target file sizes. Tiny row groups create many segments and raise transcoding overhead.

Q42 Answer: C M1 · Implement security and governance

RLS doesn't apply to workspace Admins, Members or Contributors, who can edit the model. Moving the managers to Viewer, or to app consumers, makes the existing roles take effect.

Q43 Answer: B P2 · Transform data

COALESCE returns the first non-null value, so NULL tail numbers become 'UNKNOWN'. NULLIF does the reverse: it turns 'UNKNOWN' into NULL. TRIM(ISNULL(..., '')) gives an empty string, not 'UNKNOWN'. The IIF/ISNUMERIC expression creates more NULLs instead of fewer.

Q44 Answer: C P3 · Query and analyze data

arg_max(Timestamp, *) returns the whole row with the latest timestamp for each tail number. top 1 returns only the single latest row across all aircraft. distinct returns every combination seen, not the latest. max(Timestamp) by Status groups by the wrong column and returns no other fields.

Q45 Answer: B, C S2 · Optimize enterprise-scale semantic models

Every visual sends at least one query, so fewer visuals means less work. Performance Analyzer shows which visuals cost the most. More slicers add queries, and bidirectional relationships make queries more expensive.

Core — governing what you build

Access control at every layer, ALM habits, and the modelling choices that keep numbers right.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 2 · Core

12 Maintain a data analytics solution 25–30%	21 Prepare data 45–50%	12 Implement and manage semantic models 25–30%
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Q1 SINGLE ANSWER

An external auditor must run T-SQL queries against one warehouse for a month. They must not see any other item in the workspace. What should you do, following least privilege?

- ☐ A Add the auditor to the workspace with the Contributor role
- ☐ B Share the warehouse with Read and Write permissions
- ☐ C Add the auditor to the workspace with the Viewer role
- ☐ D Share the warehouse with Read and ReadData, and no workspace role

Q2 SINGLE ANSWER

In a shared semantic model, the Payroll table and its measures must not appear in the field list for anyone outside HR. They mustn't even know the table exists. What should you implement?

- ☐ A Object-level security on the Payroll table, set in Tabular Editor
- ☐ B Hide the Payroll table and its measures in Desktop
- ☐ C A Highly Confidential label on the Payroll table
- ☐ D Row-level security that filters out every Payroll row

Q3 DRAG AND DROP — MATCH

Match each requirement to the access-control mechanism that meets it.

Choices — each may be used once, more than once, or not at all

- ☐ A Row-level security
- ☐ B Object-level security
- ☐ C Column-level security
- ☐ D OneLake folder-level security
- ☐ E Dynamic data masking

Requirement	Choice
1. Branch managers see only their branch's loan rows	
2. The CreditScore column returns an error for most analysts, but their other columns still work	
3. Agents see card numbers as XXXX-XXXX-XXXX-1234	
4. Only the Fraud team can read the Files/evidence folder in a lakehouse	
5. The Bonus measure is invisible to everyone outside Compensation	

Q4 MULTIPLE RESPONSE — CHOOSE TWO

Which two statements about endorsement in Fabric are true?

- ☐ A Any user with write permission on an item can mark it as Promoted.
- ☐ B Endorsement can only be applied to reports.
- ☐ C An item can carry both Promoted and Certified at the same time.
- ☐ D Certifying a semantic model gives every user in the tenant read access to it.
- ☐ E Only users the Fabric administrator authorises can certify items.

Q5 SINGLE ANSWER

Fabric administrators want one place to see how sensitivity labels and endorsement are used across the tenant, with links into Microsoft Purview. What should they use?

- ☐ A The lineage view of one workspace
- ☐ B The Microsoft Purview hub in Fabric
- ☐ C The Capacity Metrics app
- ☐ D A deployment pipeline

Q6 SINGLE ANSWER

Each salesperson may cover several regions, and each region has several salespeople. The mapping is in a table UserRegion(UPN, RegionKey), which relates to DimRegion. How should you implement dynamic RLS in the semantic model?

- ☐ A A role filtering DimRegion[Region] = USERNAME(), flowing to the facts
- ☐ B A role filtering UserRegion[UPN] = USERPRINCIPALNAME(), flowing through DimRegion to the facts
- ☐ C One static role per region, with users added to each role by hand
- ☐ D Object-level security that hides the other regions' columns

Q7 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about deployment pipelines, select Yes if it is true.

Statement	Yes	No
1. You can deploy only selected items from one stage to the next, not the whole workspace.	☐	☐
2. Deploying overwrites the paired items in the target stage.	☐	☐
3. A deployment pipeline keeps a line-by-line history of model changes for pull-request review.	☐	☐

Q8 SINGLE ANSWER

You select Update in a Git-connected workspace and are told there are conflicts. The same notebook was changed in the workspace and in the branch. What should you do?

- ☐ A Disconnect Git, keep the workspace version, then reconnect
- ☐ B Overwrite the notebook from a deployment pipeline, then commit
- ☐ C Delete the notebook from the workspace, then select Update again
- ☐ D Resolve the conflict in the source control pane, or in the repository, then update

Q9 SINGLE ANSWER

You must compare a development semantic model with production and deploy only the measures that changed, without touching partitions or data. Which tool is designed for this, through the XMLA endpoint?

- ☐ A ALM Toolkit
 - ☐ B Power BI Report Builder
 - ☐ C The Capacity Metrics app
 - ☐ D The OneLake file explorer
-

Q10 MULTIPLE RESPONSE — CHOOSE THREE

The skills outline asks you to perform impact analysis of downstream dependencies. Which three item types does it name as starting points?

- ☐ A Warehouses
 - ☐ B Lakehouses
 - ☐ C Individual DAX measures
 - ☐ D Report themes
 - ☐ E Semantic models
-

Q11 SINGLE ANSWER

A developer works in Power BI Desktop with a .pbip project in a local clone of the repository. How do the changes reach the Git-connected Fabric workspace?

- ☐ A They publish the .pbip file from the Publish button, which also commits to Git
 - ☐ B Deployment pipelines pull .pbip files from local disks
 - ☐ C They commit and push from the local clone, often through a pull request, then select Update in the workspace
 - ☐ D They email the folder to the workspace admin
-

Q12 SINGLE ANSWER

A data engineering team uses Spark for bronze and silver, and a BI team writes T-SQL and wants stored procedures for gold. What architecture fits both teams?

- ☐ A Warehouse for every layer
 - ☐ B Lakehouse for every layer, with stored procedures on its SQL analytics endpoint
 - ☐ C Lakehouse for bronze and silver, warehouse for gold
 - ☐ D Eventhouse for every layer
-

Q13 MULTIPLE RESPONSE — CHOOSE TWO

Which two sources can be mirrored into Fabric?

- ☐ A An Amazon S3 bucket of CSV files
- ☐ B Snowflake
- ☐ C A SharePoint folder of Excel files
- ☐ D Azure SQL Database
- ☐ E Google Cloud Storage

Q14 SINGLE ANSWER

Notebooks read the same Amazon S3 data through a shortcut many times a day, and the cloud egress charges are high. Which Fabric feature can reduce them?

- ☐ A Convert the shortcut to a mirrored database
- ☐ B Turn on shortcut caching for the workspace
- ☐ C Add more capacity units
- ☐ D Use V-Order on the S3 files

Q15 SINGLE ANSWER

A Dataflow Gen2 loads the full daily product list into a lakehouse table, and the table must always hold just the current list. Which destination update method should you choose?

- ☐ A Merge with history
- ☐ B Replace
- ☐ C Insert only new rows
- ☐ D Append

Q16 SINGLE ANSWER

Parquet files land in ADLS Gen2 every hour and must be loaded into a Fabric Warehouse table with T-SQL, at high throughput. Which statement should you use?

- ☐ A COPY INTO
- ☐ B CREATE EXTERNAL TABLE
- ☐ C INSERT ... VALUES
- ☐ D BULK INSERT from a local path

Q17 SINGLE ANSWER

An analyst wants to find semantic models that are certified and belong to the Finance domain, then request access to one. Where should they start?

- ☐ A A KQL queryset over the workspace metadata
- ☐ B The Real-Time hub, filtered by domain
- ☐ C The OneLake catalog, filtered by domain and endorsement
- ☐ D The Capacity Metrics app, filtered by workspace

Q18 MULTIPLE RESPONSE — CHOOSE THREE

You are building a Type 2 slowly changing customer dimension. Which three columns does the design need?

- ☐ A A surrogate key that's unique per version
- ☐ B The customer's total sales
- ☐ C A current-row flag such as IsCurrent
- ☐ D Validity dates such as ValidFrom and ValidTo
- ☐ E A GUID copied from the source on every load

Q19 SINGLE ANSWER

You need the product categories whose total revenue is over 1,000,000. Which clause filters on the aggregated value?

```
SELECT p.Category, SUM(f.Revenue) AS Revenue
FROM gold.FactSales f JOIN gold.DimProduct p ON p.ProductKey = f.ProductKey
GROUP BY p.Category
_____;
```

- ☐ A WHERE Revenue > 1000000
- ☐ B ORDER BY SUM(Revenue) DESC
- ☐ C WHERE SUM(Revenue) > 1000000
- ☐ D HAVING SUM(Revenue) > 1000000

Q20 SINGLE ANSWER

Each JSON order document holds an array `items`. You need one row per order item in a DataFrame. Which function should you use?

- ☐ A F.flatten("orders")
- ☐ B F.collect_list("items")
- ☐ C F.split("items", ",")
- ☐ D F.explode("items")

Q21 SINGLE ANSWER

A DataFrame read from JSON has a struct column `address` with fields `city` and `postcode`. How do you select the city as its own column?

- ☐ A F.col("address.city").alias("City")
- ☐ B F.col("city.address").alias("City")
- ☐ C F.split("address", ".").getItem(0)
- ☐ D F.explode("address").alias("City")

Q22 SINGLE ANSWER

A spreadsheet export lists Region only on the first row of each block, and the rows below have null Region. In Dataflow Gen2, how do you fill the nulls with the value above?

- ☐ A Select Region and use Replace errors
- ☐ B Select Region and use Fill down
- ☐ C Select Region and use Remove blank rows
- ☐ D Select Region and use Unpivot other columns

Q23 SINGLE ANSWER

Which data type should a gold warehouse table use for monetary amounts so that totals reconcile exactly?

- ☐ A varchar(50)
- ☐ B decimal(19,4)
- ☐ C float(53)
- ☐ D real

Q24 SINGLE ANSWER

Testers need an instant copy of a large warehouse fact table to run destructive tests against, without duplicating the data files. Which statement should you use?

- A** BACKUP TABLE gold.FactSales;
- B** CREATE VIEW test.FactSales AS SELECT * FROM gold.FactSales;
- C** CREATE TABLE test.FactSales AS CLONE OF gold.FactSales;
- D** SELECT * INTO test.FactSales FROM gold.FactSales;

Q25 SINGLE ANSWER

You need the top three products by revenue in each category, with tied products sharing a rank. Which T-SQL pattern is correct?

- A** GROUP BY Category, Product HAVING COUNT(*) <= 3, ordered by Revenue DESC
- B** SELECT TOP (3) WITH TIES ... ORDER BY Category, Revenue DESC
- C** ROW_NUMBER() OVER (PARTITION BY Category ORDER BY Revenue DESC) <= 3, in an outer query
- D** DENSE_RANK() OVER (PARTITION BY Category ORDER BY Revenue DESC) <= 3, filtered in an outer query

Q26 SINGLE ANSWER

Which KQL query returns error messages that contain the term "timeout" from the last six hours?

- A** SELECT * FROM AppLogs WHERE Level = 'Error' AND Message LIKE '%timeout%' AND Timestamp > ago(6h)
- B** AppLogs | where Timestamp > ago(6h) and Level = "Error" and Message = "timeout"
- C** AppLogs | where Timestamp between (ago(6h) .. now()) and Level == "Error" and Message has "timeout"
- D** AppLogs | where Timestamp > ago(6h) | project Level == "Error", Message has "timeout"

Q27 SINGLE ANSWER

In KQL, you need a column Status that shows "Hot" when Temperature > 80 and "OK" otherwise. Which expression is correct?

- A** | project Status = CASE WHEN Temperature > 80 THEN 'Hot' END
- B** | where Temperature > 80
- C** | summarize Status = max(Temperature)
- D** | extend Status = iff(Temperature > 80, "Hot", "OK")

Q28 SINGLE ANSWER

A DAX query using SUMMARIZECOLUMNS('Product'[Name], "Sales", [Total Sales]) returns fewer products than the Product table holds. What is the most likely reason?

- A** The query needs ORDER BY to return all rows
- B** SUMMARIZECOLUMNS returns at most 1,000 rows
- C** Products with duplicate names are merged into one row
- D** SUMMARIZECOLUMNS drops rows where all its measures return BLANK

Q29 SINGLE ANSWER

In the Visual Query Editor, an analyst needs to add each order's customer segment from the Customer table. Which step should they use?

- ☐ A Keep top rows from Customer, ordered by CustomerKey
- ☐ B Merge queries, joining Orders to Customer on CustomerKey
- ☐ C Group by CustomerKey with a Max of Segment
- ☐ D Append queries, adding Customer to Orders

Q30 SINGLE ANSWER

An auditor asks what `gold.DimCustomer` looked like at 09:00 UTC yesterday. The warehouse's data retention period covers that time. Which T-SQL feature answers this without restoring a backup?

- ☐ A `SELECT * FROM gold.DimCustomer FOR SYSTEM_TIME AS OF '<yesterday 09:00>'`
- ☐ B Restore the warehouse to a new restore point first
- ☐ C `SELECT * FROM gold.DimCustomer WHERE ModifiedAt <= '<yesterday 09:00>'`
- ☐ D Time travel: `OPTION (FOR TIMESTAMP AS OF '<yesterday 09:00 UTC>')`

Q31 SINGLE ANSWER

Every order line carries an OrderNumber that has no other attributes. How should OrderNumber be modelled in the star schema?

- ☐ A As a calculated table
- ☐ B As its own dimension table with one column
- ☐ C As a degenerate dimension: keep it on the fact table
- ☐ D Remove it, because facts shouldn't carry text

Q32 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about large semantic model storage format, select Yes if it is true.

Statement	Yes	No
1. It's needed for a model to grow past the default size limit in the service.	☐	☐
2. Microsoft recommends it for models deployed and managed through the XMLA endpoint, even small ones.	☐	☐
3. It's supported on shared (Pro-only) capacity.	☐	☐

Q33 SINGLE ANSWER

A model has two calculation groups: Time Intelligence (YTD, PY) and Currency Conversion. YTD must be computed on values that have already been converted to the selected currency. What controls the order in which they apply?

- ☐ A The precedence property of each calculation group
- ☐ B The alphabetical order of the group names
- ☐ C The order of the slicers on the page
- ☐ D The sort order of the calculation items

Q34 SINGLE ANSWER

A [Revenue] measure must display as 1.2K, 3.4M or 5.6bn depending on its size, while staying numeric for sorting. What should you implement?

- ☐ A A field parameter over three scaled measures
- ☐ B A dynamic format string on the measure, chosen by its value
- ☐ C Three separate measures, one per unit of scale
- ☐ D Wrap the measure in FORMAT() with a scaled format

Q35 SINGLE ANSWER

Which measure returns a three-month moving average of [Revenue] using DAX window functions?

- ☐ A AVERAGEX(INDEX(3, ALLSELECTED('Date'[YearMonth]), ORDERBY('Date'[YearMonth])), [Revenue])
- ☐ B AVERAGEX(WINDOW(-2, REL, 0, REL, ALLSELECTED('Date'[YearMonth]), ORDERBY('Date'[YearMonth])), [Revenue])
- ☐ C CALCULATE(AVERAGE(Sales[Revenue]), DATESINPERIOD('Date'[Date], MAX('Date'[Date]), -3, DAY))
- ☐ D AVERAGEX(OFFSET(-2, ALLSELECTED('Date'[YearMonth]), ORDERBY('Date'[YearMonth])), [Revenue])

Q36 SINGLE ANSWER

A measure SUMX(Sales, [Margin]) is slow over 400 million rows, where [Margin] = SUM(Sales[Revenue]) - SUM(Sales[Cost]). Which rewrite returns the same result faster?

- ☐ A SUMX(Sales, CALCULATE([Margin]))
- ☐ B AVERAGEX(Sales, [Margin]) * COUNTROWS(Sales)
- ☐ C SUMX(Sales, Sales[Revenue] - Sales[Cost])
- ☐ D IFERROR(SUMX(Sales, [Margin]), 0)

Q37 SINGLE ANSWER

A gold table has a TransactionDateTime column precise to the second. It's the biggest column in the Direct Lake model and is used only to filter by date and hour. What should you do?

- ☐ A Convert it to text in ISO 8601 format
- ☐ B Add a surrogate GUID key for each timestamp
- ☐ C Split it into separate Date and Hour columns in gold
- ☐ D Hide it in the model and use a measure

Q38 SINGLE ANSWER

An Import fact uses incremental refresh with a daily refresh, but the business wants today's transactions to appear in near real time without refreshing more often. What should you configure?

- ☐ A The real-time DirectQuery option, which makes a hybrid table
- ☐ B Detect data changes on a LastModified column
- ☐ C Automatic page refresh on the Import table's report
- ☐ D A scheduled refresh every minute during business hours

Q39 MULTIPLE RESPONSE — CHOOSE THREE

Which three can cause a Direct Lake on SQL analytics endpoint model to fall back to DirectQuery?

- ☐ A Row-level security defined at the SQL analytics endpoint
- ☐ B A calculation group in the model
- ☐ C A query that exceeds a capacity guardrail
- ☐ D A measure that uses variables
- ☐ E Querying a SQL view that isn't materialised

Q40 SINGLE ANSWER

After every framing, the first report queries on a Direct Lake model are slower and later ones are fast. What explains this?

- ☐ A The first query runs in Import mode to warm the cache
- ☐ B The model falls back to DirectQuery after each framing
- ☐ C The first queries transcode columns from Delta into memory
- ☐ D Framing copies all the table data into memory

Q41 SINGLE ANSWER

Most report queries filter a gold Delta table by CustomerKey, which has high cardinality. Which table optimisation groups related values into the same files, so queries can skip the files they don't need?

- ☐ A Partitioning by CustomerKey
- ☐ B VACUUM
- ☐ C Turning off V-Order
- ☐ D OPTIMIZE ... ZORDER BY (CustomerKey)

Case study · Woodgrove Bank

Questions 42–45

Existing environment

Woodgrove Bank is building a customer-analytics platform in Fabric. A card-processing partner sends nested JSON transaction files and scanned statement PDFs. The data engineering team writes PySpark. The BI team keeps a gold warehouse, WH_Customer, using T-SQL.

Customers can share accounts. Customers' home addresses change, and risk reporting must show the address that applied when each transaction happened. Workspaces exist for Dev, Test and Prod.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Requirements

1. R1. Partner files must be landed as received, including the PDFs.
2. R2. Address history must be kept in the gold customer dimension.
3. R3. Every change must be reviewable in pull requests, and promotion to Prod must not need manual rebuilds.
4. R4. Balances must be analysable by customer even though accounts are shared.

Q42 SINGLE ANSWER

Where should the raw partner files land to satisfy R1?

- ☐ A The Files area of a lakehouse
- ☐ B A warehouse staging table
- ☐ C An Import semantic model
- ☐ D An Eventhouse table

Q43 SINGLE ANSWER

Which approach to the customer dimension satisfies R2?

- A** Type 2: expire the current row and insert a new row with a new surrogate key
- B** Store the address on each fact row instead of the dimension
- C** Type 1: update the address on the current row in place
- D** Delete the old row and insert a new one with the same surrogate key

Q44 HOT AREA — YES / NO FOR EACH STATEMENT

For each proposed design for R3, select Yes if it meets the requirement.

Statement	Yes	No
1. Connect the Dev workspace to Git, and use a deployment pipeline to promote from Dev to Test to Prod.	☐	☐
2. Use a deployment pipeline alone.	☐	☐
3. Download .pbix files from Dev and upload them to Prod by hand.	☐	☐

Q45 SINGLE ANSWER

Which model design satisfies R4?

- A** A CustomerAccount bridge between DimCustomer and DimAccount, with FactBalance related to DimAccount
- B** Relate FactBalance directly to DimCustomer with a one-to-many relationship
- C** Duplicate each balance row once for every account holder
- D** Merge DimCustomer and DimAccount into one wide dimension

End of Set 04 — answers follow on the next page.

Set 04 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	D	M1		16	A	P1		31	C	S1	
2	A	M1		17	C	P1		32	Yes · Yes · No	S1	
3	1→A 2→C 3→E 4→D 5→B	M1		18	A, C, D	P2		33	A	S1	
4	A, E	M1		19	D	P2		34	B	S1	
5	B	M1		20	D	P2		35	B	S1	
6	B	M1		21	A	P2		36	C	S2	
7	Yes · Yes · No	M2		22	B	P2		37	C	S2	
8	D	M2		23	B	P2		38	A	S2	
9	A	M2		24	C	P2		39	A, C, E	S2	
10	A, B, E	M2		25	D	P3		40	C	S2	
11	C	M2		26	C	P3		41	D	S2	
12	C	P1		27	D	P3		42	A	P1	
13	B, D	P1		28	D	P3		43	A	P2	
14	B	P1		29	B	P3		44	Yes · No · No	M2	
15	B	P1		30	D	P3		45	A	S1	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: D M1 · Implement security and governance

Sharing the item gives access to that one item only, and ReadData lets the auditor query it through T-SQL. Any workspace role exposes every item in the workspace. Write gives far more than querying.

Q2 Answer: A M1 · Implement security and governance

OLS takes the table and its dependent measures out of the model's metadata for users in the role, so they can't see it or query it. RLS still shows the table, just empty. Hiding a table only affects the field list in Desktop and isn't security. Labels can't be applied to individual tables.

Q3 Answer: 1→A 2→C 3→E 4→D 5→B M1 · Implement security and governance

RLS filters rows. CLS blocks access to columns, so queries that reference them fail. Masking shows obscured values while the column can still be queried. OneLake security roles govern folders and tables in the lake. OLS removes model objects from users' view entirely.

Q4 Answer: A, E M1 · Implement security and governance

Promoted is self-service. Certified (and Master data) is limited to users the administrator authorises. Endorsement helps people find and trust an item but grants no access. An item has one endorsement at a time. Many item types can be endorsed, not just reports.

Q5 Answer: B M1 · Implement security and governance

The Purview hub shows governance insights, such as sensitivity label coverage and endorsement, with links to Purview solutions. Capacity Metrics covers compute. Lineage covers one workspace's dependencies.

Q6 Answer: B M1 · Implement security and governance

Filtering the mapping table by the signed-in user's UPN and letting that filter reach DimRegion and the facts (using bidirectional filtering applied as a security filter where needed) handles many-to-many coverage with one role. Static roles per region become an administrative burden. USERNAME compared with a region name doesn't match users to their regions.

Q7 Answer: Yes · Yes · No M2 · Maintain the analytics development lifecycle

Pipelines support selective deployment and overwrite paired items. They're for promoting content, not version control. Line-level history and pull requests come from Git integration.

Q8 Answer: D M2 · Maintain the analytics development lifecycle

Fabric reports conflicts when an item changed on both sides. You can keep one side from the workspace UI, or resolve it in Git (for example in a branch and pull request) and then update. Disconnecting Git, deleting the notebook or overwriting it from a pipeline throws away one side's changes without a decision.

Q9 Answer: A M2 · Maintain the analytics development lifecycle

ALM Toolkit compares two models' metadata and deploys only the selected differences through the XMLA endpoint. Report Builder is for paginated reports. The file explorer browses OneLake files.

Q10 Answer: A, B, E M2 · Maintain the analytics development lifecycle

The outline names lakehouses, warehouses, dataflows and semantic models. Impact analysis works at the item level, not on single measures or themes.

Q11 Answer: C M2 · Maintain the analytics development lifecycle

Local projects go through the normal Git flow: commit, push and merge. The workspace then picks up the changes with Update from the source control pane. Publishing from Desktop puts content in the workspace without creating any Git commit.

Q12 Answer: C P1 · Get data

Splitting the layers by skill set is the recommended approach. Both items store Delta in OneLake, so the split costs nothing downstream. Stored procedures on a lakehouse endpoint can't write data, and an eventhouse is for event data.

Q13 Answer: B, D P1 · Get data

Mirroring replicates databases and data warehouses, including Azure SQL Database, Snowflake, Azure Cosmos DB and Azure Database for PostgreSQL. Object stores and file folders are reached with shortcuts or ingestion instead.

Q14 Answer: B P1 · Get data

Shortcut caching keeps files read from external sources such as S3 or Google Cloud Storage in OneLake for a set period, so repeated reads don't pay egress again. V-Order is a write-time setting for Delta tables you write. You can't apply it to files you don't own.

Q15 Answer: B P1 · Get data

Replace overwrites the table on each refresh, so it always holds the latest full list. Append adds the rows again on every run and creates duplicates.

Q16 Answer: A P1 · Get data

COPY INTO is the warehouse's high-throughput command for loading files from external storage. INSERT ... VALUES inserts literal rows one statement at a time. BULK INSERT from a local path isn't the warehouse's pattern for cloud files.

Q17 Answer: C P1 · Get data

The OneLake catalog is the discovery surface for stored data items. It can be filtered by domain, item type and endorsement, and it supports access requests. The Real-Time hub is for streaming data.

Q18 Answer: A, C, D P2 · Transform data

Each version needs its own surrogate key, so facts can point at the version that was current when the transaction happened. Validity dates and a current flag make history and current-state queries simple. Totals belong in facts, not dimensions.

Q19 Answer: D P2 · Transform data

HAVING filters groups after aggregation. WHERE runs before grouping and can't use aggregate functions. Filtering individual rows over 1,000,000 answers a different question.

Q20 Answer: D P2 · Transform data

explode turns each element of an array column into its own row. split turns a string into an array. collect_list does the reverse of explode. flatten merges nested arrays but doesn't create rows.

Q21 Answer: A P2 · Transform data

Dot notation reaches into struct fields. city.address reverses the path. explode is for arrays and maps, and split works on strings, not structs.

Q22 Answer: B P2 · Transform data

Fill down copies the last non-null value into the null cells below it, which is exactly what grouped spreadsheet exports need. The other steps remove or reshape data instead.

Q23 Answer: B P2 · Transform data

decimal is exact. float and real are approximate binary types and can give rounding differences when you add up large numbers of values. Storing money as text stops arithmetic and compresses badly.

Q24 Answer: C P2 · Transform data

A zero-copy clone creates a new table that refers to the same underlying data files, so it's instant. Changes made afterwards affect only the clone. SELECT INTO copies all the data. A view can't take destructive writes. BACKUP TABLE isn't a warehouse command.

Q25 Answer: D P3 · Query and analyze data

Partitioning by category restarts the ranking in each category, and DENSE_RANK lets tied products share a rank. ROW_NUMBER with the same partition is close, but it breaks ties arbitrarily, so a tied product can be dropped. TOP (3) WITH TIES returns three rows overall, not per category. The HAVING COUNT(*) query counts rows, not ranks.

Q26 Answer: C P3 · Query and analyze data

between with a time range and has for whole-term matching is idiomatic, efficient KQL. In KQL, = isn't a comparison operator (use ==), and Message == 'timeout' would match only messages that are exactly that word. The SELECT option is SQL. project can't filter rows.

Q27 Answer: D P3 · Query and analyze data

iff() is KQL's conditional function, and extend adds the result as a column. CASE WHEN is SQL syntax. summarize aggregates. where filters rows.

Q28 Answer: D P3 · Query and analyze data

SUMMARIZECOLUMNS automatically drops rows where all the expressions return blank, which is usually what you want. To include products with no sales, make the measure return 0 or use ADDCOLUMNS over VALUES. There's no fixed row cap, ORDER BY only sorts, and products with the same name are grouped by design, not by accident.

Q29 Answer: B P3 · Query and analyze data

Merge joins two queries on a key and adds columns from the second one. The Visual Query Editor turns it into a SQL JOIN. Append unions rows.

Q30 Answer: D P3 · Query and analyze data

Warehouse time travel queries a table as it was at a past point in time, within the retention period, using the FOR TIMESTAMP AS OF query hint. A WHERE on ModifiedAt doesn't bring back rows that were updated or deleted since. Restoring to a restore point is a restore, which the stem rules out. FOR SYSTEM_TIME AS OF is temporal-table syntax, which the Fabric Warehouse doesn't use.

Q31 Answer: C S1 · Design and build semantic models

An identifier with no attributes of its own stays on the fact as a degenerate dimension. A one-column dimension adds a join for nothing. Removing it stops drill-down to individual orders.

Q32 Answer: Yes · Yes · No S1 · Design and build semantic models

Large format lifts the size ceiling to the capacity's limit and improves XMLA write performance. It needs a Fabric F, Premium P, Embedded A or Premium Per User capacity, not shared capacity.

Q33 Answer: A S1 · Design and build semantic models

When several calculation groups apply to the same measure, each group's precedence property decides the order in which they're applied. Set it on purpose so the currency conversion happens inside the YTD calculation. Names, slicer positions and item order within a group have no effect on this.

Q34 Answer: B S1 · Design and build semantic models

A dynamic format string changes only how the value displays, so the measure stays numeric for sorting, totals and charts. FORMAT() returns text, which breaks numeric behaviour. A field parameter swaps fields; it doesn't format them.

Q35 Answer: B S1 · Design and build semantic models

WINDOW(-2, REL, 0, REL, ...) returns the current month and the two before it, and AVERAGEX averages [Revenue] across them. OFFSET(-2) returns only the single row two months back. INDEX(3) returns the third month in the selection. DATESINPERIOD with -3 DAY covers three days, not three months, and averages individual rows.

Q36 Answer: C S2 · Optimize enterprise-scale semantic models

Calling a measure inside an iterator causes context transition on every row. Using column arithmetic gives the same total and lets the storage engine do the work. Adding CALCULATE explicitly is the same context transition. IFERROR blocks optimisation.

Q37 Answer: C S2 · Optimize enterprise-scale semantic models

Splitting a high-cardinality datetime cuts the number of distinct values dramatically, which shrinks dictionaries and speeds up queries. Hiding a column doesn't reduce its size. Text and GUIDs make compression worse.

Q38 Answer: A S2 · Optimize enterprise-scale semantic models

The real-time option in incremental refresh adds a DirectQuery partition for the most recent period, which makes it a hybrid table. Historical partitions stay in Import. Detect data changes cuts refresh work but doesn't make data real time.

Q39 Answer: A, C, E S2 · Optimize enterprise-scale semantic models

Views, SQL-based granular security and guardrail breaches are documented fallback triggers. Calculation groups and DAX variables are fine in Direct Lake.

Q40 Answer: C S2 · Optimize enterprise-scale semantic models

Direct Lake loads columns on demand. After framing, a column's first use triggers transcoding from Parquet into memory, and later queries reuse the cached, warm data. Framing itself copies no data.

Q41 Answer: D S2 · Optimize enterprise-scale semantic models

Z-Order puts similar values together in the same files, so filters can skip files. It works alongside V-Order. Partitioning by a high-cardinality key creates many tiny files. VACUUM only removes unreferenced files.

Q42 Answer: A P1 · Get data

A lakehouse's Files area holds files in any format, including JSON and PDF, as they arrive, and PySpark can process them from there. Warehouses and eventhouses hold structured tables.

Q43 Answer: A P2 · Transform data

Type 2 keeps every version with its own surrogate key, so each transaction joins to the address that was current at the time. Updating in place, or deleting and re-inserting with the same key, is Type 1 and loses history.

Q44 Answer: Yes · No · No M2 · Maintain the analytics development lifecycle

Git gives change history and pull-request review, and pipelines give repeatable promotion with rules. A pipeline alone keeps no change history. Manual upload is neither reviewable nor automated.

Q45 Answer: A S1 · Design and build semantic models

Shared accounts make the customer-to-account relationship many-to-many, and the standard design for that is a bridge table. Duplicating balance rows inflates totals. A direct one-to-many relationship can't express shared ownership.

Exam standard — constraints decide

Longer stems where two options both work and one qualifier rules one out. Includes repeated-scenario Yes/No series.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 3 · Exam standard

12 Maintain a data analytics solution 25-30%	21 Prepare data 45-50%	12 Implement and manage semantic models 25-30%
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Q1 SINGLE ANSWER

You share a report with a colleague who has no access to the workspace or to the report's semantic model. What access to the semantic model does sharing the report give the colleague?

- ☐ A Read permission on the semantic model
- ☐ B Read and Write permission on the semantic model
- ☐ C Build permission on the semantic model
- ☐ D No access to the model at all

Q2 HOT AREA — YES / NO FOR EACH STATEMENT

Goal: analysts must read only the tables in the gold schema of a lakehouse, through both Spark and SQL, following least privilege. For each proposed solution, select Yes if it meets the goal.

Statement	Yes	No
1. Give the analysts the workspace Viewer role and a OneLake security role that grants read on the gold schema only.	☐	☐
2. Give the analysts the workspace Contributor role.	☐	☐
3. Share the lakehouse with the ReadAll permission and no OneLake security role.	☐	☐

Q3 SINGLE ANSWER

A user exports a report labelled Confidential – Finance to PDF. What happens to the file?

- ☐ A The label is removed, but the PDF is encrypted
- ☐ B The PDF is relabelled with the tenant's default label
- ☐ C The PDF keeps the label and the label's protection
- ☐ D The export is blocked, as for every labelled item

Q4 SINGLE ANSWER

A team in another workspace must create its own reports from your certified semantic model, but must not be able to change the model. Which permission should you grant on the model?

- ☐ A Build
- ☐ B Reshare only
- ☐ C Write
- ☐ D Workspace Contributor on your workspace

Q5 SINGLE ANSWER

You have created RLS roles in a semantic model published to the service. Before going live, you want to check what a specific user in the Sales-North role will see. What should you do?

- ☐ A Check the role's members in lineage view
- ☐ B Download the .pbix and view the role in Desktop
- ☐ C Use Test as role in the model's security settings
- ☐ D Add yourself to the role and reopen the report

Q6 HOT AREA — YES / NO FOR EACH STATEMENT

You deploy a lakehouse from the Development stage to Test with a deployment pipeline. For each statement, select Yes if it is true.

Statement	Yes	No
1. The lakehouse item is created in the Test workspace.	☐	☐
2. The data in the lakehouse's Delta tables is copied to Test.	☐	☐
3. Items in the target stage are overwritten only if they're paired with the source items.	☐	☐

Q7 SINGLE ANSWER

A lakehouse is committed to a Git repository through Fabric Git integration. What does the repository contain for it?

- ☐ A The item's definition, but no table data or files
- ☐ B A .pbix file of the default semantic model
- ☐ C The definition plus a copy of every Delta table
- ☐ D A shortcut that points back to the lakehouse

Q8 BUILD LIST — SEQUENCE

You need to set up a four-stage release process: Dev, Test, UAT and Prod. Put the actions in order.

Select the 4 actions required and arrange them in the correct order. Not every action is used.

- ☐ A Set the number of stages to four and name them Dev, Test, UAT and Prod
- ☐ B Disconnect the Dev workspace from Git
- ☐ C Download each report as a .pbix file
- ☐ D Create a deployment pipeline
- ☐ E Assign the existing development workspace to the Dev stage
- ☐ F Deploy from Dev to Test

Q9 SINGLE ANSWER

An Azure DevOps release pipeline must deploy semantic model changes through the XMLA endpoint using a service principal. Apart from the XMLA endpoint being set to Read Write, what is required?

- ☐ A A personal gateway registered to the service principal
- ☐ B The service principal tenant setting turned on, and the principal given a workspace role with write access
- ☐ C Only an Azure DevOps service connection that stores the principal's secret
- ☐ D A Power BI Pro licence assigned to the service principal in Entra ID

Q10 SINGLE ANSWER

Why does the PBIR report format make team development easier than a single report.json file?

- ☐ A PBIR encrypts the report so it's safe in public repositories
 - ☐ B Each page, visual and bookmark is stored in its own file
 - ☐ C PBIR embeds the semantic model, so no live connection is needed
 - ☐ D PBIR stores a data snapshot, so tests run offline
-

Q11 SINGLE ANSWER

Reports in six other workspaces connect to your shared semantic model. You plan to remove a measure. How do you find every report that might break?

- ☐ A Run impact analysis on the semantic model
 - ☐ B Check the model's refresh history for callers
 - ☐ C Check the Capacity Metrics app for queries
 - ☐ D Open lineage view in each of the six workspaces
-

Q12 HOT AREA — YES / NO FOR EACH STATEMENT

Goal: the gold layer must support multi-table transactions and T-SQL UPDATE statements written by SQL developers. For each proposed solution, select Yes if it meets the goal.

Statement	Yes	No
1. Build the gold layer in a Fabric Warehouse.	☐	☐
2. Build the gold layer in a lakehouse and run the UPDATE statements through its SQL analytics endpoint.	☐	☐
3. Build the gold layer in an Eventhouse.	☐	☐

Q13 SINGLE ANSWER

Your company keeps curated tables in Azure Databricks Unity Catalog. Fabric users must query them without copying the data or building pipelines. What should you create?

- ☐ A A pipeline that copies the tables nightly
 - ☐ B A mirrored Azure Databricks catalog item
 - ☐ C A Dataflow Gen2 for each table
 - ☐ D An Eventstream
-

Q14 SINGLE ANSWER

Clickstream events must land continuously as Delta tables in a lakehouse, so Spark jobs can process them in batches later. Which approach should you use?

- ☐ A A KQL queryset
 - ☐ B An Eventstream with a lakehouse destination
 - ☐ C A deployment pipeline
 - ☐ D A Dataflow Gen2 scheduled every minute
-

Q15 SINGLE ANSWER

A Dataflow Gen2 that copies a 300-million-row table without changing it runs very slowly. Which feature is designed to speed up this kind of large, simple load?

- ☐ A Automatic page refresh
- ☐ B Turning off staging and adding more Power Query steps
- ☐ C Fast copy in Dataflow Gen2
- ☐ D A personal gateway

Q16 SINGLE ANSWER

An Azure SQL Database is reachable only through a private endpoint inside a virtual network. A Fabric dataflow must read from it without exposing it publicly. What should you use?

- ☐ A A OneLake shortcut
- ☐ B A virtual network (VNet) data gateway
- ☐ C A public firewall rule that allows all IP addresses
- ☐ D A personal-mode gateway

Q17 SINGLE ANSWER

Finance users want to drag CSV files from their desktop into a lakehouse's Files folder, the same way they use OneDrive. What should you give them?

- ☐ A OneLake file explorer for Windows
- ☐ B Mirroring of their OneDrive folder
- ☐ C A KQL database with one-click ingestion
- ☐ D A Spark job definition that uploads files

Q18 SINGLE ANSWER

This warehouse script maintains a Type 2 dimension. Which step is missing at the marked line?

```
-- 1) _____
-- 2) Insert new versions for changed and brand-new customers
INSERT INTO dim.Customer (CustomerID, Segment, City, ValidFrom, ValidTo, IsCurrent)
SELECT s.CustomerID, s.Segment, s.City, @LoadDate, '9999-12-31', 1
FROM   stg.Customer s
LEFT JOIN dim.Customer d ON d.CustomerID = s.CustomerID AND d.IsCurrent = 1
WHERE  d.CustomerID IS NULL;
```

- ☐ A UPDATE the changed current rows: SET ValidTo = @LoadDate, IsCurrent = 0
- ☐ B UPDATE the changed current rows: SET Segment = s.Segment, City = s.City
- ☐ C DELETE the current rows for customers whose attributes changed
- ☐ D TRUNCATE TABLE dim.Customer, then reload every version

Q19 SINGLE ANSWER

Orders carry five low-cardinality flags: IsGift, IsExpress, IsReturned, PaymentType and Channel. Adding five tiny dimensions makes the model cluttered. What is the dimensional design pattern for this?

- ☐ A A junk dimension of the flag combinations, with one key on the fact
- ☐ B Drop the flags, which reports rarely use
- ☐ C A separate factless fact table for each of the five flags
- ☐ D Add the five flags as extra columns on the date dimension

Q20 SINGLE ANSWER

A Rating column has 100 rows: 80 have a value and 20 are NULL. What does AVG(Rating) in T-SQL divide by?

- ☐ A It returns NULL
- ☐ B 80, because AVG ignores NULLs
- ☐ C 100, treating NULLs as zero
- ☐ D It raises an error

Q21 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about de-duplication in PySpark, select Yes if it is true.

Statement	Yes	No
1. df.distinct() removes only rows that are identical in every column.	☐	☐
2. df.dropDuplicates(["OrderID"]) keeps one row per OrderID, even when other columns differ.	☐	☐
3. dropDuplicates(["OrderID"]) always keeps the most recent row per OrderID.	☐	☐

Q22 SINGLE ANSWER

An event column `epoch_s` holds Unix time in seconds. Which PySpark expression converts it to a timestamp?

- ☐ A `F.col("epoch_s").cast("string")`
- ☐ B `F.timestamp_seconds("epoch_s")`
- ☐ C `F.to_date("epoch_s", "yyyy-MM-dd")`
- ☐ D `F.current_timestamp()`

Q23 SINGLE ANSWER

A warehouse view must return orders from 2026. Which filter lets the engine skip non-matching data most effectively?

- ☐ A `WHERE DATEDIFF(year, OrderDate, '2026-12-31') = 0`
- ☐ B `WHERE YEAR(OrderDate) = 2026`
- ☐ C `WHERE OrderDate >= '2026-01-01' AND OrderDate < '2027-01-01'`
- ☐ D `WHERE CONVERT(varchar(4), OrderDate, 120) = '2026'`

Q24 SINGLE ANSWER

In a notebook, a 2-billion-row fact DataFrame is joined to a 300-row currency table. Which technique avoids shuffling the large table?

- ☐ A `facts.repartition(10000).join(currency, "CurrencyCode")`
- ☐ B `facts.crossJoin(currency).filter("CurrencyCode = CurrencyCode")`
- ☐ C `facts.join(F.broadcast(currency), "CurrencyCode")`
- ☐ D `facts.toPandas().merge(currency.toPandas(), on="CurrencyCode")`

Q25 DRAG AND DROP — MATCH

Match each data-quality problem to the transformation that fixes it.

Choices — each may be used once, more than once, or not at all

- A** Drop the column
- B** Convert the data type, specifying the format or locale
- C** De-duplicate on the business key
- D** Replace null with 0
- E** Map to an Unknown member

Requirement	Choice
1. The same customer appears three times after repeated loads	
2. Discount is NULL where no discount was given	
3. A date arrives as text in dd/MM/yyyy format	
4. Order rows reference a product code missing from the dimension	

Q26 SINGLE ANSWER

For each order you need one string listing all its product names, separated by commas. Which T-SQL function should you use?

- A** COALESCE(ProductName, ', ')
- B** STRING_SPLIT(ProductName, ',')
- C** STRING_AGG(ProductName, ', ')
- D** CONCAT(ProductName, ', ')

Q27 SINGLE ANSWER

Which KQL query returns the 95th-percentile request latency in 5-minute buckets?

- A** Requests | summarize p95 = max(DurationMs) * 0.95 by bin(Timestamp, 5m)
- B** Requests | where DurationMs > 95 | summarize count() by bin(Timestamp, 5m)
- C** Requests | top 95 by DurationMs | summarize by bin(Timestamp, 5m)
- D** Requests | summarize p95 = percentile(DurationMs, 95) by bin(Timestamp, 5m)

Q28 SINGLE ANSWER

A KQL table has 40 columns. You need every column except RawPayload and DebugInfo. Which operator is most concise?

- A** project-away RawPayload, DebugInfo
- B** summarize by RawPayload
- C** project RawPayload, DebugInfo
- D** extend RawPayload = ""

Q29 SINGLE ANSWER

You want to list every measure in a semantic model, with its expression, from the DAX query view. Which query should you run?

- ☐ (A) EVALUATE SELECTEDMEASURE()
- ☐ (B) EVALUATE INFO.VIEW.MEASURES()
- ☐ (C) SELECT * FROM MEASURES
- ☐ (D) EVALUATE ALLMEASURES()

Q30 SINGLE ANSWER

Which T-SQL returns customers whose first-ever order was placed in 2026?

- ☐ (A) SELECT TOP (1) WITH TIES CustomerKey FROM gold.FactSales WHERE OrderDate >= '2026-01-01' ORDER BY OrderDate;
- ☐ (B) SELECT DISTINCT CustomerKey FROM gold.FactSales WHERE OrderDate >= '2026-01-01' AND OrderDate < '2027-01-01';
- ☐ (C) SELECT CustomerKey FROM gold.FactSales GROUP BY CustomerKey HAVING MIN(OrderDate) >= '2026-01-01' AND MIN(OrderDate) < '2027-01-01';
- ☐ (D) SELECT CustomerKey FROM gold.FactSales WHERE MIN(OrderDate) >= '2026-01-01' GROUP BY CustomerKey;

Q31 HOT AREA — YES / NO FOR EACH STATEMENT

Goal: report on a Snowflake data warehouse, where data must not be copied out of Snowflake and visuals must show values that are at most a minute old. Mirroring isn't allowed. For each proposed storage mode, select Yes if it meets the goal.

Statement	Yes	No
1. DirectQuery to Snowflake	☐	☐
2. Import with an hourly refresh	☐	☐
3. Direct Lake over a lakehouse loaded nightly	☐	☐

Q32 SINGLE ANSWER

A report page needs an Order Date slicer and a Ship Date slicer side by side, each filtering sales by its own date at the same time. What is the clearest model design?

- ☐ (A) A calculation group
- ☐ (B) Two role-playing date tables, Order Date and Ship Date, each related through an active relationship to its key
- ☐ (C) One date table with an inactive relationship, and USERELATIONSHIP in every measure
- ☐ (D) A field parameter

Q33 SINGLE ANSWER

Which measure returns the average revenue per customer, rather than per transaction line?

- ☐ (A) SUMX(Sales, Sales[Revenue]) / 2
- ☐ (B) DIVIDE([Revenue], COUNTROWS(Sales))
- ☐ (C) AVERAGEX(VALUES(Sales[CustomerKey]), [Revenue])
- ☐ (D) AVERAGE(Sales[Revenue])

Q34 SINGLE ANSWER

A measure must return Red-product sales, but must not override the user's own colour selection. If the user selects Blue, the result should be blank. Which filter argument does this?

- ☐ A CALCULATE([Sales], KEEPFILTERS('Product'[Color] = "Red"))
- ☐ B CALCULATE([Sales], 'Product'[Color] = "Red")
- ☐ C CALCULATE([Sales], REMOVEFILTERS('Product'), 'Product'[Color] = "Red")
- ☐ D CALCULATE([Sales], ALL('Product'[Color]), 'Product'[Color] = "Red")

Q35 SINGLE ANSWER

A title measure must say "Filtered by region" only when a slicer directly filters the Region column, and not when another filter only cross-filters it. Which function should you use?

- ☐ A ISCROSSFILTERED('Geo'[Region])
- ☐ B ISBLANK('Geo'[Region])
- ☐ C HASONEVALUE('Geo'[Region])
- ☐ D ISFILTERED('Geo'[Region])

Q36 SINGLE ANSWER

Report authors use a live connection to a certified shared semantic model, and they need to add their own small Import table of regional targets. What should they do?

- ☐ A Download a copy of the certified model and add the targets
- ☐ B Use a .pbids file to connect to both sources
- ☐ C Ask for Write permission and add the targets to the certified model
- ☐ D Convert to a composite model (DirectQuery to the shared model) and import the targets

Q37 HOT AREA — YES / NO FOR EACH STATEMENT

Goal: a production Direct Lake on SQL analytics endpoint model must stay available even when some queries can't run in Direct Lake mode. For each proposed DirectLakeBehavior setting, select Yes if it meets the goal.

Statement	Yes	No
1. Automatic	☐	☐
2. DirectLakeOnly	☐	☐
3. DirectQueryOnly	☐	☐

Q38 SINGLE ANSWER

A Direct Lake model fails on visuals that use DimStore. You find that StoreKey contains duplicate values. What should you do?

- ☐ A Set DirectLakeBehavior to DirectQueryOnly for the model
- ☐ B Make the DimStore relationship filter in both directions
- ☐ C Change the relationship's cardinality to many-to-many
- ☐ D De-duplicate DimStore in the gold layer so StoreKey is unique

Q39 SINGLE ANSWER

A measure divides margin by revenue with IF(ISERROR(...)) to avoid division-by-zero errors, and it's slow. What is the recommended rewrite?

- ☐ A `FORMAT([Margin] / [Revenue], "0%")`
- ☐ B `[Margin] / [Revenue]`
- ☐ C `IFERROR([Margin] / [Revenue], BLANK())`
- ☐ D `DIVIDE([Margin], [Revenue])`

Q40 SINGLE ANSWER

An incremental refresh policy is set to Archive data starting 5 years before the refresh date, and Incrementally refresh data starting 10 days before. What happens at each refresh?

- ☐ A The last 10 days are refreshed and kept, and the rest is dropped
- ☐ B Ten yearly partitions are refreshed, and 5 years are kept
- ☐ C All 5 years are refreshed, and anything older is dropped
- ☐ D The last 10 days are refreshed, 5 years are kept, and older data is dropped

Q41 MULTIPLE RESPONSE — CHOOSE TWO

A streaming job writes many tiny files to a Delta table on every micro-batch. Which two Spark Delta features reduce the small-file problem automatically, without a separate maintenance job?

- ☐ A Time travel
- ☐ B Optimize write
- ☐ C Shortcut caching
- ☐ D Auto compaction
- ☐ E VACUUM

Case study · Litware Energy

Questions 42–45

Existing environment

Litware Energy collects readings from 2 million smart meters. Each meter sends a reading every minute. Regional analysts use a Direct Lake semantic model built on the SQL analytics endpoint of the warehouse WH_Grid. Row-level security by region is defined in WH_Grid with a security policy.

The model's main fact table, FactReading, holds 9 billion rows at minute grain. The tenant has an F64 capacity. Refresh history shows queries regularly falling back to DirectQuery, and the reason given is that the table is over the row guardrail.

Requirements

1. R1. Each regional analyst must see only their region through the semantic model, enforced by the SQL security policy.
2. R2. Meter readings must be queryable within seconds of arrival for operational monitoring.
3. R3. Operations needs average consumption per meter in 15-minute buckets over the last 24 hours.
4. R4. Reports used by analysts must stop falling back to DirectQuery.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Q42 SINGLE ANSWER

How must the semantic model connect to WH_Grid to satisfy R1?

- ☐ A With a fixed identity on a shareable cloud connection
- ☐ B As an Import model refreshed nightly with SSO
- ☐ C With Microsoft Entra single sign-on (SSO) to the SQL endpoint
- ☐ D As Direct Lake on OneLake with SSO

Q43 SINGLE ANSWER

Which architecture satisfies R2?

- ☐ A An hourly Import refresh of the semantic model
 - ☐ B A Dataflow Gen2 that runs every 15 minutes
 - ☐ C A nightly pipeline that loads into WH_Grid
 - ☐ D An Eventstream that ingests into an Eventhouse
-

Q44 SINGLE ANSWER

Which KQL query over the Readings table satisfies R3?

- ☐ A Readings | summarize AvgKWh = avg(KWh) by MeterId | where Timestamp > ago(24h)
 - ☐ B Readings | where Timestamp > ago(24h) | project AvgKWh = avg(KWh) by MeterId, bin(Timestamp, 15m)
 - ☐ C Readings | where Timestamp > ago(24h) | top 15 by KWh by MeterId
 - ☐ D Readings | where Timestamp > ago(24h) | summarize AvgKWh = avg(KWh) by MeterId, bin(Timestamp, 15m)
-

Q45 SINGLE ANSWER

Which change best satisfies R4 without changing capacity size?

- ☐ A Run VACUUM on FactReading to remove old rows
- ☐ B Add pre-aggregated measures to the model for each report page
- ☐ C Set DirectLakeBehavior to DirectLakeOnly so the fallback stops
- ☐ D Point the analysts' model at an hourly gold aggregate, keeping minute detail elsewhere

End of Set 05 — answers follow on the next page.

Set 05 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	A	M1		16	B	P1		31	Yes · No · No	S1	
2	Yes · No · No	M1		17	A	P1		32	B	S1	
3	C	M1		18	A	P2		33	C	S1	
4	A	M1		19	A	P2		34	A	S1	
5	C	M1		20	B	P2		35	D	S1	
6	Yes · No · Yes	M2		21	Yes · Yes · No	P2		36	D	S1	
7	A	M2		22	B	P2		37	Yes · No · No	S2	
8	D → A → E → F	M2		23	C	P2		38	D	S2	
9	B	M2		24	C	P2		39	D	S2	
10	B	M2		25	1→C 2→D 3→B 4→E	P2		40	D	S2	
11	A	M2		26	C	P3		41	B, D	S2	
12	Yes · No · No	P1		27	D	P3		42	C	M1	
13	B	P1		28	A	P3		43	D	P1	
14	B	P1		29	B	P3		44	D	P3	
15	C	P1		30	C	P3		45	D	S2	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: A M1 · Implement security and governance

Sharing a report gives the recipient Read on the underlying semantic model so the visuals can query it, and RLS still applies. Build (needed to create new content) and Write aren't granted unless you choose to grant them.

Q2 Answer: Yes · No · No M1 · Implement security and governance

A Viewer with a OneLake security role scoped to the gold schema gets exactly that, and OneLake security is enforced across engines. Contributor can read and write everything and bypasses OneLake roles. ReadAll on its own covers all of the lakehouse's data.

Q3 Answer: C M1 · Implement security and governance

Sensitivity labels and their protection carry through to supported exports such as Excel, PowerPoint and PDF. That's the main reason labels exist. Blocking exports is a separate policy decision, not something every label does.

Q4 Answer: A M1 · Implement security and governance

Build lets users create new reports and run queries against the model. Write lets them change it. A Contributor role in your workspace would let them edit everything there.

Q5 Answer: C M1 · Implement security and governance

Test as role renders the report as a member of the role, or as a named user, so you can check filters without changing memberships. Workspace admins see all data by default, so adding yourself to the role tells you nothing.

Q6 Answer: Yes · No · Yes M2 · Maintain the analytics development lifecycle

Deployment pipelines deploy item definitions and metadata, not data. You load data into each stage's lakehouse with that stage's own pipelines or notebooks. Pairing decides whether a deployment overwrites or creates a new item.

Q7 Answer: A M2 · Maintain the analytics development lifecycle

Git integration stores item definitions as text. Data stays in OneLake. That's why each environment loads its own data, and why repositories stay small.

Q8 Answer: D → A → E → F M2 · Maintain the analytics development lifecycle

You set the pipeline's stages when you create it: between 2 and 10, renamed as you like. Then you assign workspaces and deploy stage by stage. Git connects for version control but isn't a step in creating the pipeline, and downloading .pbix files is the manual process pipelines replace.

Q9 Answer: B M2 · Maintain the analytics development lifecycle

Service principals need the tenant setting that lets them use the APIs, often limited to a security group, plus a workspace role with write access. They can't be assigned licences, and gateways have nothing to do with XMLA deployments.

Q10 Answer: B M2 · Maintain the analytics development lifecycle

PBIR breaks the report definition into separate, documented JSON files for each object. Two developers editing different visuals no longer clash on one big file. It holds no data and doesn't replace the model.

Q11 Answer: A M2 · Maintain the analytics development lifecycle

Impact analysis on a semantic model shows downstream items across workspaces, with counts of workspaces and views. You can also notify their contacts from there. Refresh history and capacity metrics don't show dependencies.

Q12 Answer: Yes · No · No P1 · Get data

Only the warehouse supports T-SQL DML with multi-table transactions. A lakehouse SQL analytics endpoint is read-only for table data, and an eventhouse is built for ingesting events.

Q13 Answer: B P1 · Get data

Mirroring an Azure Databricks catalog brings the Unity Catalog metadata into Fabric, with shortcuts to the underlying Delta data. Nothing is copied and no pipeline is needed. Copying with pipelines or dataflows duplicates the data.

Q14 **Answer: B** P1 · Get data

Eventstreams take in streaming data and can write it continuously to a lakehouse as Delta, as well as to Eventhouse or other destinations. Dataflows are batch tools. Querysets and deployment pipelines don't ingest data.

Q15 **Answer: C** P1 · Get data

Fast copy uses the pipeline Copy activity's engine for large loads with simple transformations, which is much faster than the mashup engine. Adding steps slows it down. A personal gateway doesn't support dataflows.

Q16 **Answer: B** P1 · Get data

A VNet data gateway lets Fabric reach data sources inside an Azure virtual network without a public endpoint and without a server to manage. Opening the firewall to everyone breaks the security requirement. Shortcuts don't point at SQL databases.

Q17 **Answer: A** P1 · Get data

OneLake file explorer shows OneLake as a folder in Windows File Explorer, so users can upload, open and sync files with no code. The other options are engineering tools.

Q18 **Answer: A** P2 · Transform data

Type 2 needs two steps: expire the current version of changed customers (set ValidTo and IsCurrent = 0), then insert new versions. Updating in place is Type 1. Deleting or truncating destroys history.

Q19 **Answer: A** P2 · Transform data

A junk dimension collects unrelated low-cardinality attributes into one small table of their combinations. That gives one join instead of five and keeps the fact narrow. Putting them on the date dimension would be wrong, because they have nothing to do with dates.

Q20 **Answer: B** P2 · Transform data

Aggregate functions other than COUNT(*) ignore NULLs, so AVG divides by the 80 non-null values. If NULL should count as zero, write AVG(COALESCE(Rating, 0)). That's a business decision to make explicitly.

Q21 **Answer: Yes · Yes · No** P2 · Transform data

distinct compares whole rows. dropDuplicates on a subset keeps one row per key, but doesn't guarantee which row survives. To keep the latest, use row_number over a window ordered by the timestamp.

Q22 **Answer: B** P2 · Transform data

timestamp_seconds converts seconds since the Unix epoch into a timestamp. from_unixtime followed by a cast also works. to_date with a pattern expects text. current_timestamp ignores the column completely.

Q23 **Answer: C** P2 · Transform data

A range on the bare column can be checked against file and row-group minimum and maximum values, so whole chunks of data can be skipped. Wrapping the column in a function (YEAR, CONVERT, DATEDIFF) hides the raw values, so every row has to be evaluated.

Q24 **Answer: C** P2 · Transform data

Broadcasting the small table sends a copy to every executor, so the large table is joined where it sits, with no shuffle. Converting 2 billion rows to pandas won't fit in memory. A cross join multiplies the rows.

Q25 **Answer: 1→C 2→D 3→B 4→E** P2 · Transform data

Each problem has a targeted fix. Dropping a column hides the problem and loses information, so it isn't the right answer for any of them.

Q26 Answer: C P3 · Query and analyze data

STRING_AGG concatenates values across the rows of a group. CONCAT works within a single row. STRING_SPLIT does the reverse. COALESCE replaces nulls.

Q27 Answer: D P3 · Query and analyze data

percentile() inside summarize ... by bin() gives the 95th percentile for each 5-minute bucket. 95% of the maximum isn't a percentile. top 95 keeps only the 95 slowest requests. Counting requests over 95 ms answers a different question.

Q28 Answer: A P3 · Query and analyze data

project-away removes the columns you name and keeps the rest. project keeps only the columns you list. extend and summarize don't remove columns the way you need.

Q29 Answer: B P3 · Query and analyze data

The INFO family of DAX functions returns model metadata as tables. INFO.VIEW.MEASURES (or INFO.MEASURES) lists the measures and their expressions. ALLMEASURES doesn't exist. SELECTEDMEASURE only works inside calculation items.

Q30 Answer: C P3 · Query and analyze data

The first order date is MIN(OrderDate) per customer, filtered with HAVING. The DISTINCT query returns everyone who ordered in 2026, including long-standing customers. An aggregate can't be used in WHERE. TOP (1) WITH TIES returns only the customers who ordered on the first day of 2026.

Q31 Answer: Yes · No · No S1 · Design and build semantic models

DirectQuery leaves the data in the source and queries it live. Import copies the data and is only as fresh as the last refresh. Direct Lake needs the data in OneLake, which means copying or mirroring it, and both are ruled out.

Q32 Answer: B S1 · Design and build semantic models

When users must filter by both roles at once, separate role-playing tables with active relationships are simplest. USERRELATIONSHIP switches the active relationship inside a measure, so it doesn't support two independent slicers at the same time.

Q33 Answer: C S1 · Design and build semantic models

AVERAGEX over the distinct customers evaluates [Revenue] for each customer (context transition happens because it's a measure) and then averages the results. AVERAGE and dividing by COUNTROWS both work per line.

Q34 Answer: A S1 · Design and build semantic models

A plain Boolean filter replaces the existing filter on Color, so it would show Red sales even when Blue is selected. Adding ALL or REMOVEFILTERS alongside it clears the user's selection in the same way. KEEPFILTERS combines the new filter with the existing one, so Blue AND Red gives blank.

Q35 Answer: D S1 · Design and build semantic models

ISFILTERED is TRUE only when the column is filtered directly. ISCROSSFILTERED is also TRUE when another column or table filters it indirectly. HASONEVALUE tests for a single value, which can be true without any filter.

Q36 Answer: D S1 · Design and build semantic models

DirectQuery for Power BI semantic models lets authors extend a shared model with local tables in a composite model. The governed model stays untouched and authors can add their own data. Copying the model creates the duplicates certification is meant to prevent.

Q37 Answer: Yes · No · No S2 · Optimize enterprise-scale semantic models

Automatic falls back to DirectQuery when needed, so queries keep working, if more slowly. DirectLakeOnly turns those queries into errors, which is useful in development. DirectQueryOnly gives up Direct Lake performance completely.

Q38 Answer: D S2 · Optimize enterprise-scale semantic models

Direct Lake needs the one-side key to be unique, and duplicates make queries fail. The right fix is upstream, in the gold table. Changing the relationship hides a data-quality problem, and forcing DirectQuery throws away the performance.

Q39 Answer: D S2 · Optimize enterprise-scale semantic models

DIVIDE handles a zero denominator internally and is optimised for it. ISERROR and IFERROR force extra evaluation and can stop the engine using faster plans. A plain / errors on zero. FORMAT returns text.

Q40 Answer: D S2 · Optimize enterprise-scale semantic models

The archive range sets how much history is kept, and the incremental range sets what's reprocessed. Partitions older than the archive range are dropped as time moves on.

Q41 Answer: B, D S2 · Optimize enterprise-scale semantic models

Optimize write combines data into fewer, larger files at write time. Auto compaction checks fragmentation after each write and compacts when it's needed. VACUUM removes files nobody references, time travel reads old versions, and shortcut caching reduces external egress. None of those three fix small files.

Q42 Answer: C M1 · Implement security and governance

SQL-defined RLS can only filter per user if the SQL endpoint knows who the user is. That needs SSO. A fixed identity makes every user look like the same account. Direct Lake on OneLake reads files directly and bypasses SQL RLS.

Q43 Answer: D P1 · Get data

Eventstream plus Eventhouse is Fabric's real-time pattern: continuous ingestion and KQL queries within seconds. Nightly, 15-minute or hourly batches can't meet a seconds-level requirement.

Q44 Answer: D P3 · Query and analyze data

Filter to the last 24 hours, then summarize the average by meter and 15-minute bin. project can't aggregate. Filtering on Timestamp after summarize fails, because summarize has removed that column. top doesn't take a by clause and doesn't average anything.

Q45 Answer: D S2 · Optimize enterprise-scale semantic models

The fallback happens because FactReading is over the F64 row guardrail. Cutting the rows the model reads, with a gold table at the grain analysts actually use, brings it within the guardrail. DirectLakeOnly turns fallbacks into errors. VACUUM removes unreferenced files, not rows, and measures don't change what the model scans.

Exam standard — build, release, repeat

Development lifecycle in practice (Git, pipelines, XMLA, reusable assets) alongside the full data and modelling syllabus.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 3 · Exam standard

12 Maintain a data analytics solution 25-30%	21 Prepare data 45-50%	12 Implement and manage semantic models 25-30%
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Q1 MULTIPLE RESPONSE — CHOOSE TWO

Which two can a user with the workspace Viewer role do by default?

- ☐ A Share items with other users
- ☐ B Query lakehouse and warehouse data through the SQL analytics endpoint
- ☐ C Read lakehouse files directly with Spark notebooks
- ☐ D View reports and other content in the workspace
- ☐ E Edit semantic models

Q2 SINGLE ANSWER

Most new Fabric items should start as General unless the author picks something else. Which Microsoft Purview setting does this?

- ☐ A A workspace-level default sensitivity setting
- ☐ B Certification set as the tenant default
- ☐ C A default label in the Purview label policy
- ☐ D A domain admin assigned to label items

Q3 DRAG AND DROP — MATCH

Match each scenario to the endorsement it calls for. Some scenarios don't call for endorsement.

Choices — each may be used once, more than once, or not at all	
☐ A Promoted ☐ B No endorsement: use access controls ☐ C Master data ☐ D Certified	
Requirement	Choice
1. A self-service author wants colleagues to know a dataflow is ready to reuse	
2. The data governance board has reviewed a finance model against its quality standards	
3. The product lakehouse is the single authoritative source for product data	
4. Only HR may open the salary semantic model	

Q4 SINGLE ANSWER

A developer must create and edit notebooks, lakehouses and reports in a workspace and write data. They must not be able to add users or change permissions. Which role should you assign?

- ☐ A Contributor
- ☐ B Viewer
- ☐ C Admin
- ☐ D Member

Q5 HOT AREA — YES / NO FOR EACH STATEMENT

A semantic model has a dynamic RLS role. For each statement, select Yes if it is true.

Statement	Yes	No
1. You define roles and their DAX filters in Power BI Desktop, or with external tools.	☐	☐
2. You add users or groups to roles after publishing, in the semantic model's security settings in the service.	☐	☐
3. The RLS filters apply to workspace Contributors who open the report.	☐	☐

Q6 SINGLE ANSWER

An analyst belongs to two warehouse database roles. One role has GRANT SELECT ON SCHEMA::sales. The other has DENY SELECT ON sales.Commission. What happens when the analyst queries sales.Commission?

- ☐ A Access is denied, because DENY takes precedence over GRANT
- ☐ B Access is allowed, because GRANT at schema level wins
- ☐ C The query returns only the columns granted
- ☐ D The more recently created role wins

Q7 SINGLE ANSWER

You try to add a data source rule for a semantic model in the Test stage of a deployment pipeline, but the option is greyed out. Which requirement is most likely missing?

- ☐ A The workspace must be connected to GitHub, not Azure DevOps
- ☐ B The pipeline must have exactly three stages
- ☐ C The model must be in Import mode
- ☐ D You must be the owner of the semantic model in the target stage

Q8 BUILD LIST — SEQUENCE

A gold table column must be renamed. Reports in several workspaces depend on it. Put the steps of a safe change in order.

Select the 4 actions required and arrange them in the correct order. Not every action is used.

- ☐ A Promote the change through the deployment pipeline to Test, then Production
- ☐ B Run impact analysis on the lakehouse to find the dependent items
- ☐ C Make the change in the Development workspace and test it
- ☐ D Delete the dependent reports
- ☐ E Make the change directly in Production
- ☐ F Notify the contacts of the affected items

Q9 SINGLE ANSWER

A .pbit template must ask the user which server and database to connect to when they open it. What should the template contain?

- ☐ A Power Query parameters used in the source step
- ☐ B A .pbids file embedded in the template
- ☐ C A deployment rule for the server and database
- ☐ D Hard-coded connection strings in the source step

Q10 SINGLE ANSWER

Only the latest partition of a 2-billion-row Import table needs reprocessing after a late correction. The rest of the model must not be refreshed. What should you use?

- ☐ A A deployment pipeline redeploy of the model to the same stage
- ☐ B Republishing the .pbix with only that partition's data loaded
- ☐ C An on-demand refresh of the semantic model
- ☐ D A partition-level refresh through XMLA, or the enhanced refresh REST API

Q11 DRAG AND DROP — MATCH

Match each lifecycle requirement to the Fabric capability that meets it.

Choices — each may be used once, more than once, or not at all

- ☐ A Capacity Metrics app
- ☐ B Impact analysis
- ☐ C Deployment pipeline with rules
- ☐ D Git integration
- ☐ E XMLA endpoint (read-write)

Requirement	Choice
1. Keep a full change history of notebooks and models, and review changes in pull requests	
2. Promote the same tested content from Test to Production, pointing at each stage's own lakehouse	
3. Change one measure in a shared model without republishing any report	
4. List every report that will be affected before a warehouse table is dropped	

Q12 MULTIPLE RESPONSE — CHOOSE TWO

Which two requirements point to a lakehouse rather than a warehouse?

- ☐ A Stored procedures must update rows inside multi-table transactions
- ☐ B Data must be stored in Delta-Parquet format
- ☐ C Report authors write T-SQL views only
- ☐ D The data includes images and semi-structured JSON files
- ☐ E The engineering team works mainly in PySpark

Q13 SINGLE ANSWER

Lakehouse A has a OneLake security role that limits analysts to the EU rows of the Customers table. Lakehouse B contains a shortcut to that table. An analyst reads the shortcut from Lakehouse B. What does the analyst see?

- ☐ A All rows, because Lakehouse B has no role defined
- ☐ B Only the rows created after the shortcut was made
- ☐ C Only the EU rows, because the target's security applies
- ☐ D No rows, because shortcuts can't carry row filters

Q14 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about a mirrored database in Fabric, select Yes if it is true.

Statement	Yes	No
1. The replicated data is stored in OneLake in Delta format.	☐	☐
2. You can run UPDATE statements against the mirrored tables through the SQL analytics endpoint.	☐	☐
3. The mirrored data can be used in semantic models and joined with other Fabric data.	☐	☐

Q15 SINGLE ANSWER

A pipeline must start as soon as a new file arrives in a storage folder, not on a schedule. What should you configure?

- ☐ A A schedule trigger that runs every minute
- ☐ B An event trigger on file creation in the folder
- ☐ C An Until loop that polls the folder
- ☐ D A deployment rule on the storage path

Q16 SINGLE ANSWER

In a notebook attached to a lakehouse, which path reads the Delta table behind a Tables-folder shortcut named `partner_prices`?

- ☐ A `SELECT * FROM OPENROWSET('partner_prices')`
- ☐ B `spark.read.csv("Files/partner_prices.csv")`
- ☐ C `external_table("partner_prices")`
- ☐ D `spark.read.format("delta").load("Tables/partner_prices")`

Q17 SINGLE ANSWER

Data engineers want to be alerted, and to start a clean-up job, whenever an item is deleted in a production workspace. Which Fabric surface exposes these workspace item events so you can act on them?

- ☐ A The workspace's lineage view
- ☐ B The Real-Time hub, with an Activator rule
- ☐ C The deployment pipeline history
- ☐ D The OneLake catalog's Govern tab

Q18 SINGLE ANSWER

A team is choosing a store for a new self-service relational data mart that analysts will manage with SQL in Fabric. Which item fits the current platform?

- ☐ A KQL queryset
 - ☐ B An Eventhouse
 - ☐ C A Power BI datamart
 - ☐ D A warehouse
-

Q19 SINGLE ANSWER

A notebook rebuilds the silver table `customers_clean` from scratch every night. Which write mode replaces the existing contents?

- ☐ A `.mode("append").saveAsTable("customers_clean")`
 - ☐ B `.mode("overwrite").saveAsTable("customers_clean")`
 - ☐ C `.mode("ignore").saveAsTable("customers_clean")`
 - ☐ D `.mode("errorifexists").saveAsTable("customers_clean")`
-

Q20 SINGLE ANSWER

A source has added a new column, and appending to the existing Delta table now fails with a schema-mismatch error. The new column must be added to the table. What should you do?

- ☐ A Write with `.option("mergeSchema", "true")`
 - ☐ B Convert the table to CSV
 - ☐ C Use `.mode("ignore")`
 - ☐ D Drop the new column every time
-

Q21 SINGLE ANSWER

A 3-billion-row gold table is almost always filtered by month. When does partitioning the Delta table by a Year/Month column make sense?

- ☐ A When each partition stays large, well above about 1 GB
 - ☐ B Never, because Direct Lake can't read partitions
 - ☐ C Always, by the most detailed date column
 - ☐ D When the column has millions of distinct values
-

Q22 SINGLE ANSWER

A Dataflow Gen2 imports a budget sheet with the columns Department, Jan, Feb, ..., Dec. The gold table needs Department, Month and Amount columns. Which step should you use?

- ☐ A Select Department, then Pivot column
- ☐ B Transpose, then promote headers
- ☐ C Select Department, then Unpivot other columns
- ☐ D Group by Department, summing the months

Q23 SINGLE ANSWER

Two DataFrames hold the same customer columns in a different order, and one also has an extra column, Loyalty. They must be stacked into one DataFrame. Which call is correct?

- ☐ A df1.union(df2).distinct()
 - ☐ B df1.crossJoin(df2).dropDuplicates()
 - ☐ C df1.unionByName(df2, allowMissingColumns=True)
 - ☐ D df1.join(df2, "CustomerID", "outer")
-

Q24 SINGLE ANSWER

A warehouse load must add an AgeBand attribute ("<25", "25-44", "45-64", "65+") to DimCustomer, based on BirthDate. Where should this enrichment happen?

- ☐ A In the gold load, as a CASE expression in the DimCustomer procedure
 - ☐ B In a Power Query step in each user's report
 - ☐ C In each report as a visual calculation
 - ☐ D As a DAX calculated column in each semantic model
-

Q25 SINGLE ANSWER

A table records each order's dates as it moves through its lifecycle: ordered, packed, shipped and delivered. Each row is updated as the order progresses. Which fact table type is this?

- ☐ A Transaction fact
 - ☐ B Periodic snapshot fact
 - ☐ C Accumulating snapshot fact
 - ☐ D Factless fact
-

Q26 SINGLE ANSWER

You need customers who bought both Product A and Product B, in any orders. Which T-SQL pattern is correct?

- ☐ A SELECT CustomerKey FROM gold.FactSales WHERE ProductKey IN (@A, @B) GROUP BY CustomerKey HAVING COUNT(*) = 2;
 - ☐ B SELECT CustomerKey FROM gold.FactSales WHERE ProductKey IN (@A, @B) GROUP BY CustomerKey HAVING COUNT(DISTINCT ProductKey) = 2;
 - ☐ C SELECT DISTINCT CustomerKey FROM gold.FactSales WHERE ProductKey = @A AND ProductKey = @B;
 - ☐ D SELECT DISTINCT CustomerKey FROM gold.FactSales WHERE ProductKey IN (@A, @B) ORDER BY CustomerKey;
-

Q27 SINGLE ANSWER

A KQL query ends with `| summarize count() by Country`. What is the name of the count column in the result, if you don't name it?

- ☐ A count_
 - ☐ B Count
 - ☐ C rows
 - ☐ D _count
-

Q28 SINGLE ANSWER

Which KQL filter returns only today's events, from midnight UTC onwards?

- ☐ A | where day(Timestamp) = 1
 - ☐ B | where Timestamp >= startofday(now())
 - ☐ C | where Timestamp > ago(1d)
 - ☐ D | where Timestamp == today()
-

Q29 SINGLE ANSWER

A DAX query must list every product category, including those with no sales, which should show 0. Which query does this?

- ☐ A EVALUATE SUMMARIZECOLUMNS('Product'[Category], "Sales", [Total Sales])
 - ☐ B EVALUATE TOPN(10, VALUES('Product'[Category]))
 - ☐ C EVALUATE ADDCOLUMNS(VALUES('Product'[Category]), "Sales", [Total Sales] + 0)
 - ☐ D EVALUATE FILTER(VALUES('Product'[Category]), [Total Sales] > 0)
-

Q30 SINGLE ANSWER

In the Visual Query Editor, a business user needs total quantity per warehouse location. Which operation should they add after selecting the table?

- ☐ A Group by Location, with Sum of Quantity
 - ☐ B Remove duplicates on Location
 - ☐ C Append queries, one per location
 - ☐ D Merge queries on Location
-

Q31 SINGLE ANSWER

A model has FactSales and FactReturns. Users want to compare sales and returns by Product and Date. How should the facts be connected?

- ☐ A With a direct relationship between FactSales and FactReturns on OrderID
 - ☐ B Through shared (conformed) Product and Date dimensions, each related to both facts
 - ☐ C By merging the two facts into one table
 - ☐ D With a bidirectional many-to-many relationship between the facts
-

Q32 SINGLE ANSWER

Which measure returns revenue for the previous month, using standard time intelligence over a marked date table?

- ☐ A CALCULATE([Revenue], DATEADD('Date'[Date], -1, MONTH))
- ☐ B CALCULATE([Revenue], 'Date'[MonthNumber] - 1)
- ☐ C CALCULATE([Revenue], PREVIOUSYEAR('Date'[Date]))
- ☐ D [Revenue] - 1

Q33 SINGLE ANSWER

In a calculation group, most items should keep the base measure's own format string, and only the YoY % item should show as a percentage. What should the format string expression of the other items return?

- ☐ A SELECTEDMEASURENAME()
- ☐ B FORMAT(SELECTEDMEASURE(), "General")
- ☐ C SELECTEDMEASUREFORMATSTRING()
- ☐ D "0.00%"

Q34 SINGLE ANSWER

Users choose N (5, 10 or 20) from a slicer, and a measure must show sales only for the top N products. The slicer's table must not filter the model. What should you build?

- ☐ A An N table related to Product, read with SELECTEDVALUE
- ☐ B A disconnected table of N values, read with SELECTEDVALUE in a TOPN measure
- ☐ C A field parameter over Top 5, Top 10 and Top 20 measures
- ☐ D A calculation group with Top 5, Top 10 and Top 20 items

Q35 MULTIPLE RESPONSE — CHOOSE TWO

Which two are valid reasons to turn on large semantic model storage format?

- ☐ A You want to use query scale-out for the model
- ☐ B The model is expected to grow past the default size limit after refresh in the service
- ☐ C The model is published to My workspace
- ☐ D The model is a small DirectQuery model in a shared (Pro) workspace
- ☐ E You want to add RLS roles

Q36 SINGLE ANSWER

Performance Analyzer shows a visual with a short DAX query time but a long Other time. What does this usually mean?

- ☐ A The DAX measure is evaluated twice and needs variables
- ☐ B It was waiting on other visuals, often because the page has too many
- ☐ C The storage engine is slow to scan the fact table
- ☐ D The model has outgrown small storage format

Q37 HOT AREA — YES / NO FOR EACH STATEMENT

Goal: a Direct Lake model over a central lakehouse must include calculated tables and an Import table of targets. Security is managed in the semantic model. For each proposed configuration, select Yes if it meets the goal.

Statement	Yes	No
1. Direct Lake on OneLake	☐	☐
2. Direct Lake on the SQL analytics endpoint	☐	☐
3. DirectQuery over the SQL analytics endpoint, with no Direct Lake	☐	☐

Q38 SINGLE ANSWER

A model has several many-to-many relationships set to filter in both directions, added “just in case”. Queries are slow and some totals look wrong. What should you do first?

- ☐ A Convert the slow measures into calculated columns on the facts
- ☐ B Make relationships single-direction, using CROSSFILTER in the measures that need it
- ☐ C Turn on large semantic model storage format
- ☐ D Add more bidirectional relationships so every path is equally valid

Q39 SINGLE ANSWER

An incremental refresh on a SQL-sourced table takes as long as a full refresh. What is the most likely cause?

- ☐ A The table has too few rows to split into partitions
- ☐ B Detect data changes is switched on for the table
- ☐ C The model uses large semantic model storage format
- ☐ D The RangeStart/RangeEnd filter doesn't fold to the source

Q40 SINGLE ANSWER

A Delta table gets frequent updates and deletes, which leave deletion vectors behind. Queries have slowed down, although there are few small files. Which maintenance action is designed to rewrite the affected files?

- ☐ A Refresh the Direct Lake model more often
- ☐ B Turn off V-Order so update and delete rewrites are cheaper
- ☐ C Run OPTIMIZE to rewrite files and purge their deletion vectors
- ☐ D Run VACUUM with a retention of 0 hours to clear the vectors

Q41 SINGLE ANSWER

A table visual on a report returns every one of 1.2 million transaction rows, and users only ever look at the largest ones. What change helps most?

- ☐ A Apply a Top N filter, or summarise with drill-through to detail
- ☐ B Add a slicer for every column so users can narrow the rows
- ☐ C Split the table into two side-by-side tables of half the rows
- ☐ D Turn off query caching for the report

Case study · Adventure Works Cycles

Questions 42–45

Existing environment

Adventure Works Cycles builds bicycles in three plants. A lakehouse, LH_Gold, is built with notebooks. A Direct Lake semantic model serves sales and inventory reports. The workspaces Dev, Test and Prod are linked by a deployment pipeline, and Dev is connected to an Azure DevOps repository.

Four developers currently edit the Dev workspace at the same time and overwrite each other's work. Inventory is counted at the end of every day for each product and warehouse.

Requirements

1. R1. Each developer must work in isolation, and their changes must be reviewed before they reach the shared Dev workspace.
2. R2. Daily inventory levels must be stored so trends over time can be analysed.
3. R3. A SQL report must compare each month's revenue with the same month in the previous year.
4. R4. The inventory measure must show the level on the last day of the selected period, not a sum across days.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Q42 SINGLE ANSWER

Which practice satisfies R1?

- ☐ A Everyone keeps editing the Dev workspace, but commits after every change
 - ☐ B Turn off Git integration and share .pbip folders by email
 - ☐ C Each developer branches out to their own workspace and merges through a pull request
 - ☐ D Add a Feature stage before Dev in the deployment pipeline
-

Q43 SINGLE ANSWER

Which gold-layer design satisfies R2?

- ☐ A An accumulating snapshot fact with one row per product
 - ☐ B A transaction fact of stock movements only
 - ☐ C One row per product and warehouse, overwritten daily
 - ☐ D A periodic snapshot fact: one row per product, warehouse and day
-

Q44 SINGLE ANSWER

Which T-SQL expression satisfies R3, given a monthly table with one row per month and no gaps?

- ☐ A LEAD(Revenue, 12) OVER (ORDER BY MonthStart)
 - ☐ B SUM(Revenue) OVER (ORDER BY MonthStart ROWS 12 PRECEDING)
 - ☐ C RANK() OVER (ORDER BY MonthStart)
 - ☐ D LAG(Revenue, 12) OVER (ORDER BY MonthStart) AS RevenueSameMonthLastYear
-

Q45 SINGLE ANSWER

Which measure satisfies R4?

- ☐ A CALCULATE(SUM(FactInventory[QtyOnHand]), DATESBETWEEN('Date'[Date], MIN('Date'[Date]), MAX('Date'[Date])))
- ☐ B MAXX(VALUES('Date'[Date]), CALCULATE(SUM(FactInventory[QtyOnHand])))
- ☐ C AVERAGEX(VALUES('Date'[Date]), CALCULATE(SUM(FactInventory[QtyOnHand])))
- ☐ D CALCULATE(SUM(FactInventory[QtyOnHand]), LASTNONBLANK('Date'[Date], CALCULATE(SUM(FactInventory[QtyOnHand]))))

End of Set 06 — answers follow on the next page.

Set 06 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	B, D	M1		16	D	P1		31	B	S1	
2	C	M1		17	B	P1		32	A	S1	
3	1→A 2→D 3→C 4→B	M1		18	D	P1		33	C	S1	
4	A	M1		19	B	P2		34	B	S1	
5	Yes · Yes · No	M1		20	A	P2		35	A, B	S1	
6	A	M1		21	A	P2		36	B	S2	
7	D	M2		22	C	P2		37	Yes · No · No	S2	
8	B → F → C → A	M2		23	C	P2		38	B	S2	
9	A	M2		24	A	P2		39	D	S2	
10	D	M2		25	C	P2		40	C	S2	
11	1→D 2→C 3→E 4→B	M2		26	B	P3		41	A	S2	
12	D, E	P1		27	A	P3		42	C	M2	
13	C	P1		28	B	P3		43	D	P2	
14	Yes · No · Yes	P1		29	C	P3		44	D	P3	
15	B	P1		30	A	P3		45	D	S1	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: B, D M1 · Implement security and governance

Viewers can view content and connect to SQL analytics endpoints with T-SQL, where SQL permissions and RLS apply. Reading OneLake files with Spark needs ReadAll or a OneLake security role. Editing and sharing need higher roles or extra permissions.

Q2 Answer: C M1 · Implement security and governance

A Purview label policy can set a default label that's applied to new content automatically. Users can still change it, as the policy allows. The other options are about endorsement, delegated administration or data access.

Q3 Answer: 1→A 2→D 3→C 4→B M1 · Implement security and governance

Endorsement signals trust and helps people find content. Promoted is self-service. Certified follows a review by authorised certifiers. Master data marks the authoritative source for an entity. Restricting who can open an item is a permissions question, and endorsement never controls access.

Q4 Answer: A M1 · Implement security and governance

Contributor gives full content creation and data read and write without access management. Member can add users and share. Admin controls everything. Viewer can't create items.

Q5 Answer: Yes · Yes · No M1 · Implement security and governance

Roles are defined in the model and members are assigned in the service. Admin, Member and Contributor have edit rights on the model, so RLS doesn't restrict them. Only Viewers and users with Read or Build permission are filtered.

Q6 Answer: A M1 · Implement security and governance

In SQL permissions, an explicit DENY overrides any GRANT, wherever the GRANT comes from. That's useful for carving exceptions out of broad grants. Role creation order has no effect.

Q7 Answer: D M2 · Maintain the analytics development lifecycle

Deployment rules can only be created by the owner of the item in the target stage. Take over the item first, or ask the owner to create the rule. Storage mode, Git provider and the number of stages don't affect rule creation.

Q8 Answer: B → F → C → A M2 · Maintain the analytics development lifecycle

Find what depends on the table, warn the owners, change and test in Dev, then promote. Changing Production directly skips testing. Deleting reports isn't a change process.

Q9 Answer: A M2 · Maintain the analytics development lifecycle

When you save a report with parameters as a .pbit, Desktop asks for the parameter values when the template is opened. A .pbids is a separate file type. Hard-coded strings can't prompt for anything.

Q10 Answer: D M2 · Maintain the analytics development lifecycle

With XMLA read-write you can refresh individual tables or partitions. The enhanced refresh REST API can too. A scheduled refresh processes whatever the policy says. Republishing replaces the whole model.

Q11 Answer: 1→D 2→C 3→E 4→B M2 · Maintain the analytics development lifecycle

Each lifecycle capability has a distinct job: Git for history and review, pipelines for promotion, XMLA for changing the model's metadata in place, and impact analysis for dependencies. The Capacity Metrics app reports on compute and solves none of these.

Q12 Answer: D, E P1 · Get data

Unstructured and semi-structured data and a Spark-first team both point to a lakehouse. T-SQL DML with transactions points to a warehouse. Both items store Delta-Parquet, so format doesn't decide anything. Views work in both.

Q13 Answer: C P1 · Get data

Internal shortcuts check access against the target using the user's identity, so OneLake security defined at the source applies. That stops shortcuts from becoming a way around security.

Q14 Answer: Yes · No · Yes P1 · Get data

Mirroring lands the source data in OneLake as Delta, with a read-only SQL analytics endpoint. It's a replica, so changes come only from the source, and you can't update it through the endpoint. Because it's Delta in OneLake, other engines and semantic models can use it.

Q15 Answer: B P1 · Get data

Storage event triggers, configured through the Real-Time hub and Activator, start a pipeline when a file is created. Polling on a schedule adds delay and wastes runs.

Q16 Answer: D P1 · Get data

A Tables shortcut appears as a table under Tables/, so Spark reads it like any Delta table. You can also use `spark.read.table("partner_prices").external_table` is KQL, and the CSV path assumes a file that doesn't exist.

Q17 Answer: B P1 · Get data

The Real-Time hub lists streaming sources and system events, including Fabric workspace item events and OneLake events. Activator rules can respond to them with alerts or actions. The OneLake catalog is for discovering stored data.

Q18 Answer: D P1 · Get data

Microsoft now points self-service SQL workloads to the Fabric warehouse. Power BI datamarts are the legacy option and are being replaced. Eventhouse is for event data, and a queryset isn't a store.

Q19 Answer: B P2 · Transform data

`overwrite` replaces the table's contents, and Delta keeps the old version for time travel. `append` would add duplicates every night. `ignore` skips the write if the table exists. `errorIfExists` fails.

Q20 Answer: A P2 · Transform data

Delta enforces the table schema by default. `mergeSchema` allows an additive change, such as a new column, to evolve the schema during the write. Dropping the column loses data, and `ignore` skips the write entirely.

Q21 Answer: A P2 · Transform data

Partitioning helps when queries filter on a low-cardinality column and each partition is still large. Partitioning by high-cardinality columns creates many small files, which hurts Direct Lake and Spark. Fabric supports partitioned Delta tables.

Q22 Answer: C P2 · Transform data

`Unpivot other columns` turns the 12 month columns into Month and Value rows, while Department stays as it is. It also keeps working if new months are added. `Pivot` does the opposite. `Transpose` swaps rows and columns completely. Grouping by Department sums the months away.

Q23 Answer: C P2 · Transform data

`unionByName` matches columns by name rather than position, and `allowMissingColumns` fills absent columns with null. `union` matches by position, so it would misalign the columns, and `distinct` doesn't fix that. Joins combine columns, not rows.

Q24 Answer: A P2 · Transform data

Enriching the data in the gold layer means it's calculated once, stored, and available to every consumer and tool. Calculated columns in each model repeat the logic, and a Direct Lake on SQL model can't have them anyway.

Q25 Answer: C P2 · Transform data

An accumulating snapshot has one row per process instance, with several date keys that are filled in as milestones happen. A transaction fact records events that are never updated. A periodic snapshot records state at regular intervals. A factless fact records events with no measures.

Q26 Answer: B P3 · Query and analyze data

Filtering to the two products and requiring two distinct products per customer finds the customers who bought both. A single row can't equal A AND B. IN without HAVING returns customers who bought either one. COUNT(*) = 2 counts rows, so a customer who bought A twice also qualifies.

Q27 Answer: A P3 · Query and analyze data

KQL names an unnamed aggregate by its function plus an underscore, so count() becomes count_. You'd then write | order by count_ desc. Give it a name, such as Events = count(), for clarity.

Q28 Answer: B P3 · Query and analyze data

startofday(now()) is midnight today, UTC. ago(1d) is a rolling 24 hours, which includes part of yesterday. today() isn't a KQL function, and == against a timestamp would match only one instant.

Q29 Answer: C P3 · Query and analyze data

ADDCOLUMNS over VALUES keeps every category, and + 0 turns blanks into zeros. SUMMARIZECOLUMNS drops rows where the measure is blank. The FILTER query removes categories with no sales on purpose. TOPN returns ten arbitrary categories.

Q30 Answer: A P3 · Query and analyze data

Group by with an aggregation produces one row per location with the total. The editor turns it into a GROUP BY in the generated SQL.

Q31 Answer: B S1 · Design and build semantic models

In a star schema, facts are analysed together through conformed dimensions. Relating facts directly creates ambiguous paths and grain problems. Merging facts with different grains corrupts both.

Q32 Answer: A S1 · Design and build semantic models

DATEADD shifts the current dates back by one month. PREVIOUSYEAR returns the previous year. Subtracting 1 from the measure, or from a MonthNumber column inside CALCULATE, isn't a valid way to shift the date context.

Q33 Answer: C S1 · Design and build semantic models

SELECTEDMEASUREFORMATSTRING() returns the format string of the measure being modified, so currency stays currency and counts stay counts. A fixed percentage format would mislabel everything else. FORMAT returns text.

Q34 Answer: B S1 · Design and build semantic models

A disconnected table drives the slicer without filtering any data. The measure reads the chosen value with SELECTEDVALUE and applies it with TOPN or a rank test. A relationship would filter products by the numbers 5, 10 or 20.

Q35 Answer: A, B S1 · Design and build semantic models

Large format lifts the size ceiling and is a prerequisite for query scale-out. Microsoft also recommends it for models managed through XMLA. It isn't available on shared capacity or My workspace, and it has nothing to do with RLS.

Q36 Answer: B S2 · Optimize enterprise-scale semantic models

Other is time spent waiting, mostly for other visuals to finish, along with background work. A high Other with a fast query points to too many visuals on the page, not to slow DAX.

Q37 Answer: Yes · No · No S2 · Optimize enterprise-scale semantic models

Direct Lake on OneLake supports composite models and calculated tables, and here security lives in the model rather than in SQL. The SQL endpoint variant supports neither composite models nor calculated tables. DirectQuery gives up Direct Lake.

Q38 Answer: B S2 · Optimize enterprise-scale semantic models

Bidirectional filtering adds work to queries and can create ambiguous paths. Single-direction relationships, with CROSSFILTER only where a measure needs it, are faster and more predictable.

Q39 Answer: D S2 · Optimize enterprise-scale semantic models

If the filter doesn't fold, each partition query brings back everything and filters it locally, so incremental refresh becomes slower than a full refresh. Check folding (View native query) and move steps that break folding later in the query.

Q40 Answer: C S2 · Optimize enterprise-scale semantic models

OPTIMIZE compacts files and purges deletion vectors from files where enough rows are marked deleted. Auto compaction may not trigger without small files. VACUUM removes unreferenced files and, with zero retention, puts time travel and running readers at risk.

Q41 Answer: A S2 · Optimize enterprise-scale semantic models

Returning fewer rows means less work in the query and in rendering. A Top N filter, or aggregation with drill-through, keeps the visual responsive. Extra columns make it worse.

Q42 Answer: C M2 · Maintain the analytics development lifecycle

Branch out gives each developer an isolated branch and workspace, and pull requests are where review happens. Committing more often doesn't stop people overwriting each other. Pipeline stages are for promotion, not for isolating feature work.

Q43 Answer: D P2 · Transform data

A periodic snapshot records the state at regular intervals, which is exactly daily inventory levels. Movements alone would have to be summed from the start of time for every query. Overwriting destroys the history.

Q44 Answer: D P3 · Query and analyze data

LAG with an offset of 12 returns the value from 12 rows earlier, which is the same month last year when the months are contiguous. LEAD looks forward. The SUM gives a rolling total. If months could be missing, use a date-keyed join instead.

Q45 Answer: D S1 · Design and build semantic models

Inventory is semi-additive: you can add it across products and warehouses, but not across time. LASTNONBLANK finds the last date in the period that has data and evaluates the quantity there. DATESBETWEEN over the period still adds the days together. The daily average and the highest daily level aren't the closing balance.

Exam standard — shaping the gold layer

Dimensional design in the data store, transformation idioms in T-SQL, PySpark and Power Query, plus the rest of the outline.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 3 · Exam standard

12

Maintain a data analytics solution
25–30%

21

Prepare data
45–50%

12

Implement and manage semantic
models
25–30%

Q1 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about item permissions on a shared warehouse or lakehouse, select Yes if it is true.

Statement	Yes	No
1. Read lets a user see the item and its metadata.	☐	☐
2. ReadData lets a user query the data through the SQL analytics endpoint.	☐	☐
3. ReadAll is the right choice when a user needs only T-SQL access and must not read the files.	☐	☐

Q2 SINGLE ANSWER

Which users can apply a sensitivity label to a semantic model?

- ☐ A Only Fabric administrators, from the tenant admin portal
- ☐ B Users with edit rights on the item who have the label in their policy
- ☐ C Only capacity administrators for the model's capacity
- ☐ D Any user who can view the model in the workspace

Q3 SINGLE ANSWER

Four hundred analysts need Viewer access to a workspace, and people join and leave every week. How should you grant access with the least administrative effort?

- ☐ A Share each item with every analyst directly
- ☐ B Add each analyst to the Viewer role by name
- ☐ C Give team leads Member so they can add analysts
- ☐ D Assign an Entra security group to the Viewer role

Q4 SINGLE ANSWER

Members of the Analysts database role must be able to query every current and future table in the gold schema of a warehouse, and nothing outside it. Which statement should you use?

- ☐ A GRANT SELECT ON gold.FactSales TO Analysts;
- ☐ B GRANT SELECT ON DATABASE::WH TO Analysts;
- ☐ C GRANT CONTROL ON SCHEMA::gold TO Analysts;
- ☐ D GRANT SELECT ON SCHEMA::gold TO Analysts;

Q5 SINGLE ANSWER

Compliance wants to detect semantic models that contain credit card numbers and alert their owners automatically. What should you use?

- ☐ A Row-level security on card columns
- ☐ B Certification with a review checklist
- ☐ C Purview data loss prevention (DLP) policies
- ☐ D Deployment rules that scan each release

Q6 SINGLE ANSWER

You connect a new workspace to an existing Git branch and choose Update. The semantic models appear, but their refreshes fail. What is the most likely reason?

- ☐ A The models were converted to DirectQuery on sync
- ☐ B Git syncs the definitions but not the partitions
- ☐ C Credentials aren't stored in Git, so they must be set again
- ☐ D Git integration removes the models' data source steps

Q7 SINGLE ANSWER

In a deployment pipeline, a report and its semantic model are deployed together from Dev to Test. Which model does the deployed report use in Test?

- ☐ A No model, so you must rebind it manually
- ☐ B The Dev semantic model
- ☐ C A copy of the model embedded in the report
- ☐ D The paired semantic model in the Test stage (autobinding)

Q8 BUILD LIST — SEQUENCE

A developer edits a model in Power BI Desktop as a .pbip project inside a local clone of the team's repository. Put the steps to get the change into the Git-connected Dev workspace in order.

Select the 4 actions required and arrange them in the correct order. Not every action is used.

- ☐ A In the Dev workspace, select Update in the source control pane
- ☐ B Deploy from Dev to Test in the deployment pipeline
- ☐ C Publish the .pbix to Production from Desktop
- ☐ D Commit the changed files and push the branch
- ☐ E Open a pull request and merge it into the Dev branch after review
- ☐ F Save the project in Desktop

Q9 SINGLE ANSWER

Which address do you use to connect Tabular Editor to semantic models in the Fabric workspace "Sales Analytics" through the XMLA endpoint?

- ☐ A abfss://Sales Analytics@onelake.dfs.fabric.microsoft.com
- ☐ B Data Source=localhost:2383
- ☐ C https://app.fabric.microsoft.com/groups/Sales Analytics
- ☐ D powerbi://api.powerbi.com/v1.0/myorg/Sales Analytics

Q10 SINGLE ANSWER

What does a .pbids file contain?

- ☐ A The report theme and page layout
- ☐ B The model, report, data and credentials
- ☐ C The report pages and model, without data
- ☐ D Data source connection details only, as JSON

Q11 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about lineage view and impact analysis, select Yes if it is true.

Statement	Yes	No
1. Lineage view shows how data flows from sources through Fabric items to reports in a workspace.	☐	☐
2. Impact analysis on a semantic model can include dependent items in other workspaces.	☐	☐
3. Impact analysis automatically fixes the reports that a change would break.	☐	☐

Q12 DRAG AND DROP — MATCH

Match each workload to the most appropriate Fabric data store.

Choices — each may be used once, more than once, or not at all

- ☐ A SQL database in Fabric
- ☐ B Lakehouse
- ☐ C Eventhouse
- ☐ D Warehouse

Requirement	Choice
1. A Spark team lands JSON, Parquet and images for a medallion architecture	
2. SQL developers maintain a dimensional model with stored procedures that update rows	
3. Telemetry at 50,000 events per second needs sub-second time-window queries	
4. A new order-entry application needs a transactional relational database	

Q13 SINGLE ANSWER

A finance team maintains Excel files in a SharePoint document library. Data engineers need them to appear in a lakehouse's Files area, kept in sync and not copied. What should you create?

- ☐ A A SharePoint shortcut in the lakehouse's Files area
- ☐ B A mirrored database of the document library
- ☐ C A pipeline that copies the files every hour
- ☐ D A SharePoint shortcut in the Tables folder

Q14 SINGLE ANSWER

A software vendor's proprietary database isn't a built-in mirroring source. The vendor can write change files continuously. How can Fabric still keep a near-real-time replica of the database in OneLake?

- ☐ A An external shortcut to the vendor's database
- ☐ B Open mirroring, with changes written to its landing zone
- ☐ C A Dataflow Gen2 scheduled every minute
- ☐ D A KQL database with an update policy

Q15 SINGLE ANSWER

A pipeline loads only new and changed rows from a large source table each night. The source has a reliable LastModified column. Which pattern should you implement?

- ☐ A Copy the full table every night into staging
- ☐ B Copy the rows whose LastModified falls in the last 24 hours
- ☐ C A watermark: copy rows newer than the last LastModified, then save it
- ☐ D Truncate the target and reload all rows each night

Q16 SINGLE ANSWER

Business-critical dataflows depend on one on-premises data gateway server. If that server goes down, refreshes fail. How do you add resilience?

- ☐ A Add a second gateway member to the same cluster
- ☐ B Schedule every refresh twice, an hour apart
- ☐ C Install a personal gateway as a backup
- ☐ D Move the dataflows to a VNet data gateway

Q17 SINGLE ANSWER

A notebook in Workspace A must read the Delta table `orders` from a lakehouse named `LH_Sales` in Workspace B, without a shortcut. Which path format should you use?

- ☐ A `abfss://WorkspaceB@onelake.dfs.fabric.microsoft.com/LH_Sales.Lakehouse/Tables/orders`
- ☐ B `powerbi://api.powerbi.com/v1.0/myorg/WorkspaceB/LH_Sales/Tables/orders`
- ☐ C `Tables/LH_Sales/orders`
- ☐ D `https://app.fabric.microsoft.com/groups/WorkspaceB/lakehouses/LH_Sales/Tables/orders`

Q18 SINGLE ANSWER

Which key format is common practice for a gold date dimension?

- ☐ A The full date as text, such as 14-Mar-2026, as the key
- ☐ B A GUID per date, generated at load
- ☐ C An integer `yyyymmdd` key, such as 20260314, plus a Date column
- ☐ D An IDENTITY integer assigned in the order rows are loaded

Q19 SINGLE ANSWER

In a PySpark Type 2 load, you need an efficient way to detect whether any of 12 tracked attributes changed between staging and the current dimension row. What is a common technique?

- ☐ A Compare the surrogate keys of the source and target rows
- ☐ B Compare the row counts per business key in source and target
- ☐ C Rewrite every current row on every load
- ☐ D Compare a hash of the 12 tracked columns, such as sha2 over concat_ws

Q20 SINGLE ANSWER

A staging table contains exact duplicate rows: every column is identical. You want a clean copy in the warehouse. Which statement is simplest?

- ☐ A CREATE TABLE stg.Orders_Clean AS SELECT * FROM stg.Orders GROUP BY OrderID;
- ☐ B CREATE TABLE stg.Orders_Clean AS SELECT DISTINCT * FROM stg.Orders;
- ☐ C DELETE FROM stg.Orders WHERE OrderID IN (SELECT OrderID FROM stg.Orders);
- ☐ D CREATE TABLE stg.Orders_Clean AS SELECT TOP (1) * FROM stg.Orders;

Q21 SINGLE ANSWER

You must reconcile customer lists from the CRM and the billing system, and show customers who appear in only one system as well as those in both. Which join should you use?

- ☐ A LEFT JOIN from CRM to billing
- ☐ B FULL OUTER JOIN on CustomerID
- ☐ C INNER JOIN on CustomerID
- ☐ D CROSS JOIN

Q22 SINGLE ANSWER

Test transactions are flagged by a CustomerName that starts with "TEST_". Which PySpark line removes them?

- ☐ A df = df.filter(~F.col("CustomerName").startswith("TEST_"))
- ☐ B df = df.filter(F.col("CustomerName") == "TEST_")
- ☐ C df = df.dropna(subset=["CustomerName"])
- ☐ D df = df.withColumn("CustomerName", F.lit("TEST_"))

Q23 SINGLE ANSWER

A notebook must produce one row per Region, with a column for each Month containing that month's total Amount. Which expression is correct?

- ☐ A df.select("Region", "Month").distinct()
- ☐ B df.groupBy("Region").pivot("Month").sum("Amount")
- ☐ C df.orderBy("Region", "Month")
- ☐ D df.groupBy("Month").agg(F.sum("Region"))

Q24 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about denormalising dimensions in the gold layer, select Yes if it is true.

Statement	Yes	No
1. Flattening Product, Subcategory and Category into one DimProduct removes relationship hops in the semantic model.	☐	☐
2. Denormalising dimensions usually increases storage a little but improves usability and query simplicity.	☐	☐
3. Measures from the fact table should be copied into dimension tables to speed up reports.	☐	☐

Q25 SINGLE ANSWER

A report needs revenue by month, and each order's date must be rolled up to the first day of its month. Which expression should you use?

- ☐ A EOMONTH(OrderDate, -1)
- ☐ B DATEFROMPARTS(YEAR(OrderDate), 1, DAY(OrderDate))
- ☐ C DATEFROMPARTS(YEAR(OrderDate), MONTH(OrderDate), 1)
- ☐ D DATEADD(day, 1 - MONTH(OrderDate), OrderDate)

Q26 SINGLE ANSWER

Which T-SQL splits customers into four equal-sized groups by lifetime revenue?

- ☐ A PERCENT_RANK() OVER (ORDER BY CustomerKey)
- ☐ B NTILE(4) OVER (ORDER BY LifetimeRevenue DESC)
- ☐ C COUNT(*) / 4
- ☐ D RANK() OVER (ORDER BY LifetimeRevenue DESC) / 4

Q27 SINGLE ANSWER

In KQL, you need to add the device's model from a small DeviceInfo table to each Telemetry row, keeping telemetry rows even when no match exists. Which operator is designed for this kind of dimension lookup?

- ☐ A Telemetry | union DeviceInfo
- ☐ B Telemetry | summarize by DeviceId
- ☐ C Telemetry | lookup kind=leftouter DeviceInfo on DeviceId
- ☐ D Telemetry | join kind=inner DeviceInfo on DeviceId

Q28 SINGLE ANSWER

A KQL table has a Payload column holding JSON text such as {"temperature": 21.5}. Which expression extracts the temperature as a number?

- ☐ A | extend Temp = toreal(parse_json(Payload).temperature)
- ☐ B | extend Temp = toint(Payload)
- ☐ C | extend Temp = Payload.temperature
- ☐ D | where Payload has "temperature" | project Temp = Payload

Q29 SINGLE ANSWER

In the DAX query view, you want a single-row result showing total sales and total cost. Which query is valid?

- (A)** SELECT [Total Sales] AS Sales, [Total Cost] AS Cost
- (B)** EVALUATE ROW("Sales", [Total Sales], "Cost", [Total Cost])
- (C)** EVALUATE [Total Sales], [Total Cost] ORDER BY 1
- (D)** EVALUATE {[Total Sales], [Total Cost]} + {}

Q30 SINGLE ANSWER

Analysts need a quick distinct count of visitors across 20 billion rows in a warehouse. A small error margin is acceptable, but speed matters. Which function should they use?

- (A)** COUNT(DISTINCT VisitorID) OVER ()
- (B)** SUM(VisitorID)
- (C)** APPROX_COUNT_DISTINCT(VisitorID)
- (D)** COUNT(VisitorID)

Q31 SINGLE ANSWER

Source data is in an on-premises SQL Server with no plan to move it into OneLake. The model needs heavy Power Query shaping and the full range of DAX features, and a daily refresh is acceptable. Which storage mode fits best?

- (A)** DirectQuery through the on-premises gateway
- (B)** Dual storage mode for every table
- (C)** Direct Lake through the on-premises gateway
- (D)** Import, refreshed through the on-premises gateway

Q32 SINGLE ANSWER

Which measure counts the customers whose [Sales] is over 1,000 in the current filter context?

- (A)** COUNTROWS(FILTER(VALUES(Customer[CustomerKey]), [Sales] > 1000))
- (B)** DISTINCTCOUNT(Customer[CustomerKey]) > 1000
- (C)** CALCULATE(COUNT(Customer[CustomerKey]), Sales[Amount] > 1000)
- (D)** COUNTROWS(FILTER(Sales, Sales[Amount] > 1000))

Q33 SINGLE ANSWER

Which argument restarts the ranking for each category in the measure below?

```
Rank in Category =  
RANK ( DENSE,  
        ALLSELECTED ( 'Product'[Category], 'Product'[Name] ),  
        ORDERBY ( [Revenue], DESC ),  
        DEFAULT,  
        _____ )
```

- (A)** ALLSELECTED
- (B)** PARTITIONBY('Product'[Category])
- (C)** DENSE
- (D)** ORDERBY([Revenue], DESC)

Q34 SINGLE ANSWER

After you add a calculation group to a model, report authors can no longer drag a numeric column into a visual and have it summed automatically. Why?

- ☐ A Adding a calculation group turns on Discourage implicit measures
- ☐ B Object-level security was applied to numeric columns
- ☐ C The model has switched to DirectQuery for those columns
- ☐ D Calculation groups hide numeric columns from the field list

Q35 SINGLE ANSWER

EmployeeDetails has exactly one row per employee and shares EmployeeKey with DimEmployee, and the two are always used together. What is the recommended model design?

- ☐ A Keep a one-to-one bidirectional relationship
- ☐ B Create a many-to-many relationship
- ☐ C Hide EmployeeDetails and use LOOKUPVALUE in every measure
- ☐ D Merge them into a single DimEmployee table, in the gold layer or in Power Query

Q36 SINGLE ANSWER

In the DAX query view, you tested an improved version of a measure with DEFINE MEASURE and it's faster. How do you apply it to the model without retyping it?

- ☐ A Select Update model with changes above the DEFINE block
- ☐ B Run the query again with EVALUATE SAVE
- ☐ C Export the query as a .dax file and import it
- ☐ D Republish the .pbix, which saves query view changes

Q37 MULTIPLE RESPONSE — CHOOSE TWO

Which two actions reduce the memory footprint of an Import model?

- ☐ A Reduce the precision or cardinality of columns, for example by rounding timestamps to the minute or splitting datetime into date and time
- ☐ B Turn on Auto date/time
- ☐ C Add a calculated column for every measure
- ☐ D Remove columns that no report or measure uses
- ☐ E Convert integer keys to text

Q38 SINGLE ANSWER

One Direct Lake semantic model must combine Delta tables from a lakehouse and from a warehouse in different workspaces. Which Direct Lake variant supports this?

- ☐ A Direct Lake on OneLake
- ☐ B Direct Lake on the SQL analytics endpoint
- ☐ C Neither variant; Direct Lake allows one source only
- ☐ D Both variants equally

Q39 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about refreshing Direct Lake semantic models, select Yes if it is true.

Statement	Yes	No
1. A Direct Lake table needs an incremental refresh policy to avoid reloading all its data.	☐	☐
2. With automatic updates turned on, the model can reframe when the underlying Delta tables change.	☐	☐
3. Framing (refresh) normally completes in seconds, because it processes metadata only.	☐	☐

Q40 SINGLE ANSWER

A bronze landing table is written every few seconds by streaming jobs and is never read by Direct Lake or reports. What should you do about V-Order on this table?

- ☐ A Turn on V-Order, and Z-Order by every filter column
- ☐ B Turn on V-Order only for the most recent partition
- ☐ C Leave V-Order off, since it adds write cost with no read benefit
- ☐ D Turn on V-Order and run OPTIMIZE every minute

Q41 SINGLE ANSWER

A report page has eight slicers, and every change makes all the visuals requery straight away, so pages feel sluggish. Which setting cuts the number of queries?

- ☐ A Large semantic model storage format
- ☐ B Syncing the slicers across pages
- ☐ C Query reduction, such as an Apply button on slicers
- ☐ D Turning off cross-highlighting on visuals

Case study · Wide World Importers

Questions 42–45

Existing environment

Wide World Importers sells novelty goods through 40 promotions a year, and a product can be in several promotions at once. Customer data, including phone numbers, is stored in a lakehouse, LH_Core. Support agents query LH_Core with both T-SQL and Spark notebooks.

A supplier publishes its product catalogue as Parquet files in a Google Cloud Storage bucket. A Direct Lake on OneLake semantic model is planned for sales analysis.

Requirements

1. R1. Support agents must not see the PhoneNumber column, whether they use T-SQL or Spark. Everything else in Customers must stay readable.
2. R2. The supplier catalogue must be available in LH_Core without being copied.
3. R3. The gold layer must correctly represent products that belong to several promotions.
4. R4. Sales must be analysable by promotion without double counting at the total.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Q42 SINGLE ANSWER

What should you configure for R1? The support agents are workspace Viewers.

- ☐ A A OneLake security role that excludes the PhoneNumber column
- ☐ B Object-level security on PhoneNumber in the model
- ☐ C Dynamic data masking on PhoneNumber via SQL
- ☐ D A Highly Confidential label on LH_Core

Q43 SINGLE ANSWER

Which approach to the supplier catalogue satisfies R2?

- ☐ A An external shortcut to the GCS bucket
 - ☐ B A pipeline that copies the files daily
 - ☐ C Mirroring the GCS bucket into OneLake
 - ☐ D A Dataflow Gen2 that loads into LH_Core
-

Q44 SINGLE ANSWER

Which gold-layer structure satisfies R3?

- ☐ A One DimProduct row per promotion, repeating the product
 - ☐ B A comma-separated PromotionList column on DimProduct
 - ☐ C A PromotionKey on each fact row, with no other link
 - ☐ D A ProductPromotion bridge table, one row per product per promotion
-

Q45 SINGLE ANSWER

Which modelling behaviour satisfies R4?

- ☐ A Remove the bridge and make every relationship filter both ways
- ☐ B Filter through the bridge, so each promotion is right and the total counts each sale once
- ☐ C Duplicate each fact row once for every promotion
- ☐ D Make the grand total the sum of the per-promotion rows

End of Set 07 — answers follow on the next page.

Set 07 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	Yes · Yes · No	M1		16	A	P1		31	D	S1	
2	B	M1		17	A	P1		32	A	S1	
3	D	M1		18	C	P2		33	B	S1	
4	D	M1		19	D	P2		34	A	S1	
5	C	M1		20	B	P2		35	D	S1	
6	C	M2		21	B	P2		36	A	S2	
7	D	M2		22	A	P2		37	A, D	S2	
8	F → D → E → A	M2		23	B	P2		38	A	S2	
9	D	M2		24	Yes · Yes · No	P2		39	No · Yes · Yes	S2	
10	D	M2		25	C	P3		40	C	S2	
11	Yes · Yes · No	M2		26	B	P3		41	C	S2	
12	1→B 2→D 3→C 4→A	P1		27	C	P3		42	A	M1	
13	A	P1		28	A	P3		43	A	P1	
14	B	P1		29	B	P3		44	D	P2	
15	C	P1		30	C	P3		45	B	S1	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: Yes · Yes · No M1 · Implement security and governance

Read covers the item's metadata, ReadData allows SQL queries, and ReadAll allows OneLake and Spark file access. For a T-SQL-only user, ReadData is the least privilege. ReadAll would expose the files.

Q2 Answer: B M1 · Implement security and governance

Labelling needs edit rights on the item and the label must be in a policy published to the user. Viewers can't change labels, and labelling isn't limited to administrators.

Q3 Answer: D M1 · Implement security and governance

Granting roles to groups means joiners and leavers are handled in one place. Adding people individually and sharing item by item don't scale. Contributor gives too much access.

Q4 Answer: D M1 · Implement security and governance

A schema-level grant covers every object in the schema, including tables created later. A database-level grant reaches beyond gold. CONTROL gives full ownership-like rights. A single-table grant misses the other tables.

Q5 Answer: C M1 · Implement security and governance

DLP policies scan supported Fabric items for sensitive information types, such as credit card numbers, and can show policy tips and send alerts. RLS filters rows. Endorsement shows trust. Deployment rules repoint data sources.

Q6 Answer: C M2 · Maintain the analytics development lifecycle

Git holds item definitions, not secrets. After you sync a workspace from Git, you configure connections and credentials (or bind the models to existing connections) in that workspace.

Q7 Answer: D M2 · Maintain the analytics development lifecycle

Deployment pipelines autobind: a deployed report connects to the paired model in the same stage, so Test reports use Test models. Rebinding by hand is only needed in unusual setups.

Q8 Answer: F → D → E → A M2 · Maintain the analytics development lifecycle

The local Git flow is save, commit, push and pull request. Once the change is merged, the workspace picks it up with Update. Publishing from Desktop or using pipelines for this step skips the review.

Q9 Answer: D M2 · Maintain the analytics development lifecycle

The workspace connection is `powerbi://api.powerbi.com/v1.0/myorg/<workspace name>`, which you'll find in the workspace settings. `abfss` is the OneLake storage path. The other options are a portal URL and a local Analysis Services port.

Q10 Answer: D M2 · Maintain the analytics development lifecycle

A `.pbids` is a small JSON file describing a connection. Opening it starts Get data in Desktop with the source filled in, and users still sign in themselves. The other options describe a `.pbic`, a `.pbix` and a theme file.

Q11 Answer: Yes · Yes · No M2 · Maintain the analytics development lifecycle

Lineage visualises dependencies, and impact analysis lists the affected downstream items, including those in other workspaces, so you can notify their owners. Nothing gets fixed automatically. Planning and fixing the change is still your job.

Q12 Answer: 1→B 2→D 3→C 4→A P1 · Get data

Pick the store by data type, workload and skill set. Lakehouse is for mixed data and Spark, warehouse for T-SQL with DML, eventhouse for streaming events, and SQL database for operational OLTP workloads. Data in all four ends up in OneLake as Delta.

Q13 Answer: A P1 · Get data

OneDrive and SharePoint are supported shortcut targets. Excel files belong under Files, because Tables-folder shortcuts are for Delta tables. Copying isn't syncing, and mirroring is for databases.

Q14 Answer: B P1 · Get data

Open mirroring lets any application write change data in the documented format to a landing zone. Fabric then applies it to Delta tables in OneLake. Shortcuts point at storage, and minute-by-minute dataflows are batch ETL.

Q15 Answer: C P1 · Get data

The watermark pattern (Lookup old watermark, Copy with a filter, update the watermark) moves only the changes. Full reloads waste time and capacity as the table grows.

Q16 Answer: A P1 · Get data

A gateway cluster with several members gives high availability. Requests go to an available member if one fails. Personal gateways are single-user and don't support dataflows.

Q17 Answer: A P1 · Get data

OneLake uses ADLS Gen2-compatible paths of the form `abfss://<workspace>@onelake.dfs.fabric.microsoft.com/<item>.<type>/...` A relative `Tables/orders` path resolves against the notebook's default lakehouse. The `powerbi://` address is the XMLA endpoint.

Q18 Answer: C P2 · Transform data

A `yyyyMMdd` integer key is compact and readable, and keeps facts easy to partition and debug. You still keep a real Date column for time intelligence. GUIDs and random keys add cardinality without meaning.

Q19 Answer: D P2 · Transform data

Hashing the tracked columns gives one comparable value per row. A different hash means a change and triggers the expire-and-insert step. Row counts and surrogate keys can't detect attribute changes. Rewriting everything destroys the change history.

Q20 Answer: B P2 · Transform data

When rows are completely identical, `SELECT DISTINCT` removes the copies. `ROW_NUMBER` is for picking one row out of near-duplicates. The `DELETE` removes every row. `TOP (1)` keeps a single row. `GROUP BY OrderID` with `SELECT *` is invalid.

Q21 Answer: B P2 · Transform data

A full outer join keeps matched rows and the unmatched rows from both sides, which is what a reconciliation needs. An inner join drops the mismatches. A left join drops customers who are only in billing.

Q22 Answer: A P2 · Transform data

`startswith` matches the prefix, and `~` negates it, so test rows are dropped. Equality matches only the exact string `"TEST_"`. `dropna` removes nulls. `withColumn` overwrites every name.

Q23 Answer: B P2 · Transform data

`groupBy` plus `pivot` plus an aggregate turns month values into columns with totals in them. You can pass the list of months to `pivot` to avoid an extra pass over the data. The other options don't `pivot`.

Q24 Answer: Yes · Yes · No P2 · Transform data

Flattening snowflaked dimensions is standard. The small repetition compresses well in columnar storage, and the model gets simpler. Measures belong in facts, so copying them into dimensions breaks the grain and causes double counting.

Q25 Answer: C P3 · Query and analyze data

`DATEFROMPARTS` builds the first day of the order's month, a clean key to group by. `EOMONTH(..., -1)` is the last day of the previous month. `DATEFROMPARTS(YEAR, 1, DAY)` builds a January date. The `DATEADD` expression subtracts the month number in days, which doesn't land on the first of the month.

Q26 Answer: B P3 · Query and analyze data

`NTILE(4)` spreads the ordered rows into four groups that are as equal as possible. Dividing a rank by 4 doesn't make equal groups. Ordering `PERCENT_RANK` by key ignores revenue.

Q27 Answer: C P3 · Query and analyze data

`lookup` is optimised for enriching a large fact table from a small dimension table, and `kind=leftouter` keeps unmatched rows. An inner join drops them. `union` stacks tables.

Q28 Answer: A P3 · Query and analyze data

parse_json turns the text into a dynamic value, the property is read with dot notation, and toreal converts it to a number. A plain string column can't be navigated with dot notation, and toint can't convert JSON text. The has filter only finds rows containing the word.

Q29 Answer: B P3 · Query and analyze data

ROW builds a one-row table from name and expression pairs, and EVALUATE needs a table. EVALUATE can't take a comma-separated list of scalars, the table-constructor expression isn't valid, and SELECT isn't DAX.

Q30 Answer: C P3 · Query and analyze data

APPROX_COUNT_DISTINCT uses a probabilistic algorithm that's much faster and uses far less memory at large scale, with a small error. COUNT counts rows, not distinct values. SUM adds up the IDs, which means nothing.

Q31 Answer: D S1 · Design and build semantic models

Direct Lake needs the data in OneLake as Delta and can't use a gateway. DirectQuery restricts Power Query and some DAX, and it puts load on the source. Dual only matters in composite models. Import with a daily refresh through the gateway fits the requirement.

Q32 Answer: A S1 · Design and build semantic models

FILTER over the distinct customers evaluates [Sales] per customer, using context transition, and keeps those over 1,000. Filtering the Sales table checks individual transactions, not customer totals. Comparing DISTINCTCOUNT with 1,000 returns TRUE or FALSE, not a count.

Q33 Answer: B S1 · Design and build semantic models

PARTITIONBY splits the relation into groups, here categories, and the ranking restarts in each one. ORDERBY decides the ranking order. DENSE controls how ties are handled. ALLSELECTED defines which rows are considered.

Q34 Answer: A S1 · Design and build semantic models

Calculation groups only apply to explicit measures, so the model sets Discourage implicit measures. Authors have to create proper measures, which is good practice anyway. The columns aren't deleted, hidden or secured, and the storage mode doesn't change.

Q35 Answer: D S1 · Design and build semantic models

Two tables in a one-to-one relationship describe the same entity, so merging them makes the model simpler and faster. One-to-one relationships filter in both directions and add complexity for no benefit.

Q36 Answer: A S2 · Optimize enterprise-scale semantic models

The DAX query view can write DEFINE MEASURE changes back to the model with Update model, so you can go straight from testing to applying. The other options don't update the measure.

Q37 Answer: A, D S2 · Optimize enterprise-scale semantic models

VertiPaq size depends on column count and cardinality, so removing unused columns and cutting cardinality give the biggest wins. Calculated columns, auto date tables and text keys all make the model bigger.

Q38 Answer: A S2 · Optimize enterprise-scale semantic models

Direct Lake on OneLake can read Delta tables from several Fabric items. Direct Lake on the SQL analytics endpoint is tied to one lakehouse or warehouse endpoint.

Q39 **Answer: No · Yes · Yes** S2 · Optimize enterprise-scale semantic models

Framing replaces data refresh in Direct Lake, so incremental refresh policies are for Import tables. Automatic updates keep the model in step with the Delta tables. You can turn them off and trigger framing from your ETL instead.

Q40 **Answer: C** S2 · Optimize enterprise-scale semantic models

V-Order trades write performance for read performance. That's right for read-heavy gold tables and wrong for write-heavy staging tables that aren't read analytically. Adding Z-Order or minute-by-minute OPTIMIZE adds even more write cost for no benefit.

Q41 **Answer: C** S2 · Optimize enterprise-scale semantic models

Query reduction lets users make several slicer and filter changes, then apply them once, so far fewer queries are sent. Turning off cross-highlighting changes interactions between visuals, not slicer behaviour. Syncing slicers adds work. Storage format doesn't change how many queries interaction causes.

Q42 **Answer: A** M1 · Implement security and governance

OneLake security roles can hide columns and are enforced across engines, so both T-SQL and Spark are covered. Masking is a SQL feature and doesn't cover Spark file access. OLS only applies inside a semantic model. Labels don't restrict access.

Q43 **Answer: A** P1 · Get data

Google Cloud Storage is a supported external shortcut target, so the data is read in place. Copying breaks the no-copy rule. Mirroring is for databases.

Q44 **Answer: D** P2 · Transform data

A bridge table is the standard way to model a multi-valued attribute. A delimited list can't be filtered or related. Repeating product rows breaks the one-row-per-product rule on DimProduct and the uniqueness Direct Lake needs on the one side of a relationship.

Q45 **Answer: B** S1 · Design and build semantic models

With a many-to-many path through a bridge, the engine calculates each promotion's sales correctly. A sale in two promotions shows under both, but the grand total counts it once. That non-additive behaviour is correct. Duplicating fact rows, or adding up the promotion rows, double counts.

Advanced — performance and the hard edges

Guardrails, fallback, file layout, DAX tuning and the security details that catch strong candidates.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 4 · Advanced

12

Maintain a data analytics solution
25-30%

21

Prepare data
45-50%

12

Implement and manage semantic
models
25-30%

Q1 MULTIPLE RESPONSE — CHOOSE THREE

A user has the workspace Viewer role. Which three mechanisms can limit the rows they see?

- ☐ A OneLake security role with a row filter on the lakehouse
- ☐ A sensitivity label on the warehouse
- ☐ A security policy (RLS) defined at the warehouse's SQL analytics endpoint
- ☐ Certified endorsement on the semantic model
- ☐ An RLS role in the semantic model

Q2 HOT AREA — YES / NO FOR EACH STATEMENT

A semantic model has RLS roles. For each user, select Yes if RLS filters their results when they use Analyze in Excel.

Statement	Yes	No
1. A user with Build permission on the model and no workspace role	☐	☐
2. A user with the workspace Member role	☐	☐
3. A user who consumes the model through a workspace app and is a member of a role	☐	☐

Q3 SINGLE ANSWER

An external partner company's analysts must view one report in your tenant. What is required?

- ☐ Publish the report to the web with a public link
- ☐ Invite them as Entra B2B guests, with external sharing allowed, then share
- ☐ Email them the .pbix file and a .pbids file
- ☐ Create Fabric-local accounts for them in the admin portal

Q4 SINGLE ANSWER

Notebooks and pipelines must reach an ADLS Gen2 account whose firewall blocks public access. You want to avoid a gateway. Which Fabric capability lets specific workspaces through the storage firewall?

- ☐ A personal gateway on a VM inside the storage virtual network
- ☐ Trusted workspace access, with a workspace identity and resource instance rule
- ☐ A sensitivity label on the account that permits Fabric access
- ☐ A firewall rule that allows all Azure service IPs

Q5 SINGLE ANSWER

Complete the warehouse RLS predicate function so each sales rep sees only their own rows.

```
CREATE FUNCTION sec.fn_rep_filter (@SalesRep AS varchar(128))
RETURNS TABLE WITH SCHEMABINDING
AS RETURN
    SELECT 1 AS result
    WHERE @SalesRep = _____
        OR _____ = 'sales.manager@contoso.com';
```

- ☐ (A) SESSION_USER_TABLE()
- ☐ (B) USER_NAME()
- ☐ (C) GETDATE()
- ☐ (D) CURRENT_TIMESTAMP

Q6 MULTIPLE RESPONSE — CHOOSE TWO

Which two statements about using Git integration and deployment pipelines together are true?

- ☐ (A) Git integration copies table data between workspaces
- ☐ (B) Deployment pipelines keep the commit history of every item
- ☐ (C) Deployment pipelines only work if the workspaces are connected to Git
- ☐ (D) A common pattern connects the Dev workspace to Git and promotes Dev to Test to Prod with a deployment pipeline
- ☐ (E) You can also build a Git-only process by connecting each stage's workspace to its own branch and merging between branches

Q7 SINGLE ANSWER

You add one new measure to a semantic model in Test and deploy it to Production with a deployment pipeline. What happens to the data in the Production model?

- ☐ (A) The Production model switches to DirectQuery until refreshed
- ☐ (B) Only metadata is deployed; Production keeps its data and needs no refresh
- ☐ (C) The Test model's data is copied into Production
- ☐ (D) Production data is cleared and must be fully reloaded

Q8 SINGLE ANSWER

In Power BI Desktop you need to move 40 measures into new display folders and rename them in one bulk scripted edit, with a preview of the code before you apply it. Which feature should you use?

- ☐ (A) TMDL view
- ☐ (B) Power Query editor
- ☐ (C) Model view drag and drop, one measure at a time
- ☐ (D) Performance Analyzer

Q9 SINGLE ANSWER

Your certified shared model isn't visible to users who don't yet have access, so they keep building their own. What helps them find it and request access?

- ☐ (A) Make the endorsed model discoverable so users can request access
- ☐ (B) Give every user Build permission on the model
- ☐ (C) Change the endorsement from Certified to Promoted
- ☐ (D) Email a .pbix copy of the model to every user

Q10 SINGLE ANSWER

In which format is a semantic model's definition stored in a Git repository by Fabric Git integration and in .pbip projects, so it can be diffed folder by folder?

- ☐ A TMDL text files
- ☐ B A binary .abf backup
- ☐ C A .pbix file
- ☐ D PBIR JSON files

Q11 DRAG AND DROP — MATCH

Match each requirement to the best way to get the data into or exposed in Fabric.

Choices — each may be used once, more than once, or not at all

- ☐ A Dataflow Gen2
- ☐ B Data pipeline
- ☐ C Mirroring
- ☐ D Notebook
- ☐ E Eventstream
- ☐ F Shortcut

Requirement	Choice
1. A Cosmos DB database must be replicated to OneLake continuously, with no ETL	
2. Existing Delta tables in ADLS Gen2 must be queryable in place	
3. Copy 500 tables from Oracle nightly, then run a validation notebook	
4. Analysts maintain a low-code clean-up of a SharePoint list into a warehouse	
5. IoT events must be filtered and routed in real time into an Eventhouse	

Q12 SINGLE ANSWER

Analysts in an Eventhouse need to join streaming events with a product dimension that lives as a Delta table in a lakehouse, without copying the dimension. What should you do?

- ☐ A Create a shortcut in the KQL database and query it with `external_table()`
- ☐ B Export the dimension to CSV and ingest it hourly
- ☐ C Deploy the dimension to the Eventhouse with a pipeline
- ☐ D Turn on OneLake availability on the lakehouse table

Q13 SINGLE ANSWER

You configure mirroring for an Azure SQL Database, and setup fails at the prerequisites check. Which configuration on the source is a documented prerequisite?

- ☐ A A personal gateway installed on the database server
- ☐ B Change data capture turned off on every table
- ☐ C The system-assigned managed identity on the logical server
- ☐ D Public network access turned off on the server

Q14 SINGLE ANSWER

Data scientists use notebooks, and SQL analysts need read-only T-SQL access to the same curated tables. Nobody needs T-SQL writes. Which single item serves both groups?

- ☐ A An Eventhouse with OneLake availability turned on
- ☐ B A warehouse, plus a lakehouse copy for the notebooks
- ☐ C Two warehouses, one per team, kept in sync
- ☐ D A lakehouse: notebooks for engineers, its SQL endpoint for analysts

Q15 SINGLE ANSWER

A pipeline runs a notebook and must pass it the load date. How should the notebook receive the value?

- ☐ A Read the value from a Spark environment variable set by the pipeline
- ☐ B Mark a parameter cell and set a matching base parameter in the activity
- ☐ C Set it with a parameter deployment rule on the notebook
- ☐ D Write it to a control table the notebook reads

Q16 SINGLE ANSWER

Five hundred tables must be copied as they are from a database to a lakehouse, with no transformation, as fast as possible. Which tool fits better than a Dataflow Gen2?

- ☐ A A Copy activity in a metadata-driven ForEach
- ☐ B A Dataflow Gen2 with 500 queries
- ☐ C The Visual Query Editor, saved as views
- ☐ D A KQL update policy per table

Q17 SINGLE ANSWER

A curated table must be visible in three lakehouses across three workspaces, and changes must appear everywhere immediately, with only one physical copy. What should you do?

- ☐ A A zero-copy clone in each of the other two
- ☐ B Copy the table to the other two lakehouses nightly
- ☐ C One physical table, with internal shortcuts in the other two
- ☐ D Mirror the table into the other two workspaces

Q18 SINGLE ANSWER

In this warehouse MERGE, which clause inserts staging rows that have no match in the dimension?

```
MERGE dim.Product AS t
USING stg.Product AS s ON t.ProductCode = s.ProductCode
WHEN MATCHED AND t.ListPrice <> s.ListPrice
    THEN UPDATE SET t.ListPrice = s.ListPrice
    _____
    (ProductCode, ProductName, ListPrice)
    VALUES (s.ProductCode, s.ProductName, s.ListPrice);
```

- ☐ A WHEN MATCHED THEN DELETE
- ☐ B WHEN NOT MATCHED BY SOURCE THEN DELETE
- ☐ C WHEN NOT MATCHED BY TARGET THEN INSERT
- ☐ D OUTPUT \$action

Q19 SINGLE ANSWER

In a Delta MERGE from a notebook, matched rows should be updated only when their attribute hash has changed. Which builder call does this?

- ☐ A .whenMatchedDelete(condition="t.RowHash = s.RowHash")
 - ☐ B .whenNotMatchedInsert(condition="t.RowHash <> s.RowHash")
 - ☐ C .whenMatchedUpdate(condition="t.RowHash <> s.RowHash", set={...})
 - ☐ D .whenMatchedUpdateAll(condition="t.RowHash = s.RowHash")
-

Q20 SINGLE ANSWER

A fact row arrives for ProductCode P-991 before the product is loaded into DimProduct. The product details will arrive tomorrow. What is the recommended handling?

- ☐ A Hold the fact row in staging until the product arrives
 - ☐ B Delete the fact row and reload it from source later
 - ☐ C Insert an inferred member for P-991 now, and update it when details arrive
 - ☐ D Load the fact now with a NULL product key, then fix it
-

Q21 SINGLE ANSWER

In a lakehouse notebook, which Spark SQL statement creates and fills a gold Delta table from a query in one step?

- ☐ A CREATE TABLE gold_sales_by_day USING DELTA AS SELECT ...
 - ☐ B CREATE VIEW gold_sales_by_day USING DELTA AS SELECT ...
 - ☐ C SELECT ... INTO gold_sales_by_day FROM silver_sales
 - ☐ D INSERT INTO gold_sales_by_day SELECT ...
-

Q22 SINGLE ANSWER

A gold table that a Direct Lake model reads has a column of binary type that holds a hashed identifier. What should you do?

- ☐ A Turn on V-Order for the table and re-frame
 - ☐ B Convert it to a supported type, such as hex text, or drop it
 - ☐ C Turn on DirectQuery fallback for the model
 - ☐ D Keep it; Direct Lake reads binary columns as text
-

Q23 SINGLE ANSWER

Rows in FactOrders with a NULL CustomerKey disappear when FactOrders is inner-joined to DimCustomer. Why?

- ☐ A Inner joins sort NULL keys last and truncate them
- ☐ B NULL never equals anything in a join, not even another NULL
- ☐ C DimCustomer has no Unknown member row with a NULL key
- ☐ D The hash join discards rows with NULL hashes

Q24 SINGLE ANSWER

A Dataflow Gen2 query folds until you add one step, after which everything runs in the mashup engine. Which step commonly breaks folding against a SQL source?

- ☐ A Filtering rows on a date column
- ☐ B Renaming columns
- ☐ C Removing columns
- ☐ D Adding an index column

Q25 SINGLE ANSWER

Reports mostly query sales by day, product and store, but the fact is at receipt-line grain with 5 billion rows. Which gold-layer change best supports fast Direct Lake reporting while keeping the detail?

- ☐ A Replace the line-level fact with the daily aggregate
- ☐ B Add day, product and store totals as columns on the line fact
- ☐ C Partition the line fact by ReceiptID
- ☐ D Add a day/product/store aggregate table and keep the line fact for drill-down

Q26 HOT AREA — YES / NO FOR EACH STATEMENT

A running total uses SUM(Amount) OVER (ORDER BY OrderDate ...). Several rows share the same OrderDate. For each statement, select Yes if it is true.

Statement	Yes	No
1. With RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW, all rows with the same date get the same running total.	☐	☐
2. With ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW, each row's total grows row by row, even within the same date.	☐	☐
3. ROWS and RANGE always give identical results.	☐	☐

Q27 SINGLE ANSWER

Which T-SQL filter selects the last complete calendar month, whatever today's date is?

- ☐ A WHERE MONTH(OrderDate) = MONTH(GETDATE()) - 1 AND YEAR(OrderDate) = YEAR(GETDATE())
- ☐ B WHERE OrderDate BETWEEN EOMONTH(GETDATE(), -1) AND EOMONTH(GETDATE())
- ☐ C WHERE OrderDate > DATEADD(day, -30, CAST(GETDATE() AS date))
- ☐ D WHERE OrderDate > EOMONTH(GETDATE(), -2) AND OrderDate <= EOMONTH(GETDATE(), -1)

Q28 SINGLE ANSWER

In KQL, how do you calculate each session's duration?

- ☐ A Events | project SessionId, Duration = Timestamp - prev(Timestamp)
- ☐ B Events | extend Duration = now() - Timestamp | distinct SessionId
- ☐ C Events | summarize Duration = max(Timestamp) - min(Timestamp) by SessionId
- ☐ D Events | summarize Duration = count() by SessionId

Q29 SINGLE ANSWER

Which DAX query returns the product keys in the Product table that have no rows in Sales?

- ☐ (A) `EVALUATE INTERSECT(VALUES('Product'[ProductKey]), VALUES(Sales[ProductKey]))`
- ☐ (B) `EVALUATE EXCEPT(VALUES('Product'[ProductKey]), VALUES(Sales[ProductKey]))`
- ☐ (C) `EVALUATE VALUES(Sales[ProductKey])`
- ☐ (D) `EVALUATE UNION(VALUES('Product'[ProductKey]), VALUES(Sales[ProductKey]))`

Q30 SINGLE ANSWER

Which T-SQL returns the median order amount per category on every row?

- ☐ (A) `NTILE(2) OVER (PARTITION BY Category ORDER BY Amount)`
- ☐ (B) `AVG(Amount) OVER (PARTITION BY Category ORDER BY Amount)`
- ☐ (C) `PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY Amount) OVER (PARTITION BY Category)`
- ☐ (D) `(MAX(Amount) + MIN(Amount)) / 2 OVER (PARTITION BY Category)`

Q31 SINGLE ANSWER

Which measure returns each product's sales as a share of its own category's total, respecting all other filters?

- ☐ (A) `DIVIDE([Sales], CALCULATE([Sales], ALL(Sales)))`
- ☐ (B) `DIVIDE([Sales], CALCULATE([Sales], ALLSELECTED('Product'[Name])))`
- ☐ (C) `DIVIDE([Sales], CALCULATE([Sales], ALL('Product')))`
- ☐ (D) `DIVIDE([Sales], CALCULATE([Sales], REMOVEFILTERS('Product'), VALUES('Product'[Category])))`

Q32 SINGLE ANSWER

A Time Intelligence calculation group must not change the [Margin %] measure, which is a ratio that doesn't make sense as YTD. How should the YTD item handle this?

- ☐ (A) Return `SELECTEDMEASURE()` unchanged when `SELECTEDMEASURENAME()` is "Margin %"
- ☐ (B) Move the ratio measures into a second semantic model
- ☐ (C) Replace the calculation group with a field parameter
- ☐ (D) Give the Margin % measure a lower precedence than the group

Q33 SINGLE ANSWER

In a Direct Lake on OneLake composite model, an Import table of targets relates to a Direct Lake Date table. What kind of relationship is this?

- ☐ (A) An invalid relationship that can't be created
- ☐ (B) A one-to-one relationship by definition
- ☐ (C) A regular relationship, the same as two Import tables
- ☐ (D) A limited relationship, because the tables are in different source groups

Q34 SINGLE ANSWER

Sales holds Qty, and Product holds UnitCost. There is a many-to-one relationship from Sales to Product. Which measure returns the total cost correctly?

- ☐ (A) `SUMX(Sales, Sales[Qty] * RELATED('Product'[UnitCost]))`
- ☐ (B) `SUM(Sales[Qty]) * SUM('Product'[UnitCost])`
- ☐ (C) `SUMX('Product', 'Product'[UnitCost]) * COUNTROWS(Sales)`
- ☐ (D) `RELATED('Product'[UnitCost]) * SUM(Sales[Qty])`

Q35 SINGLE ANSWER

DimEmployee contains EmployeeKey and ManagerKey (a parent-child hierarchy). Users need an Organisation hierarchy with Level 1, Level 2 and Level 3 managers in the model. What is the standard approach?

- ☐ A Flatten it into level columns with PATH and PATHITEM, then build a hierarchy
- ☐ B A field parameter over EmployeeKey and ManagerKey
- ☐ C A calculation group with one item per level
- ☐ D A self-referencing relationship from ManagerKey to EmployeeKey

Q36 SINGLE ANSWER

A Direct Lake on OneLake model runs on an F32 capacity. Its fact table has 900 million rows, and refreshes fail. What is the cause, and what is the fix?

- ☐ A It needs an incremental refresh policy to frame in chunks
- ☐ B V-Order is on; turn it off so framing has less metadata to read
- ☐ C It's over the F32 guardrail of 300 million rows; reduce rows in gold or use F64 or larger
- ☐ D The model's memory is full; turn on large semantic model storage format

Q37 SINGLE ANSWER

How can you tell whether a particular visual on a Direct Lake on SQL report is falling back to DirectQuery?

- ☐ A In lineage view, the model shows a DirectQuery icon
- ☐ B In the refresh history, each visual's fallback is listed
- ☐ C In Performance Analyzer, a Direct query duration shows on that visual
- ☐ D In the Capacity Metrics app, the visual shows as DirectQuery

Q38 SINGLE ANSWER

A query uses SUMMARIZE(Sales, 'Product'[Category], "Total", SUM(Sales[Amount])) and is slow, with confusing results under some filters. What is the recommended rewrite?

- ☐ A Store Total as a calculated column on Sales
- ☐ B Wrap the SUMMARIZE in IFERROR and CALCULATETABLE
- ☐ C Add the Total with SUMX inside the SUMMARIZE
- ☐ D Group with SUMMARIZE, add Total with ADDCOLUMNS and CALCULATE

Q39 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about Direct Lake memory behaviour, select Yes if it is true.

Statement	Yes	No
1. Direct Lake loads only the columns that queries need into memory.	☐	☐
2. If memory runs short, Direct Lake can evict less-used columns and reload them later.	☐	☐
3. Direct Lake loads the whole model into memory at framing time.	☐	☐

Q40 SINGLE ANSWER

A team wants to run OPTIMIZE (with V-Order) and VACUUM on lakehouse tables without writing any code. What can they use?

- ☐ A The Visual Query Editor on the SQL endpoint
- ☐ B A deployment pipeline's post-deploy step
- ☐ C The semantic model's refresh settings
- ☐ D The table's Maintenance option in the lakehouse explorer

Q41 SINGLE ANSWER

A composite model has a DirectQuery fact and an Import aggregation table at Date × Category grain. A visual showing Sales by Date and Product Color still sends queries to the source. Why?

- ☐ A The aggregation doesn't cover Color, so the query goes to the detail fact
- ☐ B The aggregation table is too large to stay in memory, so it's skipped
- ☐ C Aggregations only apply to measures written with SUMX
- ☐ D The Date table is in Dual mode instead of Import

Case study · Proseware Telecom

Questions 42–45

Existing environment

Proseware Telecom stores call detail records (CDRs) in an Eventhouse and billing data in a warehouse. A certified semantic model serves finance. Monthly finance reports are often exported to Excel and emailed to auditors at external firms.

Family plans are shared by up to five subscribers, and each holds an ownership percentage. The model is developed as TMDL in Azure DevOps and released through automated pipelines.

Requirements

1. R1. Exported finance files must stay unreadable to anyone outside Proseware, even if they're forwarded.
2. R2. Model releases must be scripted, quality-checked and approved before Production.
3. R3. Network operations needs dropped calls per cell tower per hour over the last 24 hours, with zero shown for hours that had no drops.
4. R4. Plan revenue must be split between subscribers by ownership percentage, without double counting.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Q42 SINGLE ANSWER

Which control over exported files satisfies R1?

- ☐ A Certification of the finance semantic model
- ☐ B Assigning the workspace to a protected Finance domain
- ☐ C RLS roles that exclude every external user's UPN
- ☐ D A sensitivity label that encrypts content for internal users only

Q43 BUILD LIST — SEQUENCE

Put the steps of the release process for R2 in order.

Select the 4 actions required and arrange them in the correct order. Not every action is used.

- A** Deploy to the Test workspace through the XMLA endpoint with a scripted tool
- B** Run automated checks in the build pipeline, such as Best Practice Analyzer rules
- C** Edit the Production model directly in the service
- D** Publish a .pbix from a developer's laptop
- E** Get approval, then deploy the same build to Production
- F** Commit the TMDL changes and open a pull request

Q44 SINGLE ANSWER

Which KQL query satisfies R3, including the zeros?

- A** CDR | where Timestamp > ago(24h) and CallResult == "Dropped" | summarize Drops = count() by TowerId, bin(Timestamp, 1h)
- B** CDR | where Timestamp > ago(24h) and CallResult == "Dropped" | make-series Drops = count() default = 0 on Timestamp from ago(24h) to now() step 1h by TowerId
- C** CDR | where CallResult == "Dropped" | make-series Drops = count() on Timestamp step 1h | take 24
- D** CDR | where Timestamp > ago(24h) | summarize Drops = countif(CallResult == "Dropped") by bin(Timestamp, 1h)

Q45 SINGLE ANSWER

Which revenue-allocation design satisfies R4?

- A** A PlanSubscriber bridge with OwnershipPct, used as a weight in a SUMX
- B** Copy the full plan revenue to every subscriber
- C** Divide each plan's revenue by five, the most subscribers allowed
- D** A bidirectional PlanSubscriber bridge with no weight column

End of Set 08 — answers follow on the next page.

Set 08 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	A, C, E	M1		16	A	P1		31	D	S1	
2	Yes · No · Yes	M1		17	C	P1		32	A	S1	
3	B	M1		18	C	P2		33	D	S1	
4	B	M1		19	C	P2		34	A	S1	
5	B	M1		20	C	P2		35	A	S1	
6	D, E	M2		21	A	P2		36	C	S2	
7	B	M2		22	B	P2		37	C	S2	
8	A	M2		23	B	P2		38	D	S2	
9	A	M2		24	D	P2		39	Yes · Yes · No	S2	
10	A	M2		25	D	P2		40	D	S2	
11	1→C 2→F 3→B 4→A 5→E	P1		26	Yes · Yes · No	P3		41	A	S2	
12	A	P1		27	D	P3		42	D	M1	
13	C	P1		28	C	P3		43	F → B → A → E	M2	
14	D	P1		29	B	P3		44	B	P3	
15	B	P1		30	C	P3		45	A	S1	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: A, C, E M1 · Implement security and governance

Rows can be filtered at three layers: OneLake security, SQL endpoint RLS and semantic model RLS. Which one applies depends on how the user reaches the data. Labels and endorsement never filter data.

Q2 Answer: Yes · No · Yes M1 · Implement security and governance

RLS applies to anyone without edit rights on the model, whatever client they use, Excel included. Members, Contributors and Admins can edit the model, so RLS doesn't restrict them.

Q3 Answer: B M1 · Implement security and governance

Fabric shares with external users through Entra B2B guest accounts, controlled by the tenant's external sharing settings. There are no Fabric-local accounts. Publish to web makes the report public to anyone, and emailing .pbix files removes all governance.

Q4 Answer: B M1 · Implement security and governance

Trusted workspace access lets a workspace's identity through the storage firewall with a resource instance rule. It works for shortcuts, pipelines and other access paths, and needs an F SKU capacity. Opening the storage to the public breaks the security requirement.

Q5 Answer: B M1 · Implement security and governance

In a Fabric Warehouse, `USER_NAME()` returns the calling user's name, which is their UPN for Entra ID users. The predicate compares it with the row's SalesRep value, and the second comparison lets the named manager see every row. The same function fills both blanks. The other options either aren't identity functions or don't exist.

Q6 Answer: D, E M2 · Maintain the analytics development lifecycle

Both processes are supported. You can combine Git with pipelines, or use Git alone with a branch per stage. Pipelines don't store commit history. Git doesn't move data. Pipelines don't need Git.

Q7 Answer: B M2 · Maintain the analytics development lifecycle

Pipelines deploy metadata, and the target keeps its data wherever the change allows. A new measure needs no refresh. Structural changes, such as new columns or tables, still need a refresh in the target to fill them.

Q8 Answer: A M2 · Maintain the analytics development lifecycle

TMDL view shows model objects as TMDL script. You can edit many objects at once and apply the changes. Performance Analyzer is a profiling tool, and Power Query shapes data, not model metadata.

Q9 Answer: A M2 · Maintain the analytics development lifecycle

When discoverability is enabled in the tenant settings and on the item, endorsed items appear to users who don't have access yet, and they can request it. That supports reuse instead of duplication.

Q10 Answer: A M2 · Maintain the analytics development lifecycle

Semantic models are serialised as TMDL: readable text files with one file per table or object. That makes diffs and merges practical. The older single-file model.bim (TMSL JSON) is also possible, but TMDL is the default and is easier to merge.

Q11 Answer: 1→C 2→F 3→B 4→A 5→E P1 · Get data

Mirroring is for continuous database replication, shortcuts for access in place, pipelines for orchestration and bulk copy, Dataflow Gen2 for low-code transforms owned by analysts, and Eventstream for real-time routing. A notebook could do several of these with code, but it isn't the best fit for any of them.

Q12 Answer: A P1 · Get data

KQL databases support OneLake shortcuts, which appear as external tables. `external_table('DimProduct')` can then be joined with streaming tables, with no copy. Hourly CSV exports duplicate the data and go out of date. OneLake availability is an Eventhouse setting that works in the other direction, and deployment pipelines don't move data.

Q13 Answer: C P1 · Get data

Mirroring Azure SQL Database needs the logical server's system-assigned managed identity. Fabric uses it to publish change data to OneLake. Mirroring doesn't depend on CDC being turned off, and a personal gateway plays no part.

Q14 Answer: D P1 · Get data

A lakehouse gives Spark read-write access and an automatic read-only SQL analytics endpoint. That's a perfect fit when T-SQL writes aren't needed. Copying tables into a second store adds cost and drift.

Q15 Answer: B P1 · Get data

Notebook activities pass base parameters into the cell marked as the parameter cell, overriding its default values. Hard-coding defeats parameterisation, and deployment rules apply at deployment time, not at run time.

Q16 Answer: A P1 · Get data

Bulk movement without transformation is what the Copy activity is built for, and it scales with parallel ForEach iterations. Dataflows suit Power Query transformations owned by analysts.

Q17 Answer: C P1 · Get data

Internal shortcuts give one physical copy and many logical views, with immediate consistency. Nightly copies create drift. Mirroring replicates external databases.

Q18 Answer: C P2 · Transform data

NOT MATCHED BY TARGET means the source row has no counterpart in the target, so it's a new member to insert. NOT MATCHED BY SOURCE handles target rows missing from the source. OUTPUT returns the action taken; it doesn't insert anything.

Q19 Answer: C P2 · Transform data

The condition argument limits updates to rows that really changed, which avoids rewriting unchanged rows. That saves time and keeps the transaction log lean. UpdateAll with an equality condition rewrites only the rows that didn't change. A not-matched clause can't compare with the target, which has no matching row. Deleting matched rows destroys data.

Q20 Answer: C P2 · Transform data

An inferred (late-arriving) member gives the fact a valid surrogate key straight away, so totals stay complete. Tomorrow's load then fills in the real attributes. NULL keys and deletes lose sales or break relationships.

Q21 Answer: A P2 · Transform data

CREATE TABLE ... USING DELTA AS SELECT (CTAS) creates the Delta table and loads it in one statement. A view stores no data. INSERT needs an existing table. SELECT INTO is T-SQL syntax, not Spark SQL.

Q22 Answer: B P2 · Transform data

Binary and GUID types aren't supported in Direct Lake semantic models. Convert the column to a supported type upstream, or leave it out if reports don't need it. V-Order and fallback settings don't change data type support.

Q23 Answer: B P2 · Transform data

In SQL, NULL = NULL is unknown, not true, so NULL keys never match. That's why gold facts should use an Unknown member key such as -1 instead of NULL.

Q24 Answer: D P2 · Transform data

Steps that need row-by-row ordering, such as adding an index column, often can't be translated to SQL. Filters, column removal and renames normally fold. Put steps that break folding as late as you can.

Q25 Answer: D P2 · Transform data

An aggregate table at the reporting grain greatly reduces the rows Direct Lake has to read and keeps the model within guardrails. Keeping the detail table preserves drill-down. Deleting the detail loses information.

Q26 Answer: Yes · Yes · No P3 · Query and analyze data

RANGE treats rows with equal ORDER BY values as peers and includes them all. ROWS counts physical rows. When the ordering column has duplicates, the frame type changes the result. That's a classic code-reading trap.

Q27 Answer: D P3 · Query and analyze data

EOMONTH(GETDATE(), -1) is the last day of last month, and EOMONTH(GETDATE(), -2) is the last day of the month before. The range between them is exactly last month (for a date column). A rolling 30 days isn't a calendar month. MONTH() - 1 gives 0 in January, and the year check is then wrong too. The BETWEEN range covers this month plus last month's final day.

Q28 Answer: C P3 · Query and analyze data

Subtracting the first event time from the last in each session gives a timespan per session. now() - Timestamp gives each event's age. count() gives the number of events, not a duration. prev() needs a serialised, sorted row set and gives gaps between events, not a session total.

Q29 Answer: B P3 · Query and analyze data

EXCEPT returns the rows of the first table that aren't in the second, which here means products with no sales. INTERSECT returns products that do have sales. UNION combines the two lists.

Q30 Answer: C P3 · Query and analyze data

PERCENTILE_CONT(0.5) gives the median, interpolated, and the OVER clause computes it per category. AVG is the mean, not the median, and adding ORDER BY turns it into a running average. NTILE makes groups. The midpoint of the range isn't the median.

Q31 Answer: D S1 · Design and build semantic models

REMOVEFILTERS('Product') clears product-level filters, and VALUES('Product'[Category]) puts back the current category, which gives the category total. ALL('Product') gives the total across all categories. ALL(Sales) removes every filter on the fact. ALLSELECTED on Name gives the total of the visible products, across categories.

Q32 Answer: A S1 · Design and build semantic models

Calculation items can branch on SELECTEDMEASURENAME() (or ISSELECTEDMEASURE) to skip particular measures. The ratio then stays as it is while additive measures get YTD. Precedence applies to calculation groups, not measures. A second model or a field parameter is excessive and doesn't solve it.

Q33 Answer: D S1 · Design and build semantic models

Relationships between tables in different source groups are limited relationships. They have different semantics: for example, no blank row is added for unmatched values, and queries are evaluated differently. Design and test measures with that in mind.

Q34 Answer: A S1 · Design and build semantic models

SUMX iterates the sales rows, and RELATED fetches each row's product cost through the relationship. Multiplying two totals is wrong. RELATED can't be used outside a row context.

Q35 Answer: A S1 · Design and build semantic models

Power BI hierarchies need fixed level columns. PATH and PATHITEM (or doing the same in the gold layer) flatten a parent-child structure into Level columns. Self-referencing relationships aren't supported for this.

Q36 Answer: C S2 · Optimize enterprise-scale semantic models

F2 to F32 share a 300-million-row-per-table guardrail. On Direct Lake on OneLake, going over it makes framing fail, because there's no DirectQuery fallback. Either reduce the rows in gold or use a larger SKU. Incremental refresh doesn't apply to Direct Lake.

Q37 Answer: C S2 · Optimize enterprise-scale semantic models

Performance Analyzer breaks a visual down into DAX query, Direct query, visual display and other. A Direct query entry for a Direct Lake visual shows that it fell back to SQL. Lineage, refresh history and the Capacity Metrics app don't show fallback for individual visuals.

Q38 Answer: D S2 · Optimize enterprise-scale semantic models

Adding calculated extension columns inside SUMMARIZE is discouraged because the semantics are complex and slow, whichever aggregate you use. Group with SUMMARIZE or SUMMARIZECOLUMNS, and add expressions with ADDCOLUMNS and CALCULATE. IFERROR and calculated columns don't address the cause.

Q39 Answer: Yes · Yes · No S2 · Optimize enterprise-scale semantic models

Direct Lake loads columns on demand and evicts (pages out) cold columns under memory pressure. That's why it can work with more data than fits in memory at once. Framing loads no data.

Q40 Answer: D S2 · Optimize enterprise-scale semantic models

The lakehouse explorer has table maintenance options that run OPTIMIZE, with optional V-Order, and VACUUM with a retention you choose. For automation, schedule a notebook that runs the same commands. The other tools don't maintain tables.

Q41 Answer: A S2 · Optimize enterprise-scale semantic models

An aggregation is only used when every column in the query is covered at its grain, directly or through relationships. Color is below the Category grain, so the detail fact is queried. Either add Color to the aggregation or accept the DirectQuery cost for that visual.

Q42 Answer: D M1 · Implement security and governance

Labels with encryption protect exported files wherever they go. Only authorised users can open them. Certification, RLS and domains have no effect once a file has left Fabric.

Q43 Answer: F → B → A → E M2 · Maintain the analytics development lifecycle

Changes are reviewed in a pull request, checked automatically, deployed to Test by script, and promoted to Production only after approval. Editing Production by hand or publishing a .pbix skips the controls R2 requires.

Q44 Answer: B P3 · Query and analyze data

make-series builds a continuous series over the range, with default = 0 filling empty hours. summarize ... by bin() leaves gaps where there were no events, and the countif version doesn't group by tower either. make-series without a from/to range and by TowerId doesn't give 24 hours per tower.

Q45 Answer: A S1 · Design and build semantic models

A weighted bridge allocates each plan's revenue by ownership share, so subscriber totals add up to plan totals. A fixed split by five is wrong for most plans, and copying the full amount double counts.

Advanced — mixed scenarios

Every sub-area at full difficulty, with questions that cross domains the way the real paper does.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 4 · Advanced

12

Maintain a data analytics solution
25–30%

21

Prepare data
45–50%

12

Implement and manage semantic
models
25–30%

Q1 SINGLE ANSWER

A lakehouse contains an external shortcut to an ADLS Gen2 container, created with a cloud connection that uses a service principal. When a workspace Viewer reads data through that shortcut, which identity does ADLS Gen2 check?

- ☐ A The identity of the workspace's Admin
- ☐ B None; the storage account must allow anonymous access
- ☐ C The credentials of the shortcut's cloud connection
- ☐ D The Viewer's own Entra identity, checked against ADLS ACLs

Q2 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about workspace roles, select Yes if it is true.

Statement	Yes	No
1. Only the Admin role can delete the workspace.	☐	☐
2. A Member can add other users as workspace Admins.	☐	☐
3. A Viewer can create a new notebook in the workspace.	☐	☐

Q3 SINGLE ANSWER

Workspace Viewers can query a lakehouse through its SQL analytics endpoint. They must be blocked from querying one table, dbo.Salaries, through SQL, while every other table stays available. What should you do?

- ☐ A REVOKE CONNECT from the Viewers' group
- ☐ B Move Salaries to the Files area of the lakehouse
- ☐ C A Highly Confidential label on the Salaries table
- ☐ D DENY SELECT ON dbo.Salaries to the Viewers' group

Q4 SINGLE ANSWER

Reports built on a semantic model labelled Highly Confidential should get the same label automatically. What should you turn on?

- ☐ A A mandatory labelling policy for reports
- ☐ B Certification of the semantic model
- ☐ C A deployment rule that copies the label
- ☐ D Downstream label inheritance in the tenant settings

Q5 SINGLE ANSWER

A lakehouse table is protected with RLS defined at its SQL analytics endpoint. A group of analysts was also given ReadAll on the lakehouse. What is the risk?

- ☐ A None; SQL RLS is enforced for Spark reads too
- ☐ B They can read all rows through Spark, bypassing the SQL RLS
- ☐ C ReadAll replaces ReadData, so SQL access stops
- ☐ D Their SQL queries fall back to DirectQuery

Q6 MULTIPLE RESPONSE — CHOOSE TWO

Which two conditions must hold before you can deploy changes to a semantic model through the XMLA endpoint?

- ☐ A The model uses DirectQuery
- ☐ B The workspace is connected to Git
- ☐ C The workspace is on a Fabric, Premium or Premium Per User capacity
- ☐ D The deploying user has the Viewer role
- ☐ E The XMLA endpoint is set to Read Write for that capacity

Q7 SINGLE ANSWER

Two workspaces, Finance-Dev and HR-Dev, must keep their items in the same Git repository without mixing their files. How should you connect them?

- ☐ A Connect each workspace to the repository with a different Git folder (directory)
- ☐ B Use one workspace for both teams
- ☐ C Create two repositories; one repository can't hold several workspaces
- ☐ D Connect both workspaces to the repository root

Q8 SINGLE ANSWER

You want to deploy a report and every item it depends on (its semantic model and the lakehouse behind it) without picking each one by hand. Which deployment pipeline feature helps?

- ☐ A Backward deployment
- ☐ B Select related
- ☐ C Deployment history
- ☐ D Compare

Q9 BUILD LIST — SEQUENCE

You need a reusable report template that asks for the server and database when it's opened. Put the steps in order.

Select the 4 actions required and arrange them in the correct order. Not every action is used.

- ☐ A Distribute the .pbit file to report authors
- ☐ B Save the report as a Power BI template (.pbit)
- ☐ C Add a deployment rule for the template
- ☐ D Change the source step to use the parameters instead of fixed values
- ☐ E Create Power Query parameters named Server and Database
- ☐ F Create a .pbids file with the server name

Q10 DRAG AND DROP — MATCH

Match each file type to its description.

Choices — each may be used once, more than once, or not at all

- A** .abf
- B** .pbids
- C** .pbip
- D** .pbit
- E** .pbix

Requirement	Choice
1. Report, model and data in one binary file	
2. A folder of text definitions (TMDL and PBIR) for source control	
3. Report and model definitions with no data, opened with parameter prompts	
4. A JSON file with data-source connection details only	

Q11 SINGLE ANSWER

A large Import semantic model must be moved, with its data, to a workspace on a capacity in another region, without a full re-refresh from source. What can you use?

- A** Export each table to CSV and re-import it
- B** A deployment pipeline between the two workspaces
- C** Git integration, syncing both workspaces to one branch
- D** XMLA backup to .abf and restore, with ADLS Gen2 storage attached

Q12 SINGLE ANSWER

Data scientists will train models in Spark on two years of semi-structured JSON telemetry kept at rest. Nobody needs sub-second interactive queries. Where should the history be stored?

- A** A lakehouse, as files or Delta tables
- B** A warehouse with JSON columns
- C** An Import semantic model
- D** An Eventhouse with a long retention policy

Q13 SINGLE ANSWER

A data owner wants one view of their own items' sensitivity label coverage, endorsement and governance recommendations. Which surface provides this?

- A** The Govern tab in the OneLake catalog
- B** The DAX query view
- C** The Real-Time hub
- D** Deployment pipeline compare

Q14 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about Eventstreams, select Yes if it is true.

Statement	Yes	No
1. An Eventstream can apply no-code transformations, such as filters and aggregations, to events in flight.	☐	☐
2. One Eventstream can send events to more than one destination, such as an Eventhouse and a lakehouse.	☐	☐
3. An Eventstream needs custom C# code to read from Azure Event Hubs.	☐	☐

Q15 SINGLE ANSWER

A production Spark batch job lives in .py files with command-line arguments and a main entry point, and the team doesn't want it in an interactive notebook. Which Fabric item should run it?

- ☐ A Spark job definition
- ☐ The Visual Query Editor
- ☐ A KQL queryset
- ☐ A Dataflow Gen2

Q16 SINGLE ANSWER

In a pipeline, a notebook must run only if the Copy activity succeeds, and a Teams message must be sent only if the Copy fails. How do you configure this?

- ☐ Notebook and Teams both on the Copy's On completion path
- ☐ All three activities in parallel, with no dependencies
- ☐ Notebook on the Copy's On success path; Teams on its On failure path
- ☐ Teams on the Notebook's On failure path

Q17 SINGLE ANSWER

An operational SQL database in Fabric runs an application. Analysts want to query its data for reporting without slowing the application. What should they use?

- ☐ Its replicated copy in OneLake, via the SQL analytics endpoint
- ☐ A nightly CSV export into a lakehouse
- ☐ The application's own database connection string
- ☐ A KQL queryset connected directly to the database

Q18 SINGLE ANSWER

A warehouse load updates FactSales and inserts audit rows into etl.LoadLog. Both changes must succeed or fail together. What should you do?

- ☐ Wrap both in BEGIN TRANSACTION ... COMMIT, rolling back on error
- ☐ Run the audit insert first, then the update
- ☐ Run them in two pipeline activities linked On success
- ☐ Run both on a lakehouse SQL analytics endpoint

Q19 SINGLE ANSWER

In PySpark, you need a column Tier that is "Gold" when Spend > 10000, "Silver" when Spend > 1000, and "Bronze" otherwise. Which expression is correct?

- ☐ A `F.expr("IF(Spend > 1000, 'Silver', IF(Spend > 10000, 'Gold', 'Bronze'))")`
- ☐ B `F.when(F.col("Spend") > 1000, "Silver").when(F.col("Spend") > 10000, "Gold").otherwise("Bronze")`
- ☐ C `F.when(F.col("Spend") > 10000, "Gold").when(F.col("Spend") > 1000, "Silver").otherwise("Bronze")`
- ☐ D `F.when(F.col("Spend") > 10000, "Gold").otherwise("Silver").otherwise("Bronze")`

Q20 SINGLE ANSWER

A string column holds dates such as "14-03-2026" (dd-MM-yyyy). Which PySpark expression converts it to a date?

- ☐ A `F.to_date("OrderDateText", "yyyy-MM-dd")`
- ☐ B `F.date_format("OrderDateText", "dd-MM-yyyy")`
- ☐ C `F.to_date("OrderDateText", "dd-MM-yyyy")`
- ☐ D `F.col("OrderDateText").cast("int")`

Q21 DRAG AND DROP — MATCH

Match each slowly changing dimension type to its behaviour.

Choices — each may be used once, more than once, or not at all

- ☐ A Type 7
- ☐ B Type 3
- ☐ C Type 1
- ☐ D Type 0
- ☐ E Type 2

Requirement	Choice
1. The attribute never changes after the row is first loaded	
2. The old value is overwritten and no history is kept	
3. A new row is added with its own surrogate key and validity dates	
4. A PreviousValue column keeps one prior value alongside the current one	

Q22 SINGLE ANSWER

In a notebook you need a 7-day rolling sum of Amount per store, ordered by date, with one row per store per day. Which window specification is correct?

- ☐ A `Window.orderBy("StoreID").rowsBetween(0, 7)`
- ☐ B `Window.partitionBy("StoreID").orderBy("Date").rowsBetween(-6, 0)`
- ☐ C `Window.partitionBy("Date").orderBy("StoreID")`
- ☐ D `Window.partitionBy("StoreID").rowsBetween(-7, 7)`

Q23 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about views, functions and stored procedures in Fabric, select Yes if it is true.

Statement	Yes	No
1. You can create stored procedures that write data in a Fabric Warehouse.	☐	☐
2. You can create views on a lakehouse's SQL analytics endpoint.	☐	☐
3. A stored procedure on a lakehouse's SQL analytics endpoint can INSERT rows into the lakehouse's Delta tables.	☐	☐

Q24 SINGLE ANSWER

Phone numbers arrive in mixed formats, such as "(020) 555-0101" and "020.555.0101". The silver table must store digits only. Which PySpark expression does this?

- (A) `F.regex_replace("Phone", "[^0-9]", "")`
- (B) `F.translate("Phone", "()", "")`
- (C) `F.regex_replace("Phone", "[0-9]", "")`
- (D) `F.regex_extract("Phone", "[0-9]+", 0)`

Q25 SINGLE ANSWER

A query must return the top five products by revenue, plus any product tied with the fifth. Which clause should you use?

- (A) `SELECT DISTINCT TOP (5) ...`
- (B) `SELECT TOP (5) PERCENT ...`
- (C) `SELECT TOP (5) WITH TIES ... ORDER BY Revenue DESC`
- (D) `SELECT TOP (5) ... ORDER BY Revenue DESC`

Q26 SINGLE ANSWER

Which KQL expression returns the percentage of failed requests per hour?

- (A) `where Status == "Failed" | summarize FailedPct = 100.0 * count() / count() by bin(Timestamp, 1h)`
- (B) `summarize FailedPct = 100.0 * dcount(Status) / count() by bin(Timestamp, 1h)`
- (C) `summarize FailedPct = 100.0 * countif(Status == "Failed") / count() by bin(Timestamp, 1h)`
- (D) `summarize FailedPct = countif(Status == "Failed") by bin(Timestamp, 1h)`

Q27 SINGLE ANSWER

Which KQL query finds the busiest hour of the day (0-23) across the last 30 days?

- (A) `Events | where Timestamp > ago(30d) | extend Hour = hourofday(Timestamp) | top 1 by Hour desc`
- (B) `Events | where Timestamp > ago(30d) | summarize Events = count() by bin(Timestamp, 1h) | top 1 by Events desc`
- (C) `Events | where Timestamp > ago(30d) | summarize Events = count() by Hour = hourofday(Timestamp) | top 1 by Events desc`
- (D) `Events | where Timestamp > ago(30d) | summarize Events = dcount(hourofday(Timestamp)) | take 1`

Q28 SINGLE ANSWER

Which DAX query returns the rows of the Customer table for customers who bought something in 2026?

- (A) `EVALUATE Customer WHERE 'Date'[Year] = 2026`
- (B) `EVALUATE CALCULATETABLE(VALUES(Customer[CustomerKey]), 'Date'[Year] = 2026, Sales)`
- (C) `EVALUATE CALCULATE(COUNTROWS(Customer), 'Date'[Year] = 2026)`
- (D) `EVALUATE FILTER(Customer, 'Date'[Year] = 2026)`

Q29 SINGLE ANSWER

You must find every column named like '%Email%' across all tables in a warehouse, for a privacy review. Which query should you run?

- (A) `SELECT TABLE_SCHEMA, TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_NAME LIKE '%Email%';`
- (B) `SELECT s.name, t.name FROM sys.tables t JOIN sys.schemas s ON s.schema_id = t.schema_id WHERE t.name LIKE '%Email%';`
- (C) `SELECT * FROM sys.dm_exec_requests WHERE command LIKE '%Email%';`
- (D) `SELECT TABLE_SCHEMA, TABLE_NAME, COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE COLUMN_NAME LIKE '%Email%';`

Q30 SINGLE ANSWER

A warehouse in Workspace A must query a table in a lakehouse in Workspace B. A three-part name fails because cross-database queries only reach items in the same workspace. What is the supported approach?

- (A) Shortcut the table into a Workspace A lakehouse, then use a three-part name
- (B) Add Workspace B's endpoint as a linked server
- (C) Use OPENROWSET with Workspace B's SQL endpoint address
- (D) Use a four-part name that starts with Workspace B's name

Q31 MULTIPLE RESPONSE — CHOOSE TWO

Which two are good reasons to choose Direct Lake for a new semantic model?

- (A) The gold data already sits as Delta tables in OneLake
- (B) Visuals must read live from an external Oracle database
- (C) You want Import-like query speed without a data-copying refresh
- (D) The data is in an on-premises database reached through a gateway
- (E) The workspace is on shared (Pro) capacity

Q32 SINGLE ANSWER

A product's price changes over time, and the PriceHistory table holds one row per product per change date. A measure must return the latest price within the selected period. Which function fits?

- (A) `CALCULATE(SUM(PriceHistory[Price]), LASTDATE('Date'[Date]))`
- (B) `LASTNONBLANKVALUE('Date'[Date], MAX(PriceHistory[Price]))`
- (C) `FIRSTNONBLANKVALUE('Date'[Date], MAX(PriceHistory[Price]))`
- (D) `AVERAGEX(VALUES('Date'[Date]), MAX(PriceHistory[Price]))`

Q33 SINGLE ANSWER

A disconnected table, SelectedRegions, drives a slicer. A measure must filter Sales by the regions chosen there, as if a relationship existed. Which function applies the filter?

- ☐ A RELATED(SelectedRegions[Region])
- ☐ B TREATAS(VALUEs(SelectedRegions[Region]), 'Geo'[Region])
- ☐ C USERELATIONSHIP(SelectedRegions[Region], 'Geo'[Region])
- ☐ D CROSSFILTER(SelectedRegions[Region], 'Geo'[Region], BOTH)

Q34 SINGLE ANSWER

A measure must number each customer's orders 1, 2, 3, ... by order date, restarting for every customer, in a table visual with Customer and OrderID. Which window function and argument should you use?

- ☐ A INDEX(1) with ORDERBY(OrderDate), PARTITIONBY(CustomerKey)
- ☐ B RANK(DENSE) with ORDERBY(OrderDate) and no PARTITIONBY
- ☐ C ROWNUMBER with ORDERBY(OrderDate), PARTITIONBY(CustomerKey)
- ☐ D OFFSET(-1) with ORDERBY(OrderDate), PARTITIONBY(CustomerKey)

Q35 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about calculation items, field parameters and dynamic format strings, select Yes if it is true.

Statement	Yes	No
1. A calculation item can change the format string of the measure it modifies.	☐	☐
2. A field parameter changes which field a visual shows. It doesn't change how a measure is calculated.	☐	☐
3. Dynamic format strings convert the measure's result to text, so visuals can no longer sort it numerically.	☐	☐

Q36 SINGLE ANSWER

A Direct Lake on SQL analytics endpoint model needs Order Date and Ship Date as separate role-playing date tables. Calculated tables aren't available in this variant. What should you do?

- ☐ A Create the second date table upstream and add it to the model
- ☐ B Import just the Ship Date table, in a composite model
- ☐ C Add a calculation group with Order and Ship items
- ☐ D Add a DAX calculated table that copies Date

Q37 SINGLE ANSWER

A measure loops over a 600-million-row fact with SUMX(Sales, IF([Customer Sales] > 1000, Sales[Amount])), where [Customer Sales] is a measure. What is the main performance problem, and a better approach?

- ☐ A Errors in IF; wrap the expression in IFERROR
- ☐ B IF inside SUMX; replace it with SWITCH(TRUE())
- ☐ C Model size; turn on large semantic model storage format
- ☐ D Context transition per fact row; iterate VALUEs(Sales[CustomerKey])

Q38 MULTIPLE RESPONSE — CHOOSE THREE

Which three statements about Delta table maintenance are true?

- ☐ A VACUUM rearranges the active files to speed up reads
 - ☐ B VACUUM removes data files that are no longer referenced and are older than the retention threshold
 - ☐ C OPTIMIZE deletes the table's time-travel history
 - ☐ D Z-ordering puts related values in the same files to improve data skipping
 - ☐ E OPTIMIZE compacts small files into larger ones
-

Q39 SINGLE ANSWER

You publish an Import model with an incremental refresh policy covering 10 years. The first refresh in the service times out. What is a recommended technique?

- ☐ A Remove the policy and use a full refresh instead
 - ☐ B Apply the policy, then refresh the partitions in batches through XMLA
 - ☐ C Shorten the archive range to one day, then widen it
 - ☐ D Switch the table to Direct Lake and keep the policy
-

Q40 SINGLE ANSWER

A Direct Lake on SQL model reads a complex SQL view, and every query against it falls back to DirectQuery. What change keeps those queries in Direct Lake?

- ☐ A Grant the users ReadAll on the warehouse
 - ☐ B Materialise the view as a gold Delta table loaded by the ETL
 - ☐ C Rewrite the view with CTEs instead of subqueries
 - ☐ D Set DirectLakeBehavior to DirectLakeOnly
-

Q41 SINGLE ANSWER

A DirectQuery model over a well-maintained warehouse uses relationships whose keys always match. Queries use outer joins. Which relationship setting lets the engine generate more efficient inner joins?

- ☐ A Mark as date table
- ☐ B Make this relationship active
- ☐ C Cross filter direction: Both
- ☐ D Assume referential integrity

Existing environment

Margie's Travel sells trips through 900 agents organised in a management hierarchy that's up to six levels deep. Bookings are written to a lakehouse, LH_Trips, by notebooks. A Direct Lake on OneLake semantic model serves agent and manager reports.

A hotel partner publishes nightly rate tables in Apache Iceberg format in an Amazon S3 bucket. Customer names arrive from three booking channels with inconsistent spelling. A nightly pipeline writes several gold tables in sequence.

Requirements

1. R1. Each manager must see bookings for themselves and everyone below them in the hierarchy.
2. R2. Partner rate tables must be queryable in LH_Trips without copying them.
3. R3. Customer records from the three channels must be matched, even when names are spelled slightly differently.
4. R4. After the nightly load, reports must show all the new data together, never a partly loaded mix of tables.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Q42 SINGLE ANSWER

Which RLS filter on DimAgent satisfies R1? The table has an AgentPath column built with PATH(AgentKey, ManagerKey).

- ☐ A PATHCONTAINS(DimAgent[AgentPath], USERPRINCIPALNAME())
- ☐ B PATHCONTAINS(DimAgent[AgentPath], LOOKUPVALUE(DimAgent[AgentKey], DimAgent[UPN], USERPRINCIPALNAME()))
- ☐ C DimAgent[UPN] = USERPRINCIPALNAME() || ISINScope(DimAgent[AgentPath])
- ☐ D DimAgent[ManagerKey] = LOOKUPVALUE(DimAgent[AgentKey], DimAgent[UPN], USERPRINCIPALNAME())

Q43 SINGLE ANSWER

Which approach to the partner rate tables satisfies R2?

- ☐ A A Dataflow Gen2 that reads them into a warehouse
- ☐ B A pipeline that converts them to Delta each night
- ☐ C Mirroring the S3 bucket into OneLake
- ☐ D An external shortcut to the Iceberg tables in S3

Q44 SINGLE ANSWER

Which Dataflow Gen2 feature best satisfies R3?

- ☐ A Append the channels, then remove duplicates
- ☐ B Merge queries with an exact match
- ☐ C Merge queries with fuzzy matching
- ☐ D Change type with locale on names

Q45 SINGLE ANSWER

Which configuration satisfies R4?

- ☐ A Switch the model to DirectQuery for live data
- ☐ B Schedule the model refresh for 06:00, after the usual finish
- ☐ C Keep automatic updates on so tables reframe as they land
- ☐ D Turn off automatic updates; refresh the model as the pipeline's last step

End of Set 09 — answers follow on the next page.

Set 09 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	C	M1		16	C	P1		31	A, C	S1	
2	Yes · No · No	M1		17	A	P1		32	B	S1	
3	D	M1		18	A	P2		33	B	S1	
4	D	M1		19	C	P2		34	C	S1	
5	B	M1		20	C	P2		35	Yes · Yes · No	S1	
6	C, E	M2		21	1→D 2→C 3→E 4→B	P2		36	A	S1	
7	A	M2		22	B	P2		37	D	S2	
8	B	M2		23	Yes · Yes · No	P2		38	B, D, E	S2	
9	E → D → B → A	M2		24	A	P2		39	B	S2	
10	1→E 2→C 3→D 4→B	M2		25	C	P3		40	B	S2	
11	D	M2		26	C	P3		41	D	S2	
12	A	P1		27	C	P3		42	B	M1	
13	A	P1		28	B	P3		43	D	P1	
14	Yes · Yes · No	P1		29	D	P3		44	C	P2	
15	A	P1		30	A	P3		45	D	S2	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: C M1 · Implement security and governance

External shortcuts pass authorisation to the connection's credential when they call the target. That's why you control end users on the Fabric side, with workspace roles, item permissions and OneLake security. Internal shortcuts are different: they check the calling user's identity.

Q2 Answer: Yes · No · No M1 · Implement security and governance

Admin is the only role that can delete the workspace or manage Admins. Members can add Members, Contributors and Viewers. Viewers can't create items.

Q3 Answer: D M1 · Implement security and governance

The SQL analytics endpoint supports granular T-SQL permissions such as GRANT and DENY on objects. Deleting the table or removing the users goes much too far, and labels don't block queries.

Q4 Answer: D M1 · Implement security and governance

With downstream inheritance, a label on an upstream item such as a semantic model is applied to downstream content, such as reports and dashboards built on it. A mandatory labelling policy makes authors pick a label but doesn't make it match the model's. Certification and deployment rules don't affect labels.

Q5 Answer: B M1 · Implement security and governance

SQL endpoint security only governs queries that go through the SQL endpoint. ReadAll gives direct OneLake access, where SQL RLS doesn't apply. Use OneLake security for rules that must hold across engines, or don't grant ReadAll.

Q6 Answer: C, E M2 · Maintain the analytics development lifecycle

XMLA write needs a supported capacity and the Read Write setting, plus enough permission on the workspace. Storage mode and Git integration don't matter. Viewers can't write.

Q7 Answer: A M2 · Maintain the analytics development lifecycle

When you connect a workspace, you choose a branch and a folder, so several workspaces can share one repository in separate folders. Connecting both to the root would mix and overwrite their files.

Q8 Answer: B M2 · Maintain the analytics development lifecycle

Select related adds an item's dependencies to the selection, so a consistent set is deployed together. Compare shows differences between stages, and history shows past deployments.

Q9 Answer: E → D → B → A M2 · Maintain the analytics development lifecycle

Parameters make the source configurable. Saving as a .pbid removes the data and keeps the parameter prompts, so authors fill in values when they open it. A .pbids or a deployment rule is a different mechanism.

Q10 Answer: 1→E 2→C 3→D 4→B M2 · Maintain the analytics development lifecycle

Each reuse and ALM format carries something different. .abf is an Analysis Services backup file, used when backing up and restoring through XMLA, and doesn't fit any of these descriptions.

Q11 Answer: D M2 · Maintain the analytics development lifecycle

Backup and restore of semantic models works through XMLA and needs the workspaces connected to an ADLS Gen2 account. The .abf file includes the data. Pipelines and Git move metadata only.

Q12 Answer: A P1 · Get data

Spark-first machine learning over semi-structured history fits a lakehouse. An Eventhouse is optimised for low-latency queries over streaming data. Warehouses hold structured tables and aren't built for Spark workloads.

Q13 Answer: A P1 · Get data

The OneLake catalog has a Govern tab with insights and recommended actions about the items you own, such as labelling and endorsement. The other surfaces are for streaming data, ALM or querying.

Q14 Answer: Yes · Yes · No P1 · Get data

Eventstreams are a no-code canvas for streaming: sources such as Event Hubs and IoT Hub, inline transformations, and several destinations. Code is optional, not required.

Q15 Answer: A P1 · Get data

A Spark job definition runs packaged Spark applications (Python, Scala or Java files) with arguments, on a schedule or from a pipeline. Notebooks are for interactive work. The other options aren't Spark runtimes.

Q16 Answer: C P1 · Get data

Activity dependency conditions (on success, on failure, on completion, on skip) control the flow in a pipeline. On completion runs whatever the outcome, and parallel activities ignore it. Hanging Teams off the notebook's failure path never fires when the Copy itself fails.

Q17 Answer: A P1 · Get data

SQL database in Fabric replicates its data to OneLake as Delta in near real time. Analytical queries can use that copy through the SQL analytics endpoint or Direct Lake, which keeps the load off the transactional database.

Q18 Answer: A P2 · Transform data

The Fabric Warehouse supports ACID transactions across several tables, so an explicit transaction gives all-or-nothing behaviour. Separate activities can partly fail, and the SQL analytics endpoint can't write.

Q19 Answer: C P2 · Transform data

Chained when and otherwise is how PySpark writes CASE logic. The conditions are checked in order, so the highest threshold has to come first. Testing > 1000 first labels every big spender Silver, in the when chain and in the nested IF alike. otherwise can only appear once.

Q20 Answer: C P2 · Transform data

to_date with a pattern that matches the input reads it correctly. The wrong pattern gives nulls. date_format converts a date to text, which is the opposite direction.

Q21 Answer: 1→D 2→C 3→E 4→B P2 · Transform data

Type 0 keeps the original value, Type 1 overwrites, Type 2 versions rows, and Type 3 keeps limited history in extra columns. Type 7 is a hybrid that combines Type 1 and Type 2 access paths, and it doesn't fit any of these descriptions.

Q22 Answer: B P2 · Transform data

Partition by store, order by date, and take the current row plus the six before it, which is seven days when there's one row per day. The other specifications partition or frame the rows wrongly.

Q23 Answer: Yes · Yes · No P2 · Transform data

The warehouse supports views, functions and stored procedures with full DML. The lakehouse SQL analytics endpoint supports views and other read-only objects, but it can't modify Delta tables. Those are written with Spark or other Delta writers.

Q24 Answer: A P2 · Transform data

regexp_replace with the pattern [^0-9] removes every character that isn't a digit. The [0-9] pattern does the opposite and deletes the digits. translate removes only the brackets. regexp_extract returns only the first run of digits, such as 020.

Q25 Answer: C P3 · Query and analyze data

WITH TIES returns extra rows that tie with the last row on the ORDER BY value. Plain TOP (5) cuts off at exactly five. PERCENT returns a share of the rows.

Q26 Answer: C P3 · Query and analyze data

countif counts the rows that meet a condition, and dividing by count() gives the failure share per hour. Using 100.0 avoids integer division. dcount counts distinct status values. Filtering to failures first makes both counts the same, so the result is always 100. countif alone gives a count, not a percentage.

Q27 Answer: C P3 · Query and analyze data

hourofday pulls out the hour number, so all days add up into 24 groups, and top 1 picks the busiest. bin(Timestamp, 1h) makes 720 separate hourly buckets, so it finds the busiest single hour, not hour of day. Sorting by Hour returns hour 23. dcount of the hour gives 24.

Q28 Answer: B P3 · Query and analyze data

CALCULATETABLE evaluates the table expression with the year filter applied. Passing Sales as a filter (an expanded-table filter) limits the result to customers that have sales rows in that context. The FILTER query references a Date column that isn't in the row context. CALCULATE returns a scalar. DAX has no WHERE clause.

Q29 Answer: D P3 · Query and analyze data

INFORMATION_SCHEMA.COLUMNS (or sys.columns) lists the column metadata for every table and view. sys.tables and INFORMATION_SCHEMA.TABLES search table names, not column names. dm_exec_requests shows running requests, not schema.

Q30 Answer: A P3 · Query and analyze data

Cross-database queries work within a workspace. A shortcut brings the other workspace's table into a local lakehouse without copying it, and from there it can be queried by name. Linked servers, OPENROWSET against another endpoint and workspace-qualified four-part names aren't supported.

Q31 Answer: A, C S1 · Design and build semantic models

Direct Lake is for Delta data in OneLake when you want fast, VertiPaq-class queries without copy-based refreshes. It needs a Fabric capacity, can't use gateways, and doesn't query external databases live.

Q32 Answer: B S1 · Design and build semantic models

LASTNONBLANKVALUE returns the expression's value at the last date that has data in context, which gives the latest price. LASTDATE takes the period's last calendar date, which is blank if the price didn't change that day. FIRSTNONBLANKVALUE gives the earliest price. An average of daily prices isn't the latest price.

Q33 Answer: B S1 · Design and build semantic models

TREATAS applies one table's values as a filter on another column, which works as a virtual relationship. USERELATIONSHIP and CROSSFILTER need an existing physical relationship. RELATED needs a row context and a relationship.

Q34 Answer: C S1 · Design and build semantic models

ROWNUMBER gives unique sequential numbers, and PARTITIONBY restarts the numbering per customer. RANK without a partition ranks across everyone. OFFSET and INDEX return rows, not sequence numbers, even with the right partition.

Q35 Answer: Yes · Yes · No S1 · Design and build semantic models

Calculation items can carry format string expressions. Field parameters swap fields. Dynamic format strings change only the display, and the value stays numeric. That's how they differ from FORMAT().

Q36 Answer: A S1 · Design and build semantic models

Direct Lake on SQL doesn't support calculated tables or composite models, so role-playing copies have to exist upstream as tables or views. A view may fall back to DirectQuery, so a small physical table is often better. Calculation groups don't create tables.

Q37 Answer: D S2 · Optimize enterprise-scale semantic models

Calling a measure inside an iterator over the fact table causes context transition on every row, which is hundreds of millions of times here. Iterating the distinct customers cuts the work to the number of customers. IF is allowed in SUMX, and SWITCH, IFERROR or storage format don't touch the real cost.

Q38 Answer: B, D, E S2 · Optimize enterprise-scale semantic models

OPTIMIZE improves the file layout, VACUUM reclaims storage, and Z-order improves data skipping. OPTIMIZE adds a new version and doesn't delete history; VACUUM is what eventually removes old files. VACUUM doesn't touch the layout of active files.

Q39 Answer: B S2 · Optimize enterprise-scale semantic models

The first incremental refresh loads all the history. With XMLA you can apply the policy to create the partitions and then refresh them in manageable batches. After that, regular refreshes only touch the incremental window.

Q40 Answer: B S2 · Optimize enterprise-scale semantic models

Direct Lake can only read Delta tables directly, so non-materialised views cause fallback, however the view is written. Persisting the result as a table makes it readable by Direct Lake. DirectLakeOnly turns the fallback into errors, and ReadAll has nothing to do with it.

Q41 Answer: D S2 · Optimize enterprise-scale semantic models

Assume referential integrity tells the engine that every fact key has a matching dimension row, so it can use inner joins in the SQL it sends to the source. Only turn it on when integrity really holds, or rows will go missing.

Q42 Answer: B M1 · Implement security and governance

PATHCONTAINS keeps every agent whose path includes the signed-in manager's key, which is the manager and everyone below them. The path holds keys, so testing it for a UPN never matches. Matching ManagerKey shows only direct reports, not the whole chain or the manager. ISINSCOPE describes visual grouping and has no place in an RLS filter.

Q43 Answer: D P1 · Get data

OneLake shortcuts can expose Iceberg tables held in supported storage, such as S3, as tables, without copying them. The metadata is virtualised so Fabric engines can read them. Copying breaks R2, and mirroring is for databases.

Q44 Answer: C P2 · Transform data

Fuzzy merge matches rows whose text values are similar rather than identical, with a similarity threshold you can tune and optional transformation tables. Exact merge and remove duplicates miss spelling variants, and locale settings affect type conversion, not matching.

Q45 Answer: D S2 · Optimize enterprise-scale semantic models

Triggering framing from the pipeline once all the tables are written moves every table to its new version together. With automatic updates on, reframing can happen part-way through the load. A fixed-time schedule guesses when the load ends. DirectQuery gives up Direct Lake and still shows partly loaded tables.

Set 10

Final mock exam

Sit this one last, under exam conditions, in one 100-minute session. It mixes every format and every sub-area at full difficulty.

45 questions

100 minutes

Pass mark used here: 34 / 45 (75%)

Level 5 · Final mock

12

Maintain a data analytics solution
25-30%

21

Prepare data
45-50%

12

Implement and manage semantic models
25-30%

Q1 MULTIPLE RESPONSE — CHOOSE TWO

A data scientist needs Spark read access to the tables of one lakehouse only, and nothing else in the workspace. Which two approaches meet the requirement with least privilege?

- ☐ A Share the lakehouse with the data scientist and grant ReadAll, with no workspace role
- ☐ B Share the lakehouse with Write permission
- ☐ C Give the workspace Member role
- ☐ D Give the workspace Viewer role plus a OneLake security role that grants read on the lakehouse's tables
- ☐ E Give the workspace Contributor role

Q2 SINGLE ANSWER

You created a OneLake security role that grants a group read on the Sales table only. Users in the group who were given ReadAll on the lakehouse can still read every table. What is the most likely reason?

- ☐ A The other tables are shortcuts, which ignore roles
- ☐ B The new role needs a Deny rule on the other tables
- ☐ C DefaultReader still grants them everything, and roles add up
- ☐ D OneLake roles only take effect after 24 hours

Q3 SINGLE ANSWER

The central governance team wants Fabric items to appear in Microsoft Purview Unified Catalog alongside Azure SQL and ADLS sources, with lineage and classification. What should they do?

- ☐ A Certify every item they want catalogued
- ☐ B Create a Fabric domain for each Azure source
- ☐ C Connect every workspace to a Git repository
- ☐ D Register and scan the Fabric tenant in Purview

Q4 DRAG AND DROP — MATCH

Match each requirement to the layer where the row filter should be defined.

Choices — each may be used once, more than once, or not at all

- ☐ A Semantic model RLS role
- ☐ B OneLake security role
- ☐ C SQL analytics endpoint security policy
- ☐ D Sensitivity label

Requirement	Choice
1. Both Spark notebooks and SQL queries on a lakehouse must enforce a region filter	
2. Only T-SQL users and a Direct Lake on SQL analytics endpoint model must honour the filter defined in the warehouse	
3. Consumers of an Import model through an app see only their own region	

Q5 SINGLE ANSWER

Support staff may query `dbo.Customer` in a warehouse but must be blocked from the `Email` column only. All other columns must stay queryable. Which statement does this most directly?

- ☐ A `DENY SELECT ON dbo.Customer TO Support;`
- ☐ B `ALTER TABLE dbo.Customer DROP COLUMN Email;`
- ☐ C `DENY SELECT ON dbo.Customer (Email) TO Support;`
- ☐ D `REVOKE CONNECT FROM Support;`

Q6 SINGLE ANSWER

Report authors must build new reports on a semantic model in the Finance workspace. They must not see or open any other item in that workspace. What should you grant?

- ☐ A Build permission on the model, with no workspace role
- ☐ B The Contributor role on the Finance workspace
- ☐ C Write permission on the model, with no workspace role
- ☐ D The Viewer role on the Finance workspace

Q7 SINGLE ANSWER

Several items in a Git-connected workspace have changed. A developer wants to commit only the semantic model and keep a half-finished notebook out of the commit. What can they do?

- ☐ A Move the notebook to another workspace first
- ☐ B Disconnect Git, commit the model, then reconnect
- ☐ C Select only the semantic model in the pane and commit
- ☐ D Commit everything; commits always include all items

Q8 SINGLE ANSWER

Notebooks deployed from Dev to Test must automatically attach to the Test lakehouse, not the Dev lakehouse. What should you configure in the deployment pipeline?

- ☐ A A Git branch per stage with its own lakehouse
- ☐ B A data source rule on the notebooks' reports
- ☐ C A default lakehouse rule for the notebooks in Test
- ☐ D A parameter rule on the Test dataflows

Q9 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about the XMLA endpoint, select Yes if it is true.

Statement	Yes	No
1. With the Read setting, client tools such as DAX Studio and Excel can query the model.	☐	☐
2. With the Read Write setting, SQL Server Management Studio can refresh individual partitions.	☐	☐
3. The XMLA endpoint is available for workspaces on shared (Pro) capacity.	☐	☐

Q10 SINGLE ANSWER

Impact analysis on a lakehouse lists 22 downstream reports. You have time to contact only the owners of the most-used ones before a change. Which information in impact analysis helps you prioritise?

- ☐ A The number of measures in each model
- ☐ B The view counts of the affected items
- ☐ C The colours used in each report
- ☐ D The items' sensitivity labels

Q11 SINGLE ANSWER

A workspace connected to Git contains some item types that Fabric Git integration doesn't support. What happens to those items?

- ☐ A The Git connection fails for the whole workspace
- ☐ B They're stored as binary files in the repository
- ☐ C They stay in the workspace but are left out of Git
- ☐ D They're deleted from the workspace on the next Update

Q12 SINGLE ANSWER

Sensor data in an Eventhouse must be fast to query for the last 30 days and kept for two years in total. Which settings should you configure on the KQL database or table?

- ☐ A A 30-day caching policy and a 730-day retention policy
- ☐ B A 730-day caching policy and a 30-day retention policy
- ☐ C V-Order on the KQL table, with a 730-day retention
- ☐ D A 30-day retention policy, with OneLake availability on

Q13 SINGLE ANSWER

A team wants to copy changed rows from a cloud database into a lakehouse every hour, incrementally, with a simple guided setup and no pipeline to design. Which Fabric item is designed for this?

- ☐ A A Spark job definition
- ☐ B A Copy job
- ☐ C A KQL queryset
- ☐ D A deployment pipeline

Q14 MULTIPLE RESPONSE — CHOOSE THREE

Which three Fabric items can be the target of an internal OneLake shortcut?

- ☐ A A deployment pipeline
 - ☐ B A lakehouse
 - ☐ C A KQL database
 - ☐ D A Power BI report
 - ☐ E A warehouse
-

Q15 SINGLE ANSWER

A pipeline must pull JSON from a REST API that returns 100 records per page with a next-page link. You want a no-code solution. What should you use?

- ☐ A A Dataflow Gen2 with a single Web call
 - ☐ B Mirroring of the API's database
 - ☐ C An external shortcut to the API URL
 - ☐ D A Copy activity with REST pagination rules
-

Q16 SINGLE ANSWER

SQL developers must INSERT and UPDATE the curated tables with T-SQL, and data scientists must read the same tables from Spark notebooks. Where should the curated tables live?

- ☐ A In a lakehouse, with the SQL developers writing through the SQL analytics endpoint
 - ☐ B In a warehouse; Spark reads its Delta tables through OneLake, for example through a shortcut or the OneLake path
 - ☐ C In an Eventhouse
 - ☐ D In two copies, one per team
-

Q17 SINGLE ANSWER

An Eventstream carries freezer temperature readings. Operations wants an email within seconds whenever a reading goes above -15 °C. What should you add?

- ☐ A An Activator destination with a threshold rule
 - ☐ B A KQL queryset with the threshold query saved
 - ☐ C A Power BI report subscription by email
 - ☐ D A pipeline that checks readings every hour
-

Q18 SINGLE ANSWER

A warehouse dimension needs integer surrogate keys for new members, continuing from the current maximum. Which pattern is reliable?

- ☐ A Use the business key as the surrogate key
- ☐ B MAX(existing key) + ROW_NUMBER() over the new rows
- ☐ C CAST(RAND() * 1000000 AS int) for each new row
- ☐ D NEWID() for each new row

Q19 SINGLE ANSWER

In PySpark, a developer generates surrogate keys with `monotonically_increasing_id()`. Which statement about the values is true?

- ☐ A They're consecutive integers starting from 1
- ☐ B They're GUIDs encoded as long integers
- ☐ C They're guaranteed identical across reruns
- ☐ D They're unique and increasing, but not consecutive

Q20 SINGLE ANSWER

A staging column `Quantity` is text, and a few rows contain values like 'N/A'. The load must convert valid values to int and turn invalid ones into NULL, without failing. Which expression should you use?

- ☐ A `ISNUMERIC(Quantity)`
- ☐ B `TRY_CAST(Quantity AS int)`
- ☐ C `CAST(Quantity AS int)`
- ☐ D `CONVERT(int, Quantity)`

Q21 SINGLE ANSWER

A notebook reads thousands of daily CSV files with `inferSchema=true`. It's slow, and column types sometimes change between runs. What should you do?

- ☐ A Set `samplingRatio` so inference reads fewer rows
- ☐ B Pass an explicit `StructType` schema to the reader
- ☐ C Read each file in a separate notebook run
- ☐ D Increase the number of executors

Q22 SINGLE ANSWER

An analyst in Dataflow Gen2 wants a new column that pulls the first name out of values like "Smith, Jane" by typing a couple of example outputs, rather than writing M code. Which feature should they use?

- ☐ A Column from examples
- ☐ B Merge queries with fuzzy matching
- ☐ C Conditional column
- ☐ D Fill down

Q23 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about star schemas in the gold layer, select Yes if it is true.

Statement	Yes	No
1. Fact tables hold foreign keys to dimensions plus numeric measures at a declared grain.	☐	☐
2. Dimension tables hold descriptive attributes used to filter and group.	☐	☐
3. For Power BI, a deeply snowflaked dimension is generally preferable to a flattened one.	☐	☐

Q24 SINGLE ANSWER

A monthly summary over a 6-billion-row fact is read by hundreds of report queries a day. As a view, it recalculates the aggregation every time and is slow. What is the better gold-layer design?

- ☐ A Move the aggregation into each report's measures
- ☐ B Turn the view into a stored procedure
- ☐ C Keep the view and add WITH (NOLOCK) hints
- ☐ D Persist the summary as a table, rebuilt after each load

Q25 SINGLE ANSWER

A daily sales fact has no rows on days a store was closed. A report must show every day, with 0 for closed days. What should the gold-layer query do?

- ☐ A Leave the gaps for the report to handle
- ☐ B Cross join DimDate with stores, LEFT JOIN the fact, COALESCE to 0
- ☐ C LEFT JOIN the fact to DimDate, with COALESCE to 0
- ☐ D INNER JOIN DimDate to the fact, with COALESCE to 0

Q26 SINGLE ANSWER

For each customer order, you need the number of days until that customer's next order. Which expression should you use?

- ☐ A DATEDIFF(day, OrderDate, LEAD(OrderDate) OVER (PARTITION BY CustomerKey ORDER BY OrderDate))
- ☐ B ROW_NUMBER() OVER (PARTITION BY CustomerKey ORDER BY OrderDate)
- ☐ C DATEDIFF(day, MIN(OrderDate), MAX(OrderDate))
- ☐ D DATEDIFF(day, OrderDate, LAG(OrderDate) OVER (ORDER BY OrderDate))

Q27 SINGLE ANSWER

Logs are split across the KQL tables Logs2025 and Logs2026. You need to query both as one, and know which table each row came from. Which query does this?

- ☐ A union withsource=SourceTable Logs2025, Logs2026 | where Level == "Error"
- ☐ B Logs2025 | union Logs2026 | where Level == "Error"
- ☐ C Logs2025 | join kind=fullouter Logs2026 on Timestamp
- ☐ D Logs2025 | lookup Logs2026 on Level | extend Source = "both"

Q28 SINGLE ANSWER

Which DAX query returns the single best-selling product in each region?

- ☐ A EVALUATE VALUES('Product'[Name])
- ☐ B EVALUATE TOPN(1, 'Product', [Total Sales])
- ☐ C EVALUATE SUMMARIZECOLUMNS('Geo'[Region], "Top", MAX('Product'[Name]))
- ☐ D EVALUATE GENERATE(VALUES('Geo'[Region]), TOPN(1, VALUES('Product'[Name]), [Total Sales], DESC))

Q29 SINGLE ANSWER

A table has 1,000 rows. MiddleName is NULL in 400 of them. What do COUNT(*) and COUNT(MiddleName) return?

- ☐ A 1,000 and 600
 - ☐ B 600 and 600
 - ☐ C 1,000 and 1,000
 - ☐ D 400 and 600
-

Q30 SINGLE ANSWER

A user built a query in the Visual Query Editor and wants to hand-edit the generated T-SQL to add a window function. What should they do?

- ☐ A View the generated SQL and open it as a new SQL query
 - ☐ B Add a window step from the editor's toolbar
 - ☐ C Save it as a view, then edit the view in the editor
 - ☐ D Open it in the DAX query view to edit
-

Q31 SINGLE ANSWER

A Direct Lake model needs an AgeBand attribute derived from BirthDate. The solution must work for both Direct Lake variants and must not depend on preview features. Where should AgeBand be created?

- ☐ A As a column in the gold Delta table
 - ☐ B As a calculated table of age bands
 - ☐ C As a DAX calculated column
 - ☐ D As a report-level measure
-

Q32 SINGLE ANSWER

A card must show 0 instead of blank when a region has no sales in the selected period. Which measure is the most concise?

- ☐ A [Total Sales] * 0
 - ☐ B COALESCE([Total Sales], 0)
 - ☐ C IF(ISERROR([Total Sales]), 0)
 - ☐ D BLANK()
-

Q33 SINGLE ANSWER

A measure must return the revenue of the first month in the selected period, as a baseline for an index chart. Which window function should you use?

- ☐ A CALCULATE([Revenue], OFFSET(1, ALLSELECTED('Date'[YearMonth])))
- ☐ B RANK(DENSE, ALLSELECTED('Date'[YearMonth]))
- ☐ C CALCULATE([Revenue], INDEX(1, ALLSELECTED('Date'[YearMonth]), ORDERBY('Date'[YearMonth], ASC)))
- ☐ D CALCULATE([Revenue], WINDOW(0, REL, 0, REL, ALLSELECTED('Date'[YearMonth])))

Q34 SINGLE ANSWER

In a Direct Lake model, a relationship joins FactSales[ProductKey] (bigint) to DimProduct[ProductKey] (string). Visuals that use the relationship fail. What should you do?

- ☐ A Set DirectLakeBehavior to Automatic for fallback
- ☐ B Make both columns the same type in gold, ideally integer
- ☐ C Make the relationship filter in both directions
- ☐ D Change the column's data type in the model view

Q35 SINGLE ANSWER

An analyst wants a slider from 0% to 20% in steps of 1%, which a measure then uses to project price increases. What does creating a numeric range (what-if) parameter generate?

- ☐ A A table related to the fact by percentage
- ☐ B A calculation group with 21 items
- ☐ C A field parameter table that uses NAMEOF
- ☐ D A GENERATESERIES table plus a SELECTEDVALUE measure

Q36 MULTIPLE RESPONSE — CHOOSE TWO

Which two of these are DAX window functions?

- ☐ A OFFSET
- ☐ B RELATED
- ☐ C EARLIER
- ☐ D LOOKUPVALUE
- ☐ E WINDOW

Q37 SINGLE ANSWER

You need to confirm whether V-Order is turned on for an existing gold Delta table in a lakehouse. What should you check?

- ☐ A The workspace's Spark pool settings
- ☐ B The table's row count in DESCRIBE DETAIL
- ☐ C The semantic model's DirectLakeBehavior
- ☐ D The table property delta.parquet.vorder.enabled

Q38 SINGLE ANSWER

A measure defines VAR TotalAll = [Sales] and then returns CALCULATE(TotalAll, ALL('Product')), expecting the all-products total. It returns the same value as [Sales]. Why?

- ☐ A ALL needs REMOVEFILTERS alongside it to work
- ☐ B ALL only works on the fact table, not on dimensions
- ☐ C A VAR is evaluated where it's defined, so CALCULATE can't change it
- ☐ D CALCULATE needs a measure reference, not a column

Q39 SINGLE ANSWER

In a composite model, slicers on DimCustomer (which relates to an Import aggregation and to a DirectQuery fact) send a DirectQuery query every time the page loads. Which change helps most?

- ☐ A Set DimCustomer to Dual storage mode
- ☐ B Remove the relationship to the aggregation
- ☐ C Turn off query reduction
- ☐ D Set DimCustomer to DirectQuery

Q40 HOT AREA — YES / NO FOR EACH STATEMENT

For each statement about Direct Lake guardrails, select Yes if it is true.

Statement	Yes	No
1. Capacities from F2 to F32 share the same per-table row guardrail.	☐	☐
2. Most guardrails are checked per query, while the maximum model size in memory is checked at the model level.	☐	☐
3. A Direct Lake model can use an on-premises data gateway to reach its data.	☐	☐

Q41 SINGLE ANSWER

A slicer on CustomerName lists 5 million distinct values, and the page loads slowly. What is the best design change?

- ☐ A Make the Customer relationship bidirectional
- ☐ B Turn the slicer into a table visual of all customers
- ☐ C Sort the slicer by name and turn on Select all
- ☐ D Filter on a lower-cardinality attribute, or use the filter pane

Case study · Coho Winery

Questions 42-45

Existing environment

Coho Winery operates vineyards in France, Spain and Italy. EU regulation requires all its analytics data to stay in the EU. One central semantic model is used by 30 thin reports owned by regional teams, each in its own workspace.

Sales by varietal (Merlot, Tempranillo, Sangiovese and others) are stored in a warehouse. A Direct Lake on OneLake model reads the gold lakehouse. Its fact table has grown past the capacity's row guardrail, and someone has set DirectLakeBehavior to Automatic, expecting that to keep reports working.

Once you move past this case study you cannot return to it. Read the requirements before the questions.

Requirements

1. R1. All Fabric data must remain in the EU.
2. R2. The central model must be changed and released independently of the 30 reports, and the reports must always use the governed model.
3. R3. A SQL report must show monthly sales with one column per varietal.
4. R4. Reports must work again after the fact table outgrew the guardrail.

Q42 SINGLE ANSWER

Which configuration satisfies R1?

- ☐ A Assign the workspaces to an EU domain
 - ☐ B Set the tenant's home region in each workspace
 - ☐ C Assign the workspaces to a capacity in an EU region
 - ☐ D Apply an EU-only sensitivity label
-

Q43 SINGLE ANSWER

Which model-and-report architecture satisfies R2?

- ☐ A Merge all 30 reports into the model's .pbix
 - ☐ B Give each region its own copy of the model
 - ☐ C Embed a copy of the model in each of the 30 reports
 - ☐ D One model in its own workspace; thin reports connect to it live
-

Q44 SINGLE ANSWER

Which T-SQL approach to the varietal columns satisfies R3?

- ☐ A A recursive CTE that walks the varietals
 - ☐ B `SELECT Month, Varietal, SUM(Amount) ... GROUP BY Month, Varietal`
 - ☐ C `SUM(CASE WHEN Varietal = 'Merlot' THEN Amount END) AS Merlot, ... GROUP BY Month`
 - ☐ D `SELECT Month, STRING_AGG(Varietal, ',') ... GROUP BY Month`
-

Q45 SINGLE ANSWER

Which statement about R4 is correct?

- ☐ A OneLake has no fallback; reduce the table's rows or use a larger capacity
- ☐ B Run `VACUUM`, then `OPTIMIZE`, to cut the row count
- ☐ C Set `DirectLakeBehavior` to `DirectLakeOnly` to raise the limit
- ☐ D Automatic already falls back to `DirectQuery`; nothing else is needed

End of Set 10 — answers follow on the next page.

Set 10 · Answer key

Q	Answer	Area	✓	Q	Answer	Area	✓	Q	Answer	Area	✓
1	A, D	M1		16	B	P1		31	A	S1	
2	C	M1		17	A	P1		32	B	S1	
3	D	M1		18	B	P2		33	C	S1	
4	1→B 2→C 3→A	M1		19	D	P2		34	B	S1	
5	C	M1		20	B	P2		35	D	S1	
6	A	M1		21	B	P2		36	A, E	S1	
7	C	M2		22	A	P2		37	D	S2	
8	C	M2		23	Yes · Yes · No	P2		38	C	S2	
9	Yes · Yes · No	M2		24	D	P2		39	A	S2	
10	B	M2		25	B	P2		40	Yes · Yes · No	S2	
11	C	M2		26	A	P3		41	D	S2	
12	A	P1		27	A	P3		42	C	P1	
13	B	P1		28	D	P3		43	D	M2	
14	B, C, E	P1		29	A	P3		44	C	P3	
15	D	P1		30	A	P3		45	A	S2	

Sub-area	Questions	Your score
M1 · Implement security and governance	6	
M2 · Maintain the analytics development lifecycle	6	
P1 · Get data	7	
P2 · Transform data	8	
P3 · Query and analyze data	6	
S1 · Design and build semantic models	6	
S2 · Optimize enterprise-scale semantic models	6	
Total — 34 or more is a pass on this paper	45	

Scoring: one mark per question. For multiple-response, Yes/No, matching and build-list items, score the fraction of parts you got right (the real exam gives partial credit on multi-part items and never penalises a wrong answer).

Explanations

Q1 Answer: A, D M1 · Implement security and governance

Item sharing with ReadAll limits access to that lakehouse and allows Spark reads. A Viewer with a OneLake security role gets the same through roles. Contributor, Member and Write all grant write access well beyond reading.

Q2 Answer: C M1 · Implement security and governance

OneLake security roles add up. DefaultReader gives ReadAll users access to everything, so a narrower role makes no difference until DefaultReader is changed or those users are taken out of it. Roles grant access; there are no Deny rules to add. They apply straight away, and shortcuts don't bypass them.

Q3 Answer: D M1 · Implement security and governance

Scanning the Fabric tenant from Purview brings Fabric metadata into the enterprise catalogue and Data Map, next to other sources. Certification and Git don't publish metadata to Purview. Domains organise workspaces within Fabric.

Q4 Answer: 1→B 2→C 3→A M1 · Implement security and governance

Define the filter at the layer every access path goes through. OneLake security covers all engines, SQL policies cover SQL access (including Direct Lake on SQL with SSO), and model RLS covers model consumers. Labels never filter rows.

Q5 Answer: C M1 · Implement security and governance

A column-level DENY blocks that column, and any query that references it, while the rest of the table stays available. Denying the whole table or revoking CONNECT goes too far. Dropping the column destroys the data.

Q6 Answer: A M1 · Implement security and governance

Build on the item is enough to create reports, including in their own workspaces, and exposes nothing else. Any workspace role shows every item. Write would let them change the model.

Q7 Answer: C M2 · Maintain the analytics development lifecycle

The source control pane lets you choose which changed items to commit, so unfinished work stays uncommitted. You don't need to delete or disconnect anything.

Q8 Answer: C M2 · Maintain the analytics development lifecycle

Notebook deployment rules can change the default lakehouse per stage, so the same notebook runs against each stage's lakehouse. Data source and parameter rules apply to other item types.

Q9 Answer: Yes · Yes · No M2 · Maintain the analytics development lifecycle

Read allows queries and metadata browsing. Read Write adds management operations, such as processing partitions and deploying. The XMLA endpoint needs Fabric, Premium or Premium Per User capacity.

Q10 Answer: B M2 · Maintain the analytics development lifecycle

Impact analysis shows usage, such as views, alongside the affected workspaces and items, so you can focus on the reports that matter most. Labels and model contents don't tell you about usage.

Q11 Answer: C M2 · Maintain the analytics development lifecycle

Git integration syncs only supported item types. Other items stay in the workspace untouched and don't appear in the repository. Check the current list of supported items before relying on Git for a given item type.

Q12 Answer: A P1 · Get data

The caching policy decides how much recent data is kept in the hot cache for the fastest queries. The retention policy decides how long data is kept at all. Swapping the two values deletes data after 30 days. V-Order applies to Delta writes from Spark, not to KQL tables, and OneLake availability doesn't extend retention.

Q13 Answer: B P1 · Get data

Copy job is a simplified item for data movement that supports full and incremental copies on a schedule, without building a pipeline. Spark job definitions run code. The other two don't move data.

Q14 Answer: B, C, E P1 · Get data

Internal shortcuts point at data in Fabric items, such as lakehouses, warehouses, KQL databases and mirrored databases. Reports and pipelines don't hold tabular data that a shortcut can reference.

Q15 **Answer: D** P1 · Get data

The Copy activity's REST source supports pagination rules that follow next-page links until there are none left. Shortcuts and mirroring don't connect to REST APIs. A single web call returns only the first page.

Q16 **Answer: B** P1 · Get data

T-SQL writes need a warehouse. Warehouse tables are Delta in OneLake, so Spark can read them without a copy. A lakehouse SQL endpoint can't write data.

Q17 **Answer: A** P1 · Get data

Activator watches streaming data and fires actions, such as email or Teams messages, when conditions are met, in real time. Hourly pipelines, saved querysets and report subscriptions can't react within seconds.

Q18 **Answer: B** P2 · Transform data

Adding ROW_NUMBER to the current maximum gives compact, sequential integer keys for the new rows. NEWID gives GUIDs, which are wide, unsupported in Direct Lake and compress poorly. RAND can collide. Business keys tie the model to the source system.

Q19 **Answer: D** P2 · Transform data

monotonically_increasing_id encodes the partition ID in the upper bits, so the values are unique and increasing, but with large gaps between partitions. If you need consecutive keys, use row_number over a window, plus an offset from the existing maximum.

Q20 **Answer: B** P2 · Transform data

TRY_CAST returns NULL when the conversion fails, instead of raising an error. CAST and CONVERT fail the whole statement on 'N/A'. ISNUMERIC returns a flag, not a converted value.

Q21 **Answer: B** P2 · Transform data

An explicit schema avoids the extra pass inferSchema needs and pins the column types, so they don't drift from run to run. More executors don't fix inconsistent types.

Q22 **Answer: A** P2 · Transform data

Column from examples works out the transformation from sample outputs you type, then writes the M expression for you. A conditional column needs rules you define yourself, and the other features join or fill values.

Q23 **Answer: Yes · Yes · No** P2 · Transform data

Facts hold keys and measures, and dimensions hold descriptive context. Power BI guidance favours flattened (star) dimensions over snowflakes, for simpler models and fewer relationship hops.

Q24 **Answer: D** P2 · Transform data

A view recalculates on every query. When the same expensive aggregate is read often, storing it once per load is cheaper and faster. Direct Lake can also read a table but not a view. A stored procedure can't be queried like a table, and locking hints don't make the aggregation cheaper.

Q25 **Answer: B** P2 · Transform data

Building the full date-by-store grid first, then left joining the facts, makes sure every day appears, and COALESCE turns the missing sales into zeros. An inner join keeps the gaps. Left joining the fact to DimDate keeps only the days that have sales, and DimDate alone has no store dimension.

Q26 Answer: A P3 · Query and analyze data

LEAD gets the next order date within the customer's partition, and DATEDIFF measures the gap. LAG looks backwards, and without a partition it mixes customers. MIN and MAX give one span per group, not one per order.

Q27 Answer: A P3 · Query and analyze data

union stacks tables, and withsource adds a column naming each row's source table. A plain union works but doesn't say where each row came from. Joins and lookups combine columns, not rows.

Q28 Answer: D P3 · Query and analyze data

GENERATE evaluates TOPN once per region, in that region's context, which gives the top product in each. TOPN on its own returns one product overall. MAX of the name returns the alphabetically last name, not the best seller.

Q29 Answer: A P3 · Query and analyze data

COUNT(*) counts rows. COUNT(column) counts the non-NULL values in that column. That difference often explains reconciliation gaps.

Q30 Answer: A P3 · Query and analyze data

The Visual Query Editor shows the T-SQL it generates. Users can open that as a SQL query and extend it by hand, which is useful when the visual steps can't express a window function. The visual editor has no window step, and the DAX query view works on semantic models, not the warehouse.

Q31 Answer: A S1 · Design and build semantic models

Calculated columns and tables aren't supported in Direct Lake on SQL, and their support in Direct Lake on OneLake has been limited or in preview. Creating the attribute in the gold layer works everywhere and for every consumer.

Q32 Answer: B S1 · Design and build semantic models

COALESCE returns the first non-blank argument, so blank becomes 0. ISERROR checks for errors, not blanks. Multiplying by 0 returns 0 even when there are sales.

Q33 Answer: C S1 · Design and build semantic models

INDEX(1, ...) returns the first row of the ordered relation, which is the first selected month, whatever the current row is. OFFSET(1) is the next row. WINDOW(0, REL, 0, REL) is the current row. RANK returns a number, not a revenue.

Q34 Answer: B S1 · Design and build semantic models

Direct Lake needs related columns to have matching data types. Fix the type upstream, preferably as integers for compression. Filter direction and fallback settings don't fix a type mismatch.

Q35 Answer: D S1 · Design and build semantic models

A what-if parameter creates a disconnected table from GENERATESERIES and a measure such as SELECTEDVALUE(Parameter[Value], default). Your measures then use it. It doesn't relate to the facts.

Q36 Answer: A, E S1 · Design and build semantic models

The window function family is INDEX, OFFSET, WINDOW, RANK and ROWNUMBER, and they share the relation, ORDERBY and PARTITIONBY parameters. EARLIER refers to an outer row context, RELATED follows a relationship, and LOOKUPVALUE finds a value by matching columns.

Q37 Answer: D S2 · Optimize enterprise-scale semantic models

V-Order is controlled at session, table or write level, and the table property records the setting for the table. The other settings aren't related to V-Order.

Q38 Answer: C S2 · Optimize enterprise-scale semantic models

A VAR is a constant that's already been evaluated in the outer context, so CALCULATE's filters can't affect it. Write CALCULATE([Sales], ALL('Product')), either directly or inside the variable's definition. This trap catches performance tuners and correctness reviewers alike.

Q39 Answer: A S2 · Optimize enterprise-scale semantic models

Dual lets the engine answer slicer and aggregation queries from memory, while still joining to the DirectQuery fact when it needs to. DirectQuery storage would send every slicer query to the source.

Q40 Answer: Yes · Yes · No S2 · Optimize enterprise-scale semantic models

Guardrails step up from F64. Most limits apply per query, but model memory is a model-level limit. Direct Lake reads OneLake directly and can't use any gateway.

Q41 Answer: D S2 · Optimize enterprise-scale semantic models

A slicer has to fetch its distinct values, and millions of them are expensive to query and render. Filtering by lower-cardinality attributes, or using search and the filter pane, keeps pages responsive.

Q42 Answer: C P1 · Get data

Data residency in Fabric follows the region of the capacity a workspace is assigned to. Labels and domains are governance metadata and don't control where data is stored.

Q43 Answer: D M2 · Maintain the analytics development lifecycle

The thin-report pattern keeps one governed model with its own lifecycle, and reports connect to it live. Copies of the model drift apart, and a single .pbix ties every report's release to the model's.

Q44 Answer: C P3 · Query and analyze data

Conditional aggregation turns rows into columns with plain SUM(CASE ...) expressions, which work in any SQL engine. GROUP BY Month, Varietal keeps varieties as rows, and STRING_AGG makes one text list. No static SQL can add columns by itself, so a new varietal means extending the column list, or pivoting in the semantic model instead.

Q45 Answer: A S2 · Optimize enterprise-scale semantic models

Only Direct Lake on the SQL analytics endpoint can fall back to DirectQuery. Direct Lake on OneLake has no fallback, so a guardrail breach still fails whatever DirectLakeBehavior says, and no setting raises the guardrail. VACUUM and OPTIMIZE don't reduce the rows in active files.